CS 700 Midterm Notes Mon 10/26, open book and notes, no electronics

1. Metrics (Lecture 2)

Good metrics are linear, reliable, repeatable, easy to use, consistent and independent. Reliable, repeatable, consistent and independent are required. Linear and easy to use are nice but not necessary.

MIPS $= \dfrac{n}{T_e \times 10^6}$, MFLOPS $= \dfrac{f}{T_e \times 10^6}$. MIPS fails reliable and consistent (CISC vs RISC, CPI). MFLOPS gives an integer program 0 and a FLOP isn't defined the same everywhere. SPEC normalizes to a basis machine and takes a geometric mean, which is the wrong mean for times.

Means metrics (clock rate, MIPS, MFLOPS) count what was done. Ends metrics (execution time) count what got accomplished. Execution time is the fundamental metric. Report wall clock and CPU time.

Speedup of system 2 over 1 is $S_{2,1} = \dfrac{T_1}{T_2} = \dfrac{R_2}{R_1}$. Relative change is $\Delta_{2,1} = \dfrac{T_1 - T_2}{T_2} = S_{2,1} - 1$. Precision $= \dfrac{\text{relevant retrieved}}{\text{retrieved}}$, recall $= \dfrac{\text{relevant retrieved}}{\text{all relevant}}$.

2. Summarizing data (Lecture 4)

Center

Mean $\bar{x} = \frac{1}{n}\sum x_i$, median is the middle sorted value (average the two middle ones when n is even), mode is the most frequent value or bin. Use the median when the data is skewed or has outliers, the mode for categorical data, and the mean when the total matters.

Harmonic $\bar{x}_H = \dfrac{n}{\sum 1/x_i}$ for rates whose numerator (work) is fixed.

Geometric $\bar{x}_G = \left(\prod x_i\right)^{1/n}$ for ratios that compound or normalized values. Arithmetic for times, sizes and anything whose sum means something.

The mean of a rate should track total work over total time. If every program does the same work F, then the right average rate is $\dfrac{nF}{\sum t_i}$, which is the harmonic mean of the rates.

Weighted forms, with $\sum w_i = 1$, $\bar{x}_A = \sum w_i x_i$, $\bar{x}_H = 1/\sum \frac{w_i}{x_i}$, $\bar{x}_G = \prod x_i^{w_i}$. Always $\bar{x}_H \le \bar{x}_G \le \bar{x}_A$, equal only when all values are equal.

Spread

Range $= x_{\max} - x_{\min}$. Sample variance $s^2 = \dfrac{\sum (x_i - \bar{x})^2}{n-1} = \dfrac{\sum x_i^2 - n\bar{x}^2}{n-1}$. Standard deviation s. Coefficient of variation COV $= s/\bar{x}$ (unitless). IQR $= Q_3 - Q_1$, SIQR $= \dfrac{Q_3 - Q_1}{2}$. Midhinge $= \dfrac{Q_1 + Q_3}{2}$. Use s with the mean, SIQR or IQR with the median.

Quantiles, course convention

Sort $x_{(1)} \le \cdots \le x_{(n)}$ and give $x_{(i)}$ the plotting position $f_i = \dfrac{i - 0.5}{n}$. For a quantile q, find i with $f_i \le q \le f_{i+1}$ and interpolate linearly between $x_{(i)}$ and $x_{(i+1)}$. Histogram bins by Sturges, $k = \lfloor 1 + \log_2 n \rfloor$ on the slide (HW3 rounded instead, so say which you used). Excel's PERCENTILE on the slides interpolates at $1 + (n-1)p$, which gives slightly different quartiles.

Normal Q-Q plot

Plot $\left(z_i,\ x_{(i)}\right)$ with $z_i = \Phi^{-1}(f_i)$, or rescale to $m + s\,z_i$ so both axes share units. A straight line means normal. Ends bending away from the line mean heavy tails, and a curve bending one way means skew.

3. Confidence intervals (Lecture 6)

Mean

Large n (slides say $n > 30$) $\quad \bar{x} \mp z_{\alpha/2}\dfrac{s}{\sqrt{n}}$.

Small n, normal population $\quad \bar{x} \mp t_{[\alpha/2,\,n-1]}\dfrac{s}{\sqrt{n}}$.

All decks use upper-tail notation, $z_{\alpha/2}$ or $z[\alpha/2]$ is the value with area $\alpha/2$ above it, which the CI deck also calls the $(1-\alpha/2)$-quantile. Same number, e.g. 1.96 for 95%. The p.50 table labels them $z_{0.95}$, $z_{0.975}$. For $n = 15$ use df 14, $t = 2.145$.

Common levels $\quad 90\% \to 1.645$, $95\% \to 1.960$, $99\% \to 2.576$.

Batch means split a correlated series into batches and treat batch means as the sample. A 95% CI means 95% of intervals built this way cover μ. It does not mean μ has a 95% chance of being in this particular interval, and it says nothing about 95% of the data. The width shrinks like $1/\sqrt{n}$, so halving it takes four times as many runs.

How many measurements

For a half-width of e as a fraction of the mean, $n = \left(\dfrac{z_{\alpha/2}\,s}{e\,\bar{x}}\right)^2$, rounded up. Slide example $s = 2.14$, $\bar{x} = 7.94$, 90%, $e = 0.035$ gives $n = 160.5 \to 161$. The slide's "within 7%" is the full width, so the half-width is 3.5%.

Proportion

$p = m/n$, $\quad p \mp z_{\alpha/2}\sqrt{\dfrac{p(1-p)}{n}}$, valid when $np \ge 10$ (slide) and $n(1-p) \ge 10$ (study guide). Wald interval, poor for small n. Sample size with e as a RELATIVE error (slides) $n = \dfrac{z_{\alpha/2}^2\,p(1-p)}{(e\,p)^2}$. Slide example $e = 0.005$, $p = 0.1097$ gives 1.25 million 10 ms samples, 3.46 hours. With an absolute half-width e (measurement study guide) drop the p in the denominator. Slide example $m = 658$ of $n = 6000$, so $p = 0.1097$ and the 95% CI is 10.2% to 11.8%.

Variance (normal data only)

$\left(\dfrac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}},\ \dfrac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}}\right)$. The larger χ^2 quantile goes in the left denominator.

4. Hypothesis testing (Lecture 9)

Steps state H_0 and H_1, choose α and one or two sides BEFORE the data, compute the statistic, compare with the critical value or find p, then report the effect with an interval.

$Z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$, $\quad t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ with $n-1$ df. Two-sided, reject if $|Z| > z[\alpha/2]$. One-sided, reject if $Z > z[\alpha]$. A two-sided test at α rejects exactly when the $(1-\alpha)$ CI excludes μ_0.

p is the probability of a statistic at least this extreme IF H_0 is true. It's not the probability that H_0 is true.

	H_0 true	H_0 false
Reject	Type I (α)	correct (power $1-\beta$)
Don't reject	correct	Type II (β)

Runs needed to detect a difference δ $\quad n = \left(\dfrac{(z[\alpha/2] + z[\beta])\,\sigma}{\delta}\right)^2$. At $\alpha = 0.05$ and power 0.80, $(1.96 + 0.84)^2 \approx 7.8$, so $n \approx 7.8\,\sigma^2/\delta^2$.

Not rejecting is not proof of no difference. With m tests at α each, expect about $m\alpha$ false positives, and Bonferroni uses α/m per test. Holm sorts the p values and compares the k-th smallest with $\alpha/(m-k+1)$, stopping at the first failure. Benjamini-Hochberg compares the k-th smallest with $k\alpha/m$ and controls the false discovery rate. Correlated samples count for less, e.g. 400 samples with $n_{\text{eff}} = 100$ halve Z. Statistical significance says nothing about whether the effect matters, so give the effect size and its CI.

5. Comparing two alternatives (Lecture 10)

Pick the test

1. Same unit (workload, benchmark) under both? Yes means paired, no means unpaired.
2. Paired with roughly normal differences, paired t on the differences.
3. Paired with outliers, Wilcoxon signed-rank if symmetric, sign test or bootstrap if skewed.
4. Unpaired and roughly normal, Welch's t.
5. Unpaired and skewed or ordinal, Mann-Whitney U or bootstrap.

For paired tests, normality is about the differences, not A or B.

Paired

$d_i = a_i - b_i, \quad \bar{d} \mp t[\alpha/2,\, n-1]\,\dfrac{s_d}{\sqrt{n}}, \quad n$ is the number of PAIRS.

Pairing removes the shared variation $2\rho\sigma_A\sigma_B$ from the variance of the difference, so it helps whenever $\rho > 0$.

Unpaired, Welch

$$s = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}, \quad (\bar{x}_A - \bar{x}_B) \mp t[\alpha/2,\, \nu]\, s,$$

$$\nu = \frac{\left(\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}\right)^2}{\frac{(s_A^2/n_A)^2}{n_A - 1} + \frac{(s_B^2/n_B)^2}{n_B - 1}}$$

Use ν as is or round DOWN (more conservative). It lies between $\min(n_A, n_B) - 1$ and $n_A + n_B - 2$. Pooled t assumes equal variances and uses $n_A + n_B - 2$ df. Welch is the default.

Reading results

If the CI for the difference contains 0, you can't say they differ at that level. Overlapping single-system CIs do NOT mean no difference (HW6 Miller example), and checking overlap is not a test. Runs needed paired $n = \big((z[\alpha/2] + z[\beta])\, s_d/\delta\big)^2$, unpaired per group $n = 2\big((z[\alpha/2] + z[\beta])\, s/\delta\big)^2$.

6. Measurement (Lecture 11)

A CI describes sampling error and nothing else. Repetition shrinks random error but never systematic error such as bias, a wrong clock or perturbation. Strategies event counting gives exact totals with tiny overhead but no context, tracing gives full detail and order but the most perturbation and data, sampling sets the overhead you choose but is statistical and misses rare events. Exactness, context and low overhead, you get two. Report the clock and its resolution. Finer observation means more perturbation. Timer rollover after $(2^n - 1)\, T_c$ for an n-bit counter with tick T_c. A single reading is off by up to one tick, so time k repetitions and divide, $(T_k - T_{\text{loop}})/k$. Overhead share of a measurement is about $2c/(T + 2c)$ for read cost c.

7. Plagiarism and authorship (full research topics in Part 2)

Plagiarism per Luke's TR includes copied text without quotes even when cited, close paraphrase, reused figures, and self-plagiarism of your own published text. Cite ideas, quote words. Authorship goes to people who made an intellectual contribution.

8. More formulas from the handouts and study guides

Metrics

$T_e = \dfrac{n \times \text{CPI}}{f_{\text{clock}}}$ (not on the slides). GFLOPS and TFLOPS divide by 10^9 and 10^{12}. Jain's fairness $J = \dfrac{(\sum x_i)^2}{n \sum x_i^2}$ (HW2 only).

Summarizing

Quantile q sits at position $r = nq + 0.5$ in the sorted data, interpolate between $x_{(\lfloor r \rfloor)}$ and the next value. Tukey fences $Q_1 - 1.5\,\text{IQR}$ and $Q_3 + 1.5\,\text{IQR}$, beyond them is an outlier. Work-weighted harmonic mean uses $w_i = F_i / \sum F_j$. Inverse CDFs for Q-Q plots, $u = f_i$ exponential with mean a, $-a \ln(1-u)$ (rate $\lambda = 1/a$) Pareto, $k(1-u)^{-1/a}$ geometric, $\lceil \ln(1-u)/\ln(1-p) \rceil$ (Summ p.64 writes $\ln u$, which is the random-variate form) normal, $z \approx 4.91\,[q^{0.14} - (1-q)^{0.14}]$, then $m + s\,z$. Feitelson puts the data on the x-axis, Setia and HW3 on the y-axis, so slope rules read reversed. $\ln 0.9 = -0.1054$, $\ln 0.7 = -0.3567$, $\ln 0.5 = -0.6931$, $\ln 0.3 = -1.2040$, $\ln 0.1 = -2.3026$.

Confidence intervals

Batch means with m batches $\bar{b} \mp t[\alpha/2,\, m-1]\, s_b/\sqrt{m}$. Sample size scales as $n_2 = n_1 (h_1/h_2)^2$ for half-widths h. Proportion size with ABSOLUTE error e $n = z_{\alpha/2}^2\, p(1-p)/e^2$, use $p = 0.5$ with no pilot (95%, $e = 0.03$ gives 1068). Log transform build the CI on $\ln x$, then (e^L, e^U) is an interval for the median, centred on the geometric mean. Bootstrap percentile 95% CI is the 2.5th and 97.5th percentiles of the B resampled statistics. Monte Carlo $\bar{Y}(n) = (b-a)\frac{1}{n}\sum g(X_i)$, $\text{Var} = \text{Var}(Y)/n$.

Hypothesis testing

Two proportions $Z = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}}$, pooled $\hat{p} = \dfrac{m_1 + m_2}{n_1 + n_2}$. McNemar (paired yes/no) $Z = \dfrac{b - c}{\sqrt{b + c}}$, only the disagreements b and c count. Paired $t = \bar{d}/(s_d/\sqrt{n})$. Power $\approx P\big[Z > z[\alpha/2] - |\delta|\sqrt{n}/\sigma\big]$. Family-wise error $1 - (1-\alpha)^m$, e.g. 23% at $m = 5$, 92% at $m = 50$. Simultaneous intervals use $z[\alpha/(2m)]$. Correlated samples $n_{\text{eff}} = n\dfrac{1-\rho}{1+\rho}$. Pocock cutoffs for peeking 2.18 (2 looks), 2.41 (5 looks). Chi-square goodness of fit $\sum (O-E)^2/E$ with $k - m$ df (m includes the 1 for the total plus fitted parameters).

Comparing alternatives

Pooled $s_p^2 = \dfrac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}$, Cohen's $d = (\bar{x}_A - \bar{x}_B)/s_p$, paired $d_z = \bar{d}/s_d$. Signed-rank, large n $\mu_W = n(n+1)/4$, $\sigma_W^2 = n(n+1)(2n+1)/24$. Sign test $S \sim \text{Bin}(n, \frac{1}{2})$, $z = (S - n/2)/(\sqrt{n}/2)$. Mann-Whitney $U_1 = W_1 - n_1(n_1+1)/2$, $\mu_U = n_1 n_2/2$, $\sigma_U^2 = n_1 n_2 (n_1 + n_2 + 1)/12$, $A_{12} = U_1/(n_1 n_2)$. All pairs of c systems $1 - (1-\alpha)^{c(c-1)/2}$ chance of a false positive.

Measurement

Wrapped counter ticks = end + 2^n − start. Instrumentation overhead $(T_{\text{inst}} - T_{\text{base}})/T_{\text{base}}$. Ratio error $(\sigma_R/R)^2 \approx (\sigma_A/A)^2 + (\sigma_B/B)^2$. Unpaired SE of a difference $\sqrt{SE_A^2 + SE_B^2}$. Nested runs, p processes with k iterations each $\text{Var}(\bar{x}) = \sigma_p^2/p + \sigma_i^2/(pk)$, cost $c_p p + c_i pk \le B$. Extra processes beat extra iterations.

Student t critical values, upper-tail area

$t[\alpha/2,\, \nu]$ for a two-sided $(1-\alpha)$ CI, e.g. 95% uses the .025 column, 90% uses .05.

df	.10	.05	.025	.01	.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
35	1.306	1.690	2.030	2.438	2.724
40	1.303	1.684	2.021	2.423	2.704
50	1.299	1.676	2.009	2.403	2.678
60	1.296	1.671	2.000	2.390	2.660
80	1.292	1.664	1.990	2.374	2.639
100	1.290	1.660	1.984	2.364	2.626
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

Square roots

2	1.414	27	5.196	52	7.211	77	8.775
3	1.732	28	5.292	53	7.280	78	8.832
4	2.000	29	5.385	54	7.348	79	8.888
5	2.236	30	5.477	55	7.416	80	8.944
6	2.449	31	5.568	56	7.483	81	9.000
7	2.646	32	5.657	57	7.550	82	9.055
8	2.828	33	5.745	58	7.616	83	9.110
9	3.000	34	5.831	59	7.681	84	9.165
10	3.162	35	5.916	60	7.746	85	9.220
11	3.317	36	6.000	61	7.810	86	9.274
12	3.464	37	6.083	62	7.874	87	9.327
13	3.606	38	6.164	63	7.937	88	9.381
14	3.742	39	6.245	64	8.000	89	9.434
15	3.873	40	6.325	65	8.062	90	9.487
16	4.000	41	6.403	66	8.124	91	9.539
17	4.123	42	6.481	67	8.185	92	9.592
18	4.243	43	6.557	68	8.246	93	9.644
19	4.359	44	6.633	69	8.307	94	9.695
20	4.472	45	6.708	70	8.367	95	9.747
21	4.583	46	6.782	71	8.426	96	9.798
22	4.690	47	6.856	72	8.485	97	9.849
23	4.796	48	6.928	73	8.544	98	9.899
24	4.899	49	7.000	74	8.602	99	9.950
25	5.000	50	7.071	75	8.660	100	10.000
26	5.099	51	7.141	76	8.718		

Standard normal $\Phi(z) = P(Z \le z)$

$P(Z > z) = 1-\Phi(z)$ and $\Phi(-z) = 1-\Phi(z)$. Critical values $z[.05] = 1.645$, $z[.025] = 1.960$, $z[.01] = 2.326$, $z[.005] = 2.576$, $z[.20] = 0.842$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

χ^2 quantiles, $P(\chi^2_\nu \le x)$

df	0.005	0.025	0.05	0.95	0.975	0.995
1	0.00	0.00	0.00	3.84	5.02	7.88
2	0.01	0.05	0.10	5.99	7.38	10.60
3	0.07	0.22	0.35	7.81	9.35	12.84
4	0.21	0.48	0.71	9.49	11.14	14.86
5	0.41	0.83	1.15	11.07	12.83	16.75
6	0.68	1.24	1.64	12.59	14.45	18.55
7	0.99	1.69	2.17	14.07	16.01	20.28
8	1.34	2.18	2.73	15.51	17.53	21.95
9	1.73	2.70	3.33	16.92	19.02	23.59
10	2.16	3.25	3.94	18.31	20.48	25.19
11	2.60	3.82	4.57	19.68	21.92	26.76
12	3.07	4.40	5.23	21.03	23.34	28.30
13	3.57	5.01	5.89	22.36	24.74	29.82
14	4.07	5.63	6.57	23.68	26.12	31.32
15	4.60	6.26	7.26	25.00	27.49	32.80
16	5.14	6.91	7.96	26.30	28.85	34.27
17	5.70	7.56	8.67	27.59	30.19	35.72
18	6.26	8.23	9.39	28.87	31.53	37.16
19	6.84	8.91	10.12	30.14	32.85	38.58
20	7.43	9.59	10.85	31.41	34.17	40.00
21	8.03	10.28	11.59	32.67	35.48	41.40
22	8.64	10.98	12.34	33.92	36.78	42.80
23	9.26	11.69	13.09	35.17	38.08	44.18
24	9.89	12.40	13.85	36.42	39.36	45.56
25	10.52	13.12	14.61	37.65	40.65	46.93
26	11.16	13.84	15.38	38.89	41.92	48.29
27	11.81	14.57	16.15	40.11	43.19	49.64
28	12.46	15.31	16.93	41.34	44.46	50.99
29	13.12	16.05	17.71	42.56	45.72	52.34
30	13.79	16.79	18.49	43.77	46.98	53.67
40	20.71	24.43	26.51	55.76	59.34	66.77
50	27.99	32.36	34.76	67.50	71.42	79.49
60	35.53	40.48	43.19	79.08	83.30	91.95

Advisors, CS 701, publishing and ethics

These four lectures have no math and no homework. Expect true/false, "is this acceptable and why", classify-the-misconduct and one h-index calculation. Every fact below has a cite like (CS701 p.4), which is the PDF page of that deck. Luke's plagiarism tech report GMU-CS-TR-2013-8 is cited as (Luke TR p.N). A myth-vs-truth table for all four decks is at the end.

0. Why these lectures are in CS 700 (Intro deck)

The big idea

CS 700 used to be only the "systems" course on experimental methods and statistics. In 2018 the PhD program was restructured to cut time to completion, so the course now also teaches how research works and helps match you with a faculty member for CS 701 (Intro p.2, p.3). That's why advisor, publishing and ethics lectures sit next to confidence intervals.

Facts

- CS 700 is the required "systems" counterpart to the required theory courses CS 583 and CS 600. Its subject is experimental methods and quantitative techniques in CS (Intro p.2).
- Course topics are an overview of the research process (meeting CS faculty, matching for CS 701) plus the "old CS 700" of measurement, simulation and statistical methods (Intro p.3).
- PhD-process topics are what a PhD is and why get one, finding an advisor and working well with one, picking a dissertation topic, finding a job (academia vs industry), research ethics and integrity, how and when to use AI, and tools like LaTeX, plotting and scripting. Reading papers, reviewing papers and giving talks are left to the advanced seminar courses, CS 800 (Intro p.4).
- Early weeks use the second half of class to discuss one of these topics, with pointers to CRA workshop talks. Faculty research talks run through the semester to help you find a 701 match. Guest lectures at the end cover experimental techniques in areas like software engineering and ML (Intro p.5).
- Scientific method has four steps. Identify a problem and form a hypothesis that is testable and refutable, design an experiment, conduct it, then do hypothesis testing with statistics (Intro p.9).
- Feitelson (2006) gives three meanings of experimental CS. Building systems (hardware or software) as the counterpart to theory, experimentation as a feedback step in the engineering loop, and evaluating computer systems with the rigorous methods of the natural sciences. The third one is the focus of this class. Feitelson also argues there's a place for just observing the real world, like analyzing measured network traffic (Intro p.10).
- The figure on Intro p.11 compares the scientific method loop (observation, hypothesis or model, concrete prediction, experimental test) with the system design loop (idea, system design, concrete implementation, experimental evaluation).
- Course goals include understanding the trade-offs between simulation, measurement and analytical modeling, rigorously comparing systems in the presence of noise (often "quality" of output matters more than speed, like face recognition), and deciding if a change has a statistically significant impact (Intro p.13). Also using statistics to cut the number of simulations and designing experiments to get the most information for the effort (Intro p.14), drawing conclusions from noisy data (Intro p.15) and aggregating large data for a big-picture view (Intro p.16).
- Grading is 40% homework, 35% midterm, 25% term project. All work is individual under the Mason Honor Code. Lost homework points can be won back by resubmitting within two weeks. The term project should ideally use experimental methods tied to your 701 research (Intro p.8).
- Office hours Wednesday 1.30 to 2.30 PM in Nguyen 5305, GTA Mohammadreza Noei, Piazza for assignment discussion, material on Canvas (Intro p.8).
- Textbook is Lilja, "Measuring Computer Performance". Jain is the alternative and somewhat dated (Intro p.7).
- LaTeX is required for all homework submissions (Intro p.26).

Likely exam questions

1. **Q.** What are the three definitions of experimental CS, and which one is this course about?
   A. Building systems, experimentation as feedback in the engineering loop, and evaluating systems with the rigorous methods of the natural sciences. This course is about the third one (Intro p.10).
2. **Q.** You get 90 on homework, 80 on the midterm, 85 on the project. What's your grade?
   A. Rule. Grade is the weighted mean $0.40\,H + 0.35\,M + 0.25\,P$ (weights from Intro p.8). Plug in $H = 90$, $M = 80$, $P = 85$. Step 1, homework $0.40 \times 90 = 36$. Step 2, midterm $0.35 \times 80 = 28$. Step 3, project $0.25 \times 85 = 21.25$. Step 4, add $36 + 28 + 21.25 = 85.25$. Result 85.25. Check, it must land between 80 and 90 since the weights sum to 1.
3. **Q.** Why was CS 700 changed in 2018?
   A. The PhD program was restructured to reduce time to completion, so the course now also gives a broader introduction to the research process (Intro p.2).
4. **Q.** What must a hypothesis be in the scientific method?
   A. Testable and refutable (Intro p.9).

1. Selecting your PhD advisor (Advisor deck)

The big idea

Picking your research advisor is the biggest decision of your PhD. You can't rush it. Most professors want to work with you informally for a semester or more first, so both sides can see if it's a good fit. Start early, do your homework on their papers, and don't open with "will you be my advisor?".

Facts

- The talk is based on a presentation by Prof. Hakan Aydin, director of the CS PhD program (Advisor p.2).
- Every PhD student gets an **academic** advisor at admission, usually someone with compatible interests. The actual **research** advisor is declared by separate paperwork within the first **24 credits**, which is a hard deadline (Advisor p.3).
- It's the most important decision of the PhD. It may decide your success in the program, how long the PhD takes and your long-term career path. Choosing the advisor and choosing the research area go hand in hand (Advisor p.4).
- "Easy" case. You were already in close contact with a faculty member during admission, so they're probably your academic advisor and maybe already your GRA boss. You know each other. You still go through an initial research phase before officially declaring, and the checklist still helps (Advisor p.5).
- Common case. You have a broad idea of your interests but should stay open-minded. Find faculty in those or close areas, read their recent projects and papers, and contact them after doing that homework (Advisor p.6).
- Three misconceptions are on Advisor p.7. Delaying the search to the last semester because the deadline is 24 credits. Thinking faculty X will agree just because you want X. Thinking faculty will answer "can you be my advisor?" right away or within a few weeks. All three are wrong.
- **PRCP** means Preliminary Research Co-Operation Phase. It's an unofficial early research involvement before a professor agrees to advise you, unless they already know you from prior publications or class work. It may take one semester or more and lets both sides judge fit (Advisor p.8).

- PRCP is a term Prof. Aydin made up for the talk. It's not an official procedure, but nearly all faculty agree an initial phase matters (Advisor p.9).
- PRCP parts are weekly or biweekly meetings, discussing old and new papers, discussing ideas, group research meetings and working on a real (maybe limited) project. CS 701 in spring, mandatory for PhD students, is a great formal PRCP (Advisor p.10).
- Fall timing. Use Fall to plan your first research involvement, attend research talks (including the CS 700 ones), request meetings with faculty, explain your background and interests, ask how to get involved. Don't ask "can you be my PhD advisor?" this early (Advisor p.11).
- Faculty side. Dedicated students with good backgrounds and work ethic are the most important asset for research-active faculty. They still may decline because they're already over-committed with advisees (advising takes a lot of time) or they saw a mismatch in the PRCP (Advisor p.12).
- Positive signs faculty look for (the more the better) are good background in the area, willingness to learn by reading and discussing papers, being independent and open to brainstorming, proposing new good ideas, and being able to join their actual research right away even in a limited way (Advisor p.13).
- Student side. During the PRCP look for answers to key questions before committing. Don't expect all answers after one meeting or a few weeks (Advisor p.14).
- You'll work closely with this person for the rest of your PhD. Ask if their research approach and style match you, and if not, whether you can adapt (Advisor p.15).
- "Making progress in research" is vague. Peer-reviewed research is the ultimate objective. Ask what submission or publication numbers and pace they expect and if you're comfortable with it, but don't underestimate how productive you'll become (Advisor p.16).
- Ask how they interact with students (regular meetings, as needed, guidance by a postdoc), whether you want a hands-on or hands-off advisor (each has pros and cons), and funding odds if becoming a GRA next year matters to you (Advisor p.17).
- Ask how they pick students' research projects and dissertation topics, and how much the student gets a say (Advisor p.18).
- Co-advising. Two faculty can co-advise, but only one is declared the official advisor. The thesis names both. Their areas are usually related but complementary. Ask if they're open to it and have done it before. It has pros and cons (Advisor p.19).
- Changing advisor is technically possible and sometimes unavoidable, but it's a last resort. That's why the informal early phase is so important (Advisor p.20).
- Wrap-up. Most faculty take their time before accepting new students. Think carefully yourself, do your homework and show you're committed once you start talking with someone (Advisor p.21).
- More advice is in the CRA-W Grad Cohort workshop talks online ("Resources from Past Years"). The department may sponsor you to attend (Advisor p.22).

Likely exam questions

1. **Q.** True or false. The academic advisor assigned at admission is your official research advisor.
   A. False. The research advisor is declared with separate paperwork within the first 24 credits (Advisor p.3).
2. **Q.** True or false. Since the deadline is 24 credits, I can wait until my last semester to look for an advisor.
   A. False. It's the first misconception on Advisor p.7. The informal PRCP alone can take a semester or more (Advisor p.8), so waiting makes you miss the deadline.
3. **Q.** What is the PRCP and is it an official GMU procedure?
   A. It's an informal phase of working together, one semester or more, before a professor agrees to advise you, so both sides can check fit (Advisor p.8). It's not official. Prof. Aydin made up the name (Advisor p.9).
4. **Q.** Give two reasons a professor may say no to a good student.
   A. They're already over-committed with PhD advisees, since advising takes a lot of time, or they noticed a mismatch during the PRCP (Advisor p.12).
5. **Q.** A student emails a professor in week 2 asking "Can you be my PhD advisor?" and waits for a reply before contacting anyone else. What's wrong?
   A. Asking that this early is what Advisor p.11 says to avoid. Instead read their papers and ask for a meeting about getting involved (Advisor p.6, p.11). Expecting a quick reply is a misconception (Advisor p.7), and faculty take their time (Advisor p.21).
6. **Q.** With co-advising, how many official advisors are there?
   A. One. Only one is declared official, but the thesis names both (Advisor p.19).

2. CS 701 and the CS 700 term project (CS701 deck)

The big idea

CS 701 is a required spring course where each first-year PhD student does a semester-long research project with one faculty supervisor. There are no classes. You and your supervisor sign a form that works like a contract with the department, then you do real hands-on research, write a report and give a short talk. The form is due Oct 31, so you need a supervisor now.

Facts about CS 701

- Full name is Research Experience in Computer Science. Mandatory for all first-year PhD CS students. Offered **only in Spring**. Must be passed with **B or better**. Prerequisite is **B or better in CS 700** (CS701 p.2).
- Each student pairs with a CS faculty supervisor for a semester-long project. The point is to interact with faculty and get research experience (CS701 p.3).
- At the end the student submits a **report to the supervisor**, every 701 student gives a short public talk of about **10 minutes** to the whole department, and the **supervisor submits the grade** to the registrar (CS701 p.3).
- Registration. Find a CS faculty member willing to supervise. Fill out the CS 701 form with the project details (description and deliverables). Student and supervisor sign it. Submit it to `csphd@gmu.edu` and `aydin@cs.gmu.edu`. The department must approve it and may ask for revisions (CS701 p.4).
- The form is a **3-way contract** among the student, the supervisor and the CS department (CS701 p.4).
- After approval the department gives you a **CRN** (course reference number), and you use it to register on **PatriotWeb**. The form, revisions and getting the CRN all take time (CS701 p.5).
- Who can supervise. Any **tenure-track or tenured** CS faculty, meaning assistant, associate or full professors. Not people whose directory title says "Teaching" or "Instructor" (CS701 p.6).
- You can have a **co-supervisor**, who co-signs the form. CS-affiliated faculty can co-supervise. Even with a co-supervisor, the registrar records the course under one CS faculty member (CS701 p.6).
- Email the form to `csphd@gmu.edu` with the subject line "[CS701 form]" and cc `aydin@cs.gmu.edu`. Submitting on time is critical. **Deadline Oct 31, 2026** (CS701 p.7).
- **No regular class meetings.** Student and supervisor meet periodically or as needed. Any required meeting pattern (weekly, biweekly) goes on the form. Periodic meetings help you stay in touch and discuss progress and unclear points (CS701 p.8).
- The project doesn't have to be groundbreaking or immediately publishable. But the effort must match a **3-credit semester-long course** for first-year PhD students. It's much more than a

regular course's term project. Don't expect to finish by "sprinting" in the last few weeks. Timely planning and progress in spring is critical (CS701 p.9).
- Literature review or paper reading is likely needed first and is an essential part of many projects. But the project **can't be only** a literature review or survey (CS701 p.10).
- A **hands-on** (implementation) part is required beyond the literature review. Software, simulations or experiments. For a Theoretical CS project, working on a proof for a non-trivial open problem or a non-trivial novel complexity analysis is also OK (CS701 p.11).
- Implementing a set of techniques from possibly several papers and evaluating them thoroughly, possibly with new performance metrics, is fine. Proposing new techniques or extending existing ones is also reasonable. Discuss details with the supervisor (CS701 p.12).
- Form field "Description of the research activity". Must be **more than a title**, about one paragraph (attach sheets if needed), and must state the hands-on part (CS701 p.13).
- Form field "Deliverables and other requirements". A **project report is the default and mandatory** deliverable. List other requirements like software or hardware artefacts and periodic meetings. The report goes to the supervisor at the end of spring. Ask the supervisor about format and length (CS701 p.14).
- A conference or journal submission completed by the end of spring **can't** be a deliverable. "A draft paper to be revised and submitted for publication in the future" **is** OK, and the report can then be that draft paper (CS701 p.15).
- A successful 701 can be the first step to an advisor relationship, but that's not the primary intent. Many students end up with a different area and advisor than their 701 (CS701 p.18).
- Suggested steps (CS701 p.27). (1) Identify a few CS faculty in your preferred areas. (2) Contact them and request meetings about possible projects. (3) Collect information from the ones who are available. (4) Decide, fill out the form, get the faculty signature and submit it to `csphd@gmu.edu` and `aydin@cs.gmu.edu`. If nobody in the first group is available, pick a second group (maybe a different area) and repeat.

The 701 misconceptions, myth then truth

- *Myth.* My 701 supervisor automatically becomes my PhD advisor. *Truth.* No. Declaring the PhD advisor is a separate process with a separate form within the first 24 credits (CS701 p.16).
- *Myth.* My dissertation will be in the area of my 701 project. *Truth.* Not necessarily. You can do 701 in area X and the dissertation in area Y (CS701 p.17).
- *Myth.* My first-choice professor will agree. *Truth.* Not necessarily, they may be unavailable for various reasons (CS701 p.19).
- *Myth.* After I contact a professor, we'll meet and decide quickly. *Truth.* No guarantee (CS701 p.20).
- *Myth.* I should wait for the first professor's answer before contacting a second. *Truth.* You shouldn't. Contact several faculty and learn about different projects, to save time (CS701 p.21).
- *Myth.* I can work with only one faculty member. *Truth.* You can have co-supervisors, but only one is the official primary supervisor, and that one must be tenured or tenure-track CS faculty. The registrar records the primary. The co-supervisor can be CS-affiliated faculty, is named next to the primary on the form and co-signs it. You can acknowledge the co-supervisor in the report and talk (CS701 p.22).
- *Myth.* I can wait until late October to search. *Truth.* That would be a big mistake (CS701 p.23).
- *Myth.* If I'm a GRA, my sponsor will be my 701 supervisor. *Truth.* That's most natural but not required (CS701 p.24).
- *Myth.* If I can't find a project in my favorite area, I'll take 701 next year. *Truth.* Not possible, it's a catalog rule (CS701 p.25).
- *Myth.* If I can't find a supervisor, the department will assign one. *Truth.* No. A CS faculty member has to agree to supervise (CS701 p.26).

Facts about the CS 700 term project

- **Individual** project. Pick an experimental CS problem with quantitative analysis of a computer system, technique, algorithm or method. Comparing several techniques quantitatively is also fine. Do the experiments or analysis and write a report (CS701 p.28).
- Must include **experiments and/or simulations**, and you must apply class statistics like confidence intervals and regression (CS701 p.29).
- Ideally pick a topic in your research area, even better tied to your 701, but that's **not required and not graded** (CS701 p.29).
- **Not acceptable** are a literature survey of quantitative or statistical techniques, and applying statistics to polls or surveys (CS701 p.29).
- **1-page proposal** as a PDF on Canvas. It briefly covers the problem or algorithms, the experimental techniques and the methodology. The slide says due midnight Sunday Oct 25, but it was moved to **Nov 2** (CS701 p.30).
- **Status update by Nov 15**, reporting any changes or unexpected problems (CS701 p.30).
- **Dec 7** class is for **5-minute** presentations covering the problem, the techniques you implemented or simulated, and preliminary results (CS701 p.31).
- Final report is a **6-8 page** technical report in an **ACM or IEEE** conference template, submitted on Canvas **by Dec 7** (CS701 p.32).

Likely exam questions

1. **Q.** When is CS 701 offered, what grade is needed, and what's the prerequisite?
   A. Spring only, B or better, and B or better in CS 700 (CS701 p.2).
2. **Q.** A draft form lists an "Assistant Professor, Teaching" as supervisor, the description "Federated learning", the work "read and summarize 15 papers", and the deliverable "paper submitted to INFOCOM by May". Fix every problem.
   A. The supervisor must be tenured or tenure-track, and "Teaching" titles don't qualify (CS701 p.6). That person could only co-supervise and co-sign next to a tenure-line CS primary (CS701 p.22). The description is just a title and must be a paragraph naming the hands-on part (CS701 p.13). A literature review alone isn't allowed (CS701 p.10), so add implementation, simulation or experiments (CS701 p.11). A conference submission can't be a deliverable, but "a draft paper to be revised and submitted in the future" can (CS701 p.15). The mandatory report must be listed (CS701 p.14).
3. **Q.** Who gets the CS 701 form, and what happens after approval?
   A. Email it to csphd@gmu.edu with "[CS701 form]" in the subject and cc aydin@cs.gmu.edu (CS701 p.7). The department approves it or asks for a revision (CS701 p.4), then gives you a CRN to register on PatriotWeb (CS701 p.5).
4. **Q.** True or false. The 701 project must be publishable, so it can be small.
   A. False both ways. It doesn't need to be groundbreaking or publishable, but the effort must match a full 3-credit course, much more than a term project (CS701 p.9).
5. **Q.** Which is acceptable as a CS 700 term project? (a) CIs on a survey of 40 classmates' IDE preference. (b) A review of how papers use regression. (c) Timing two hash tables 30 runs each at four load factors, compared with CIs and regression.
   A. Only (c). (a) applies statistics to a poll, and (b) is a literature survey of statistical techniques, both banned (CS701 p.29). (c) is an individual quantitative comparison with experiments

and class methods (CS701 p.28-29).

6. **Q.** Give the term project deadlines.
   A. Proposal Nov 2 (slide says Oct 25), status update Nov 15, 5-minute talk Dec 7, 6-8 page ACM or IEEE report Dec 7 (CS701 p.30-32).

3. Publication practices in CS (Pub deck)

The big idea

A PhD means doing original research good enough to pass peer review, where other experts judge your paper before it's published. In CS the main venues are conferences, and they're very competitive. Good results aren't enough, because there are a fixed number of slots and writing and organization count too. Rejection is normal, so you fix the paper and resubmit. Never send the same paper to two places at once.

Facts

- A PhD needs original research ending in a dissertation you write and defend. The research must be worthy of publication in peer-reviewed journals and/or conference proceedings (Pub p.2).
- **Peer review** means experts ("peers") review articles before they're officially published or presented. The aim is to keep science honest by filtering out invalid or poor-quality work. Reviewers evaluate independently. Journal editors or conference chairs make the final decision, sometimes after discussion (Pub p.3).
- Why publish. It shows other experts, not just you, your advisor or your group, approve your work. It makes the work widely accessible. Your publication record is your main track record. The slide flags quality vs quantity (Pub p.4).
- Long-term impact is often correlated with the number of **citations**. Important papers can get hundreds or thousands. Citation counts are used to judge researchers and even venues (Pub p.5).
- **h-index** is the largest h such that an author (or journal) has at least h papers each cited at least h times. Slide example, citations 10, 9, 8, 6, 3, 1 (already sorted). Rank 1, $10 \geq 1$ yes. Rank 2, $9 \geq 2$ yes. Rank 3, $8 \geq 3$ yes. Rank 4, $6 \geq 4$ yes. Rank 5, $3 \geq 5$ no. So $h = 4$, since four papers have 4 or more but there aren't five with 5 or more. PhD students' h-index usually isn't high even at the end, but it grows as you publish more good papers (Pub p.6).
- How to compute h. Sort citations from high to low. Go down the list. h is the last rank i where the count is still $\geq i$. The answer is a number of papers, not a citation count.
- Most PhD students start publishing peer-reviewed papers during or after their **3rd year**. It takes time to build background, find a problem, solve it, learn to write a scholarly paper and beat other researchers (Pub p.7).
- Knowing the process early helps you appreciate the competition and build reading, evaluating, writing and presenting skills alongside your research (Pub p.8).
- Venue types are conferences, journals and workshops (Pub p.10).
- Conferences solicit papers on a topic. The **call for papers (CFP)** gives dates, topics, formatting rules and the organizing committee. Authors submit by a deadline and wait for accept or reject. The decision includes comments from **anonymous reviewers** who are on the **program committee (PC)** (Pub p.11).
- Format per the CFP, meaning page limits, font size limits and IEEE or ACM templates. Deadlines might be extended, less likely for first-tier conferences, so never rely on it (Pub p.12).
- **Acceptance rate** is the percent of papers accepted, so rate = accepted ÷ submitted ×100%. For example 60 accepted out of 400 submitted is $60/400 = 0.15$, which is 15%, inside the first-tier range. Lower rate generally means more prestige. First-tier conferences are usually **10-25%**, sometimes under 10%. Your advisor can tell you how competitive venues are, since picking the right venue isn't trivial (Pub p.13).
- Decision process. The PC chair assigns papers to PC members (professors or industry researchers). Each writes an independent review scoring **writing quality, organization, technical contribution and novelty**, with a final recommendation of Strong Accept, Accept, Neutral, Reject or Strong Reject. Note the impact of non-technical factors (Pub p.14).
- The PC meets to discuss all papers and decide, which can take all day or several days. Each conference has a **fixed number of slots** (about 30 to 1000+). Often not all above-threshold papers fit, so some quite reasonable papers get rejected. The PC in a sense "orders" the good papers (Pub p.15).
- Big conferences have **Area Chairs (ACs)** appointed by the PC chair, each overseeing reviews in a sub-area (Pub p.16).
- **Rebuttal**. Increasingly, reviews are released before the final decision so authors can clear up misunderstandings. Not every paper gets this, many are directly accepted or rejected. Author replies are considered in the decision (Pub p.17).
- Accepted papers are resubmitted as a **camera-ready copy** by another deadline, with reviewer comments included when possible. Authors present them at the conference, which is a chance to network and travel (Pub p.18).
- On rejection, don't take it personally or treat it like failing an exam. Use the comments to improve, and talk with your advisor and co-authors about revising and resubmitting to another venue or the same conference's next edition (Pub p.19).
- **Dual submission** is sending the same or mostly the same paper to several venues at once, usually to raise acceptance odds, or trying to get a second paper from the same material. Not allowed. It's often caught and authors get severe sanctions (Pub p.20).
- **Anonymity**. Authors never know the reviewers. In **double-blind** review, increasingly common, reviewers don't know the authors either. The CFP then asks authors to leave off names and affiliations and avoid indirect hints. A non-compliant paper is rejected right away (Pub p.21). The slide asks why it's spreading. The usual answer is so a famous name or school can't sway the reviewers and the paper is judged on content.

The 8 publication misconceptions (Pub p.9), all false

1. A publication is basically a graduate-level project report.
2. If the results are correct, my paper will be accepted.
3. If the results are correct, important and novel, it will be accepted.
4. If my advisor says "it's good", it will be accepted.
5. If my paper is rejected, the conference made an error or is biased.
6. My paper was rejected, so all that time and effort was wasted.
7. Only publishing matters, the venue isn't important.
8. The number of publications matters, not quality.

Why they're false. Slots are fixed so good papers get rejected (Pub p.15), non-technical things like writing and organization are scored (Pub p.14), rejection comments help you improve and resubmit (Pub p.19), venue prestige differs (Pub p.13), and quality counts over quantity (Pub p.4).

Likely exam questions

1. **Q.** Find the h-index for citations 5, 15, 2, 6, 11, 0, 6, 1.
   A. Rule. h is the largest h with at least h papers cited at least h times (Pub p.6). Step 1, sort high to low to get 15, 11, 6, 6, 5, 2, 1, 0. Step 2, compare each count with its rank. Rank 1, $15 \geq 1$ yes. Rank 2, $11 \geq 2$ yes. Rank 3, $6 \geq 3$ yes. Rank 4, $6 \geq 4$ yes. Rank 5, $5 \geq 5$ yes. Rank 6, $2 \geq 6$ no, so stop. Step 3, the last rank that passed is 5. Result $h = 5$. Five papers have at least 5 citations, but there aren't six with at least 6.
2. **Q.** An author has 3 papers with 500 citations each. What's h?
   A. Rule as above (Pub p.6). Step 1, sorted list is 500, 500, 500.

Step 2, rank 1, $500 \geq 1$ yes. Rank 2, $500 \geq 2$ yes. Rank 3, $500 \geq 3$ yes. There's no rank 4 paper, so h can't be 4. Result $h = 3$. Huge citation counts can't push h past the number of papers.

3. **Q.** True or false. If my results are correct, important and novel, a good conference will accept it.
   A. False. It's a listed misconception (Pub p.9). There's a fixed number of slots so reasonable papers get rejected (Pub p.15), and writing quality and organization are scored too (Pub p.14).
4. **Q.** Name the four review criteria and the five recommendations.
   A. Writing quality, organization, technical contribution and novelty. Strong Accept, Accept, Neutral, Reject, Strong Reject (Pub p.14).
5. **Q.** What must authors do for double-blind review, and what if they don't?
   A. Leave names and affiliations off and avoid indirect hints about who they are. A non-compliant paper is rejected right away (Pub p.21).
6. **Q.** A student submits the same paper to two conferences to improve the odds, planning to withdraw one. Is that OK?
   A. No. That's dual submission, which isn't allowed, is often caught and gets severe sanctions (Pub p.20, Ethics p.14).

4. Research integrity and ethics (Ethics deck, plus Luke TR)

The big idea

You must do research honestly. The three big sins are making up data (fabrication), changing or dropping data (falsification) and using others' words or ideas without credit (plagiarism). Coursework falls under the Honor Code, but research and your dissertation fall under GMU Policy 4007. Citing doesn't make copied words OK, you need quotes or your own words. AI is usually allowed if you disclose it and take full responsibility.

Facts on rules and who governs what

- The PhD goal is original research and contributions to your field, but done ethically and with the highest integrity (Ethics p.2).
- Coursework and academic activities are governed by the **GMU Honor Code**. Research activities, including dissertation research, are governed by **GMU Policy 4007** (Misconduct in Research and Scholarship) (Ethics p.3).
- IEEE, ACM and IFIP each have a Code of Ethics and Professional Conduct for their submissions and publications (Ethics p.4). Much of the deck is adapted from the IEEE Author Ethics Guidelines (Ethics p.5).
- IEEE keeps a list of authors penalized for ethics violations. They can't publish for extended periods, and IEEE informs their institutions, which usually add their own sanctions (Ethics p.6).
- Policy 4007 misconduct is monitored and investigated by GMU's **Office of Research Integrity and Assurance (ORIA)** (Ethics p.15).
- Policy 4007 covers all full-time and part-time employees and anyone affiliated with GMU, graduate students doing research or scholarship producing reports, conference papers, publications or creative works that list GMU affiliation, undergraduates in sponsored research, and anyone at GMU who teaches, researches or does scholarship (Ethics p.16).
- For graduate students, academic misconduct goes **solely through the Honor Code**, except (1) research activities, whether sponsored or not, and (2) master's theses and doctoral dissertations. Those two go through Policy 4007 (Ethics p.16).
- All research misconduct allegations are handled under 4007. The university aims for an impartial, unbiased process. Investigators are chosen for relevant expertise and screened beforehand for bias or personal, professional or financial conflicts of interest. Everyone must keep it confidential and cooperate (Ethics p.17).
- **Research misconduct** means fabrication, falsification or plagiarism in proposing, performing or reviewing research, or in reporting results. Fabricating or falsifying credentials on funding applications and manuscripts can count. It **does not include honest error or differences of opinion** (Ethics p.18).
- 4007 sets rules to receive and review allegations, runs inquiries and investigations (committees, hearings) and communicates results (Ethics p.19).
- GMU offers in-person **Responsible Conduct in Research (RCR)** training through ORIA (Ethics p.21).

FFP definitions (Policy 4007, Ethics p.20)

- **Fabrication**. Making up data or results and recording or reporting them. Slide short form is "inventing or making up data or results" (Ethics p.13).
- **Falsification**. Manipulating research materials, equipment or processes, or changing or omitting data or results, so the research isn't accurately represented. Slide short form adds manipulating equipment or software (Ethics p.13).
- **Plagiarism**. Taking another person's ideas, processes, results or words without appropriate credit.
- Even if you can't explain some trends right away, don't omit or manipulate them. Another group will likely have a similar implementation and get the real results (Ethics p.13).
- Quick test. Never measured it? Fabrication. Measured it but changed, hid or tweaked it? Falsification. Someone else's words or ideas without credit? Plagiarism.

Facts on plagiarism and citing

- IEEE defines plagiarism as using another researcher's ideas, results or words without explicitly acknowledging the original author and source. Plagiarism in any form is unacceptable and a serious breach of professional conduct with possibly severe ethical and legal consequences (Ethics p.7).
- Always cite. For a **direct quotation**, put the verbatim text in quotation marks and cite the source (Ethics p.8).
- Usually better is to **paraphrase or summarize** with an in-text citation, for ideas, processes, arguments or conclusions (Ethics p.9).
- Copying the sentence with no quotes and no citation is plagiarism (Ethics p.10).
- Data, results, information, graphics and tables also need a citation to the original whenever you refer to, adapt or reuse them (Ethics p.11).
- **Self-plagiarism**. The same rules apply to your own published work. Rephrase instead of cut and paste. Exception is extending your own conference paper to a journal submission (Ethics p.12).
- **Multiple or concurrent submissions**. A submission must not be under review or already published elsewhere, and that includes papers with substantial overlap. Check with organizers and your advisor if in doubt (Ethics p.14).
- **Conference to journal**. The journal version needs substantially more material, typically **30% more for IEEE**, and must cite and include as an attachment the conference version, explaining the differences (Ethics p.14).

Luke's plagiarism report

- Plagiarism is the **second worst** thing in research. The worst is faking your experiments. Plagiarists are often fired, publicly denounced by coauthors, and PhD students get disowned by advisors and thrown out. Avoid even the appearance of plagiarism (Luke TR p.1).
- You can plagiarize someone's **words** or their **descriptive idea**. Light rewording isn't enough unless the idea is well-known and general (Luke TR p.1).
- It harms two groups, the source who loses credit and readers misled into thinking you came up with it. Plagiarism is an ethical offense, copyright violation is a separate legal one you can

be sued for. It's not allowed anywhere in a paper, including abstract, intro, conclusion, tutorial material and especially related work (Luke TR p.1).
- Pasted text is plagiarism (Ex. 1). Pasted text plus a citation is still plagiarism (Ex. 2). Pasted text naming the authors and citing is still plagiarism, since the words are still stolen (Ex. 3) (Luke TR p.2).
- Using less of their text, more than a phrase, is still open to accusation. Under a phrase is relatively safe (Ex. 4). Light rewording with the same structure plagiarizes the idea. Never reword "to avoid plagiarizing", write in your own words from the start (Ex. 5). Rewriting in your own words without looking at the source is much better, but a quote is better still (Ex. 6) (Luke TR p.2).
- An offset block quote with citation is fine, but don't quote huge chunks without reason (Ex. 7). An inline quote in quotation marks is OK, but offset larger ones (Ex. 8) (Luke TR p.3).
- An equation copied without attribution is plagiarism, since math is dense text (Ex. 9). "As Luke and Panait have shown [1], the update rule is…" is dangerously close (Ex. 10). "The update rule (from Luke and Panait [1]) is" is acceptable. Standard notation like the Bellman equation needn't be cited, but when in doubt cite (Ex. 11). More than a few equations needs permission from the author and maybe the publisher (Ex. 12) (Luke TR p.3).
- A copied figure with no citation is plagiarism (Ex. 13). A copied figure with citation isn't plagiarism but is **copyright violation**, because figures are the author's creation and take more effort (Ex. 14). Use a figure only with permission, stated in the caption, however small (Ex. 15). Copying the original caption is plagiarism, treat captions like body text (Ex. 16) (Luke TR p.4).
- An identical bibliography entry is fine since formats are rigid (Ex. 17). A small algorithm clearly marked as from the source is OK, though rewriting it is better (Ex. 18). A large algorithm (a third of a page or more) is like a figure and needs author and publisher permission, and still looks like padding (Ex. 19) (Luke TR p.5).

Facts on AI use

- In general AI tools are OK in submissions if the use is acknowledged in detail and authors take full responsibility, including for any resulting plagiarism or falsification. Check the venue's current rules, which keep changing (Ethics p.22).
- **IEEE**. Any AI-generated content (text, figures, images, code) must be disclosed in the **acknowledgments**. Name the AI system, identify the sections that used it and briefly explain how much. Editing and grammar help is generally outside the policy, but disclosure is still **recommended** (Ethics p.23).
- **ACM**. Generative AI like ChatGPT **can't be listed as an author**. Using it to create content is permitted but must be fully disclosed, for example in the Acknowledgements. When unsure, disclose. Basic word processing that suggests text, spelling and grammar checks, and translation tools are **exempt and need not be disclosed**. The policy will be updated as the line with tools like MS-Word or Grammarly blurs (Ethics p.24).
- **Elsevier**. AI only to improve readability and language, not to replace key tasks like producing scientific insights or drawing conclusions. Disclose it in the manuscript and a statement appears in the published work. AI can't be an author or co-author or be cited as an author (Ethics p.25).
- **ICML 2024 CFP**. LLMs are allowed as general writing aids. Authors take full responsibility, including for LLM content that's plagiarism or fabrication. LLMs can't be authors (Ethics p.26).
- Always check the latest CFP or journal website. If it says nothing, the organization's general rules (IEEE, ACM) apply. Ask the organizers and your advisor if unsure (Ethics p.27).

Likely exam questions

1. **Q.** Classify each. (a) Two of 30 timing runs are slow and unexplained, so you delete them. (b) A simulator crashed, so you fill the table cell with an interpolated value and don't say so. (c) You copy a sentence from a paper without quotes or citation. (d) You report a wrong mean because of a spreadsheet typo you didn't notice.
   A. (a) Falsification, since omitting data misrepresents the research (Ethics p.20, p.13). (b) Fabrication, a result that was never measured is reported (Ethics p.20). (c) Plagiarism (Ethics p.10, p.20). (d) Not misconduct, it's honest error (Ethics p.18).
2. **Q.** A grad student is accused of misconduct on a homework and, separately, in their dissertation. Which rules apply?
   A. The homework goes through the Honor Code. The dissertation goes through Policy 4007, investigated by ORIA (Ethics p.16, p.15).
3. **Q.** You copy a paragraph word for word, write "As Luke and Panait have pointed out" and add [1]. Plagiarism?
   A. Yes. The idea is credited but the words are passed off as yours (Luke TR p.2, Ex. 3). Use quotation marks or a block quote (Luke TR p.3, Ex. 7-8, Ethics p.8), or paraphrase with a citation (Ethics p.9).
4. **Q.** You put another paper's figure in yours with "from [1]" in the caption. Plagiarism? OK?
   A. Not plagiarism but not OK. It's copyright violation (Luke TR p.4, Ex. 14). Get author and maybe publisher permission and write "reproduced with permission from [1]" in the caption (Ex. 15).
5. **Q.** You turn your 10-page conference paper into an IEEE journal paper. What's required?
   A. Rule. The journal version needs typically 30% more material for IEEE (Ethics p.14), so new length is old length $\times 1.30$. Plug in 10 pages. Step 1, $10 \times 1.30 = 13$ pages' worth. Step 2, new material is $13 - 10 = 3$ pages. Result, about 3 pages of genuinely new content. Also cite and attach the conference version and explain the differences (Ethics p.14). Reusing your own conference text is the allowed self-plagiarism exception (Ethics p.12).
6. **Q.** Can ChatGPT be a co-author on an ACM paper? Must you disclose a grammar checker?
   A. No, AI can't be an author, and content it creates must be fully disclosed (Ethics p.24). A grammar checker is exempt under ACM. Under IEEE it's outside the policy but disclosure is recommended (Ethics p.23).

Myth vs truth across all four lectures

Myth (false)	Truth	Cite
My academic advisor from admission is my research advisor	Research advisor is declared separately within 24 credits	Advisor p.3
The 24-credit deadline lets me search in my last semester	Too late, PRCP alone can take a semester or more	Advisor p.7-8
If I want to work with X, X will agree	Not guaranteed, they may be over-committed or see a mismatch	Advisor p.7, p.12
Faculty answer "be my advisor?" within weeks	Most faculty take their time	Advisor p.7, p.21
PRCP is an official GMU procedure	Aydin's own term, not official	Advisor p.9
Co-advised students have two official advisors	One official advisor, thesis names both	Advisor p.19

Myth (false)	Truth	Cite
Changing advisor is a normal step	Possible but a last resort	Advisor p.20
My 701 supervisor becomes my PhD advisor	Separate form within 24 credits	CS701 p.16
My dissertation will be in my 701 area	Can do 701 in X and dissertation in Y	CS701 p.17
My first-choice professor will agree	They may be unavailable	CS701 p.19
A meeting and a quick decision will follow	No guarantee	CS701 p.20
Wait for one professor before asking another	Contact several at once	CS701 p.21
Only one faculty member can work with me	Co-supervisors allowed, one official tenure-line CS primary	CS701 p.22
I can start searching in late October	Big mistake, form due Oct 31	CS701 p.23, p.7
My GRA sponsor must be my 701 supervisor	Natural but not required	CS701 p.24
I can take 701 next year instead	Not possible, catalog rule	CS701 p.25
The department will assign a supervisor	Faculty must agree to supervise	CS701 p.26
701 can be just a literature review	Needs a hands-on part	CS701 p.10-11
A conference submission can be a 701 deliverable	No, but a draft paper for later submission can	CS701 p.15
701 needn't be publishable so it can be small	Effort of a 3-credit course, no sprinting	CS701 p.9
A publication is like a grad project report	False, peer reviewed in a competition	Pub p.9, p.3
Correct results get accepted	False, fixed slots and writing is scored	Pub p.9, p.14-15
Correct, important and novel gets accepted	Still not guaranteed	Pub p.9, p.15
My advisor says it's good, so it'll be accepted	False	Pub p.9
Rejection means the conference erred or is biased	Reasonable papers are rejected for lack of slots	Pub p.9, p.15
Rejection wastes all the effort	Use comments to improve and resubmit	Pub p.9, p.19
Venue doesn't matter	Lower acceptance rate means more prestige	Pub p.9, p.13
Number of papers matters, not quality	Quality vs quantity	Pub p.9, p.4
Every paper gets a rebuttal	Many are directly accepted or rejected	Pub p.17
Dual submission is fine if I withdraw one later	Not allowed, severe sanctions	Pub p.20
Citing a source makes pasted text OK	Still plagiarism without quotes	Luke TR p.2, Ethics p.8
A cited figure is fine	Copyright violation, need permission	Luke TR p.4
Light rewording avoids plagiarism	Same structure still plagiarizes the idea	Luke TR p.2
Reusing my own published text is fine	Self-plagiarism, except conf to journal	Ethics p.12
Dropping unexplained data is OK	That's falsification	Ethics p.13, p.20
An honest error is research misconduct	Not misconduct	Ethics p.18
Dissertation misconduct goes to the Honor Code	Policy 4007	Ethics p.16
AI can be a co-author if disclosed	Never an author (ACM, Elsevier, ICML)	Ethics p.24-26
Disclosing AI shifts responsibility to the tool	Authors take full responsibility	Ethics p.22

Every homework problem from HW0 and HW2 to HW6, in the form the exam is likely to copy. Each one starts with **In plain words**, which says what the question really wants. Then every calculation goes in the same four beats. *Formula* gives the formula in symbols with each symbol named. *Plug in* puts the real numbers in. *Work it out* does the arithmetic one step at a time. *Result* gives the number. The final answer is marked **Answer**. Every number below was recomputed in Python from the data files. Where the homework needed a computer there's a small version you can do with a pencil. Places where the submitted answer drifts from the slides or the source are marked **Check.** When an interval is built from rounded pieces the last digit can be 1 off from the listed answer, which came from unrounded values. Either is fine on the exam.

Table values used in this section

$z[0.05]$	1.645	$t[0.05, 15]$	1.753
$z[0.025]$	1.960	$t[0.025, 15]$	2.131
$z[0.01]$	2.326	$t[0.005, 15]$	2.947
$z[0.005]$	2.576	$t[\cdot, 3]$ at .05/.025/.005	2.353/3.182/5.841
$\Phi(0.33)$	0.6293	$t[0.025, 8]$	2.306
$\Phi(0.5)$	0.6915	$t[0.05, 8]$	1.860
$\Phi(0.83)$	0.7967	$t[0.05, 6]$	1.943
$\Phi(0.9)$	0.8159	$t[0.05, 37]$	1.687
$\Phi(1.3)$	0.9032	$t[0.025, 2]$	4.303
$\Phi(1.8)$	0.9641	$\Phi(0.21)$	0.5832
$\Phi(2.0)$	0.9772	$\Phi(0.67)$	0.7486
$\Phi(2.5)$	0.9938	$\Phi(1.38)$	0.9162
$t[0.05, 5]$	2.015	$t[0.025, 4]$	2.776
$t[0.05, 24]$	1.711	$t[0.10, 37]$	1.304
$z[0.20]$	0.84	$t[0.025, 7]$	2.365
$\Phi(2.4)$	0.9918	$t[0.025, 5]$	2.571

$z[a]$ is the upper-tail value with area a above it, so $z[0.025] = 1.960$ (CI p.30). $\Phi(z)$ is the area below z, which is the chance a standard normal lands below z. Handy roots are $\sqrt{50} = 7.071$, $\sqrt{12} = 3.464$, $\sqrt{3} = 1.732$, $\sqrt{6} = 2.449$, $\sqrt{5} = 2.236$.

Which homework to open for which exam question

If the exam asks you to	Model
compute a probability, mean or variance from a table	HW0 Q5 to Q9
judge a metric against the six criteria	HW2
pick and compute a mean (times, rates, ratios)	HW3 Q1
find quartiles, a histogram, a Q-Q plot	HW3 Q2
build a z or t CI and compare widths	HW4 P1
fix a CI for skewed data	HW4 P2
run a one-sample test, p-value, threshold	HW5 P1
set up H_0 and H_1 for a claim	HW5 P2
compare two systems, paired or Welch	HW6 P1, P2
audit a described experiment	HW4 to HW6 P3

Answer key at a glance (his submitted answers, all recomputed)

HW0 Q4	1/16, 15/16, 3/8
HW0 Q5	discrete, $\mu = 2.84$, Var 5.134, SD 2.27, COV 0.80
HW0 Q6	0.75, 0.425, $4/17 = 0.235$, $17/23 = 0.739$
HW0 Q7	mean 50, SD 23.09, 0.125, 0.375, 0.125
HW0 Q8	0.0873, 0.1841, 0.8159
HW0 Q9	0.321, 0.203, 0.309
HW3 Q1	GM 1.4294 (42.94%), AM 203.33 ms, HM 2485.66 tps
HW3 Q2	means 2.0142, 0.2083, s^2 1.0457, 0.0440, Q-Q r 0.9993, 0.904
HW4 P1	$\mu = 2.0142$, $\sigma^2 = 1.0447$, all six CIs contain μ, 95% widths 0.52 vs 1.21
HW4 P2	$\mu = 0.2083$, $\sigma^2 = 0.0440$, batched widest, bootstrap narrowest, 99% batched dips to -0.158
HW5 P1	$Z = 1.80$, fail to reject, $p = 0.0718$, threshold 2.843 h
HW5 P2	$H_1: \mu < 740$, $Z = -2.50$, reject, $p = 0.0062$
HW6 P1	paired, $\bar{d} = 0.611$, CI $(-10.77, 11.99)$, not shown
HW6 P2	Welch, $\nu = 37.89$, CI $(-1.540, 0.781)$, not shown

HW0. Probability refresher (ungraded, solved here)

HW0 Q1 and Q2. Independent events, and $P[A$ and $B]$

Question. *1. Explain what is meant by "independent events" in Probability theory. 2. If A and B are independent events, what is $P[A$ and $B]$?*

In plain words. Two events are independent when one happening tells you nothing about the other, like two separate coin flips. Then the chance that both happen is just the two chances multiplied. This is the rule that lets us treat repeated runs as separate pieces of evidence later in the course.

1. Q1. Independent means knowing B happened doesn't change the chance of A. In symbols $P[A \mid B] = P[A]$, where $P[A \mid B]$ is the chance of A once you already know B happened.
2. Q2. *Formula.* Multiplication rule $P[A \cap B] = P[A \mid B]\, P[B]$, where $A \cap B$ means A and B both happen.
3. *Plug in.* Independence lets you swap $P[A \mid B]$ for $P[A]$.
4. *Result.* $P[A \cap B] = P[A]\, P[B]$.
5. Example. Two fair coins. $P[\text{both heads}] = P[H_1]\, P[H_2] = 0.5 \times 0.5 = 0.25$.

Answer. Q1, independent means $P[A \mid B] = P[A]$, so one event carries no information about the other. Q2, $P[A \cap B] = P[A]\, P[B]$.

Usual mistake. Mixing up independent with mutually exclusive. Exclusive events with nonzero probability are never independent, since $P[A \cap B] = 0$. In the course, independence is what makes $\sigma/\sqrt{n}$ the right standard error (HT p.20, CI p.65).

HW0 Q3. $E[kX + mY]$ for constants k, m

Question. *If the expected values of two random variables X and Y are $E[X]$ and $E[Y]$, respectively, what is the expected value of $k \cdot X + m \cdot Y$ (i.e., $E[k \cdot X + m \cdot Y]$), if k and m are constants?*

In plain words. Expected value is the long-run average. Averages are linear, so scaling a variable scales its average, and the average of a sum is the sum of the averages. You don't need independence for this, only for the variance version.

1. *Formula.* $E[X] = \sum_x x\, P[X = x]$, each value times its chance, added up. It's a weighted sum (Summ p.10).
2. *Plug in.* $E[kX + mY] = \sum (kx + my)\, P[x, y]$ over all pairs of values.
3. *Work it out.* Split the sum into $k \sum x\, P[x, y] + m \sum y\, P[x, y]$. Each piece collapses to $E[X]$ and $E[Y]$.
4. *Result.* $E[kX + mY] = k\, E[X] + m\, E[Y]$.
5. Example. $E[X] = 3$, $E[Y] = 5$, $k = 2$, $m = -1$ gives $2(3) + (-1)(5) = 6 - 5 = 1$.

Answer. $k\,E[X] + m\,E[Y]$, with or without independence.

Usual mistake. Doing the same for variance. $\text{Var}(X - Y) = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$ needs the correlation, and that term is exactly what pairing buys (Comp p.6). With $k = 1$, $m = -1$ you get $E[A - B] = E[A] - E[B]$, which is why $\bar{d} = \bar{x}_A - \bar{x}_B$ (Comp p.8).

HW0 Q4. Fair coin tossed 4 times. P(all heads), P(at least one tail), P(exactly 2H 2T)

Question. *A fair coin is tossed 4 times. (i) What is the probability that all four tosses will be "heads"? (ii) What is the probability that at least one toss will give "tail"? (iii) What is the probability that there will be exactly 2 heads and 2 tails? Show your work (i.e., explain how you obtained your answers).*

In plain words. With a fair coin every sequence of 4 tosses is equally likely. So each probability is the number of sequences you want divided by the total number of sequences. For "at least one" it's easier to count the opposite, no tails at all, and subtract from 1.

1. *Formula.* $P[\text{event}] = \dfrac{\text{sequences in the event}}{\text{all sequences}}$. All sequences $= 2^n$, where n is the number of tosses.
2. *Plug in.* $2^4 = 2 \times 2 \times 2 \times 2 = 16$ sequences.
3. All heads. Only HHHH, so 1 sequence. $P = 1/16 = 0.0625$.
4. At least one tail. *Formula.* $P[\text{at least one}] = 1 - P[\text{none}]$. *Plug in.* $1 - 1/16$. *Work it out.* $16/16 - 1/16 = 15/16$. *Result.* 0.9375.
5. Exactly 2 heads. *Formula.* $\text{count} = \binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$, the number of ways to choose which k of the n tosses are heads. *Plug in.* $\binom{4}{2} = \dfrac{4!}{2!\,2!}$. *Work it out.* $\dfrac{24}{2 \times 2} = 6$ sequences (HHTT, HTHT, HTTH, THHT, THTH, TTHH). *Result.* $6/16 = 3/8 = 0.375$.

Answer. $1/16 = 0.0625$, $15/16 = 0.9375$, $3/8 = 0.375$.

Usual mistake. Adding probabilities instead of using the complement for "at least one". The same coin is the error model on CI p.12. Two $\pm E$ sources give $x - 2E, x, x + 2E$ with $1/4, 1/2, 1/4$. With 3 sources, $P[x + E] = \binom{3}{2}/8 = 3/8$, and many sources turn the binomial into a normal (CI p.15).

HW0 Q5. Requests per ms with $P_0, P_1, P_2, P_4, P_5, P_6, P_7$ = .18, .22, .14, .18, .12, .10, .06. Type? Why is $P_7 = 0.09$ invalid? Mean, variance, SD, COV

Question. *On a certain server, the number of client requests per millisecond varies between 0 and 7, with the following probabilities (P_i is the probability that there are i requests per millisecond). $P_0 = 0.18$, $P_1 = 0.22$, $P_2 = 0.14$, $P_4 = 0.18$, $P_5 = 0.12$, $P_6 = 0.10$, $P_7 = 0.06$. (i) What type of probability distribution is this (discrete or continuous)? (ii) If P_7 was 0.09, this would not be a valid probability distribution. Why? (iii) Let X denote the random variable that shows the number of client requests per millisecond (i.e., $P_i = P(X = i)$). Find the mean, variance, standard deviation, coefficient of variation for X. Show your work.*

In plain words. You get a table of values and their chances. Discrete means it only takes separate values you can count, like 0, 1, 2 requests. The mean is each value times its chance, added up. Variance says how spread out it is, the SD is its square root, and COV is the SD divided by the mean, so it's the spread relative to the size.

1. Whole-number values only, so it's discrete.
2. Probabilities must add to exactly 1. Listed values give $.18 + .22 + .14 + .18 + .12 + .10 + .06 = 1.00$, so $P_3 = 0$. With $P_7 = 0.09$ the sum is $1.00 - 0.06 + 0.09 = 1.03 > 1$, so it's invalid.
3. Mean. *Formula.* $\mu = E[X] = \sum x\,p(x)$, where x is a number of requests and $p(x)$ its probability. *Plug in.* $0(.18) + 1(.22) + 2(.14) + 4(.18) + 5(.12) + 6(.10) + 7(.06)$. *Work it out.* $0 + 0.22 + 0.28 + 0.72 + 0.60 + 0.60 + 0.42$. *Result.* $\mu = 2.84$.
4. Average of squares. *Formula.* $E[X^2] = \sum x^2\,p(x)$. *Plug in.* $0(.18) + 1(.22) + 4(.14) + 16(.18) + 25(.12) + 36(.10) + 49(.06)$. *Work it out.* $0 + 0.22 + 0.56 + 2.88 + 3.00 + 3.60 + 2.94$. *Result.* 13.20.
5. Variance. *Formula.* $\text{Var} = E[X^2] - \mu^2$, the average of the squares minus the square of the average. *Plug in.* $13.20 - 2.84^2$. *Work it out.* $2.84^2 = 8.0656$, so $13.20 - 8.0656$. *Result.* 5.1344.
6. SD. *Formula.* $\sigma = \sqrt{\text{Var}}$. *Plug in.* $\sqrt{5.1344}$. *Result.* 2.266.
7. COV. *Formula.* $\text{COV} = \sigma/\mu$ (Summ p.41). *Plug in.* $2.266/2.84$. *Result.* 0.798.

Answer. Discrete, mean 2.84, variance 5.134, SD 2.27, COV 0.80.

Usual mistake. Dividing by $n - 1$ here. These are known probabilities of a population, so there's no $n - 1$. Also forgetting the missing P_3 and putting P_4's value at $x = 3$.

Small version. $X \in \{0, 1, 2\}$ with $.25, .5, .25$. $E[X] = 0(.25) + 1(.5) + 2(.25) = 0 + 0.5 + 0.5 = 1$. $E[X^2] = 0(.25) + 1(.5) + 4(.25) = 0 + 0.5 + 1 = 1.5$. $\text{Var} = 1.5 - 1^2 = 0.5$, $\text{SD} = \sqrt{0.5} = 0.707$, $\text{COV} = 0.707/1 = 0.707$. COV is meaningless if the mean is 0 or negative (Summ p.41).

HW0 Q6. Cohort with Spring 10 (6 BS, 4 MS) and Fall 30 (17 BS, 13 MS). P(Fall), P(MS), P(Spring|MS), P(Fall|BS)

Question. *The 2019 PhD CS cohort has the following student profiles. Spring 2019 Start Term, 10 students started. 6 have only BS degree, 4 have MS degrees. Fall 2019 Start Term, 30 students started. 17 have only BS degree, 13 have MS degrees. If a random student is selected from the 2019 cohort, (i) What is the probability that the student started in Fall 2019? (ii) What is the probability that the student has MS degree? (iii) If we randomly select a student and the student has MS degree, what is the probability that the student started in Spring'19? (iv) If we randomly select a student and the student has only BS degree, what is the probability that the student started in Fall'19?*

In plain words. This is counting from a table. A plain probability divides by all 40 students. A conditional one, written $P[A \mid B]$ and read "A given B", only looks at the students in group B, because you already know the student is in B.

1. Totals. Students $= 10 + 30 = 40$. MS $= 4 + 13 = 17$. BS-only $= 6 + 17 = 23$.
2. P[Fall]. *Formula.* $P[A] = \dfrac{\text{students in } A}{\text{all students}}$. *Plug in.* 30/40. *Result.* 0.75.
3. P[MS]. *Plug in.* 17/40. *Result.* 0.425.
4. P[Spring | MS]. *Formula.* $P[A \mid B] = \dfrac{P[A \text{ and } B]}{P[B]}$, the chance of A once you know B happened. Here A is started in Spring and B is holds an MS. *Plug in.* P[Spring and MS] = 4/40 (4 of the 40 are Spring MS) and P[MS] = 17/40, so $\dfrac{4/40}{17/40}$. *Work it out.* The /40 cancels, leaving 4/17. *Result.* 0.235. Shortcut, just count inside the MS group, 4 of 17.
5. P[Fall | BS]. *Plug in.* P[Fall and BS] = 17/40, P[BS] = 23/40, so $\dfrac{17/40}{23/40}$. *Work it out.* 17/23. *Result.* 0.739.

Answer. 0.75, 0.425, 0.235, 0.739.

Usual mistake. Dividing by the wrong total, for example 4/10,

which is $P[\text{MS} \mid \text{Spring}]$. The same direction mix-up is the p-value trap, since $P[\text{data} \mid H_0] \neq P[H_0 \mid \text{data}]$ (HT p.13 to 14).

HW0 Q7. $X \sim U(10, 90)$**. Mean, SD,** $P[X < 20]$**,** $P[X > 60]$**,** $P[20 < X < 30]$

Question. *Assume a random variable has uniform distribution between 10 and 90. What is the mean? What is the standard deviation? What is the probability that the random variable will take a value (i) less than 20? (ii) more than 60? (iii) between 20 and 30?*

In plain words. Uniform means every value between 10 and 90 is equally likely, so the density is a flat line. Any probability is just the length of the stretch you care about divided by the full length, 80. The mean is the middle, and the SD has a fixed formula.

1. Mean. *Formula.* $\mu = (a+b)/2$, where $a = 10$ is the low end and $b = 90$ the high end. *Plug in.* $(10 + 90)/2$. *Work it out.* $100/2$. *Result.* 50.
2. Variance. *Formula.* $\sigma^2 = (b-a)^2/12$. *Plug in.* $(90-10)^2/12$. *Work it out.* $80^2 = 6400$, then $6400/12$. *Result.* 533.3.
3. SD. *Formula.* $\sigma = \sqrt{\sigma^2} = (b-a)/\sqrt{12}$. *Plug in.* $80/\sqrt{12}$. *Work it out.* $80/3.464$. *Result.* 23.09.
4. Probabilities. *Formula.* $P[c < X < d] = \dfrac{d-c}{b-a}$, length of the stretch over 80.
5. $P[X < 20]$. *Plug in.* $(20-10)/80$. *Work it out.* $10/80$. *Result.* 0.125.
6. $P[X > 60]$. *Plug in.* $(90-60)/80$. *Work it out.* $30/80$. *Result.* 0.375.
7. $P[20 < X < 30]$. *Plug in.* $(30-20)/80$. *Work it out.* $10/80$. *Result.* 0.125.

Answer. Mean 50, SD 23.09, then 0.125, 0.375, 0.125.

Usual mistake. Using $(b-a)/12$ or forgetting the square. Rounding error of a timer is uniform on $\pm\Delta$, so its SD is $2\Delta/\sqrt{12}$ (quantization, CI p.9 to 10).

HW0 Q8. $Z \sim N(0, 1)$**.** $P[0.9 < Z < 1.3]$**,** $P[Z > 0.9]$**,** $P[Z > -0.9]$

Question. *Assume a random variable has standard normal distribution $N(0, 1)$. What is the probability that this random variable will take on a value (i) between 0.9 and 1.3, (ii) greater than 0.9, (iii) greater than -0.9? Explain how you obtained your results.*

In plain words. Z is the standard bell curve with mean 0 and SD 1. The table gives $\Phi(z)$, the area to the left of z, which is the chance of landing below z. Every question is a left area, a right area (1 minus the left area), or the gap between two left areas.

1. Between. *Formula.* $P[a < Z < b] = \Phi(b) - \Phi(a)$. *Plug in.* $\Phi(1.3) - \Phi(0.9)$. *Work it out.* $0.9032 - 0.8159$. *Result.* 0.0873.
2. Above. *Formula.* $P[Z > a] = 1 - \Phi(a)$. *Plug in.* $1 - \Phi(0.9)$. *Work it out.* $1 - 0.8159$. *Result.* 0.1841.
3. Above a negative. *Formula.* Symmetry gives $\Phi(-a) = 1 - \Phi(a)$. *Plug in.* $P[Z > -0.9] = 1 - \Phi(-0.9) = 1 - (1 - \Phi(0.9))$. *Work it out.* That's just $\Phi(0.9)$. *Result.* 0.8159.

Answer. 0.0873, 0.1841, 0.8159.

Usual mistake. Answering $P[Z > -0.9]$ with 0.1841. Sketch the curve and shade before subtracting.

HW0 Q9. Highs $\sim N(85, 6^2)$**.** $P[83 < X < 88]$**,** $P[X > 90]$**,** $P[X < 82]$ **with z-scores**

Question. *Over the course of several years, in a certain city, it is observed that the highest daily temperature in August, in Fahrenheit, is normally distributed with mean = 85 and standard deviation = 6. What is the probability that a randomly selected August day's highest temperature is (i) between 83 and 88, (ii) greater than 90, (iii) smaller than 82? Solve problem 9 using z scores. Show your work.*

In plain words. Turn each temperature into a z-score, which counts how many SDs it sits away from the mean. Then it's a standard normal question and you use the same table as Q8.

1. *Formula.* $z = (x - \mu)/\sigma$, where x is the temperature, $\mu = 85$ the mean and $\sigma = 6$ the SD.
2. Between 83 and 88. *Plug in.* $z_1 = (83 - 85)/6 = -2/6 = -0.33$ and $z_2 = (88 - 85)/6 = 3/6 = 0.5$. *Work it out.* $\Phi(0.5) - \Phi(-0.33) = 0.6915 - (1 - 0.6293) = 0.6915 - 0.3707$. *Result.* 0.3208.
3. Above 90. *Plug in.* $z = (90 - 85)/6 = 5/6 = 0.83$. *Work it out.* $1 - \Phi(0.83) = 1 - 0.7967$. *Result.* 0.2033.
4. Below 82. *Plug in.* $z = (82 - 85)/6 = -3/6 = -0.5$. *Work it out.* $\Phi(-0.5) = 1 - \Phi(0.5) = 1 - 0.6915$. *Result.* 0.3085.

Answer. About 0.321, 0.203, 0.309 (unrounded z gives 0.322 and 0.202).

Backwards version. Where do the hottest 5% of days start? *Formula.* $x = \mu + z\sigma$. *Plug in.* $85 + z[0.05] \times 6 = 85 + 1.645 \times 6$. *Work it out.* $1.645 \times 6 = 9.87$, so $85 + 9.87$. *Result.* 94.9.

HW2. Beyond performance metrics (report)

HW2. Pick metrics from your area, evaluate them on the lecture's criteria, propose extra criteria (Metrics p.33)

Question. *HW 2, Beyond Performance Metrics. (i) Identify one or more metrics commonly used in your area of research. If you haven't yet decided on your research area, pick one of the areas you might be interested in. Look at one or more seminal research articles in your area and see what metrics were used. (ii) Evaluate the metric(s) based on the criteria discussed in today's lecture. What additional criteria might be useful in evaluating metrics in your research area? (iii) Submit a short typed report (at most 4 pages), using LaTeX.*

In plain words. Pick the numbers people in your research area use to say "my system is better", then grade each one against the six properties a good metric should have. The key idea is that a metric can be easy to measure and still mislead, like MIPS. It's graded on argument, so every yes or no needs a concrete example of how the metric fails.

1. List the six criteria (Metrics p.7 to 12). Linear (2x metric means 2x performance). Reliable (A scores higher means A really is better). Repeatable (same experiment, same value). Easy to use (easy to measure correctly). Consistent (same units and definition on every system). Independent (vendors can't bend it).
2. Say which are required. Reliable, repeatable, consistent and independent are required, while linear and easy to use are nice to have (Metrics p.13).
3. Classify means versus ends. Means-based counts what was done, useful or not. Ends-based counts what got accomplished (Metrics p.27 to 28).
4. For each metric, give a verdict per criterion with one concrete failure.
5. Propose extra criteria your area needs.

His answer. Area is low-latency video streaming. Metrics were average throughput and the QoE score (Yin et al. 2015, chunk

quality minus a switch penalty minus a stall penalty, as a weighted sum).

Criterion	Throughput	QoE score
Linear	Yes	No
Reliable	No	No
Repeatable	Partly	Partly
Easy to use	Yes	Yes
Consistent	Partly	No
Independent	No	No

- Throughput isn't reliable because a sender can inflate it by filling a deep queue (bufferbloat), the same failure as MIPS (Metrics p.17). It isn't independent because ISPs advertise it. Repeatability is partial because Pantheon found algorithm rankings change with the path. It's a means metric used for ends claims, since 8 Mbps with two stalls beats 5 Mbps with none.
- QoE isn't consistent because each paper picks its own weights (Yin's stall weight 3000, Pensieve's three variants), the MFLOPS problem. It isn't linear since it adds unlike units. It isn't reliable because the linear form was never validated against subjective tests (Peroni and Gorinsky 2024). It isn't independent because authors set the weights they're judged under.
- Extra criteria. Externality, meaning what a flow does to others on the link, which Jain's index hides. Report the distribution and the tail, not just the mean. State the conditions the metric was checked over.

By hand, Jain's index. *Formula.* $J = \dfrac{(\sum x_i)^2}{n \sum x_i^2}$, where x_i is flow i's throughput and n the number of flows. It's 1 when all flows get the same and $1/n$ when one flow gets everything. *Plug in.* Flows of 90 and 10 Mbps, $n = 2$, so $\dfrac{(90+10)^2}{2(90^2+10^2)}$. *Work it out.* $\dfrac{100^2}{2(8100+100)} = \dfrac{10000}{16400}$. *Result.* 0.61.

By hand, precision and recall (Metrics p.32). *Formula.* Precision $= \dfrac{\text{relevant retrieved}}{\text{all retrieved}}$ and recall $= \dfrac{\text{relevant retrieved}}{\text{all relevant}}$. *Plug in.* 30 retrieved, 20 of them relevant, 40 relevant ones missed. Precision $= 20/30$, recall $= 20/(20+40) = 20/60$. *Result.* Precision 0.67, recall 0.33.

Check. The repeatability paragraph says Puffer's 95% CI on stall ratio was "around 10 to 17 percent of the mean" after "1.75 years" per scheme. Those numbers are from the arXiv preprint. The NSDI 2020 version he cites (and quotes in HW4) says ±13% to ±21% with at least 2.5 years per scheme. The argument stands, but the two homeworks quote different versions.

Usual mistake. Saying a metric "fails" without naming the failure. On the exam, MIPS is repeatable, easy and independent but not linear, reliable or consistent (Metrics p.17).

How to write it on the exam.

1. Name the metric and what it counts, then say means or ends.
2. Go down the six criteria in order, one line each, with a yes, no or partly and a concrete reason.
3. Flag which failures matter most, the required four (Metrics p.13).
4. Add one criterion your area needs and say why the six miss it.

Exam variant, MIPS. *Formula.* MIPS $= \dfrac{n}{T_e \times 10^6}$ (Metrics p.16), where n is the number of instructions run and T_e the run time in seconds. *Plug in.* A CISC machine runs 50M instructions in 1 s, so $\dfrac{50 \times 10^6}{1 \times 10^6} = 50$. A RISC machine runs 100M simpler instructions for the same program in 0.8 s, so $\dfrac{100 \times 10^6}{0.8 \times 10^6} = 125$. *Work it out.* MIPS ratio $125/50 = 2.5$. Real speedup is old time over new time, $1/0.8 = 1.25$. *Result.* MIPS ranks RISC 2.5x better, but it's really 1.25x faster. So MIPS isn't linear, and with different instruction sets it isn't consistent either. Execution time is the ends-based alternative (Metrics p.27 to 28).

HW3. Choosing a mean, quartiles, histograms, Q-Q plots

HW3 Q1a. Five stacked features improve quality 10, 50, 70, 85, 15%. Mean improvement per feature, and which mean?

Question. *Q1 (3 pts). For each of the following sub-questions, i) indicate what type of mean (arithmetic, geometric, or harmonic) should be used, with explanation, ii) Compute the mean value based on your choice. a. A researcher considers adding five new features to her optimization project to improve the quality of the solution. Each feature is implemented on top of the previous one (i.e., Feature 2 is implemented on top of Feature 1, Feature 3 is implemented on top of Feature 2, and so on.) She observes the following quality improvements after adding each feature. Feature 1, 10%. Feature 2, 50%. Feature 3, 70%. Feature 4, 85%. Feature 5, 15%. What is the mean improvement per feature? What type of mean did you use and why?*

In plain words. The mean you pick depends on how the numbers combine. Here each feature's gain stacks on top of the previous one, so the gains multiply instead of adding. The geometric mean is the one number that, multiplied by itself 5 times, gives the same total gain as the real five.

1. Each feature builds on the last, so effects multiply. Use the geometric mean (Summ p.30, p.32).
2. *Formula.* GM $= (x_1 x_2 \cdots x_n)^{1/n}$, where $x_i = 1 + \text{gain}_i$ is the ratio new over old and $n = 5$ features (Summ p.23).
3. *Plug in.* Ratios 1.10, 1.50, 1.70, 1.85, 1.15, so GM $= (1.10 \times 1.50 \times 1.70 \times 1.85 \times 1.15)^{1/5}$.
4. *Work it out.* $1.10 \times 1.50 = 1.65$, $\times 1.70 = 2.805$, $\times 1.85 = 5.18925$, $\times 1.15 = 5.9676$. Fifth root by hand with logs, $\ln 5.9676 = 1.786$, $1.786/5 = 0.357$, $e^{0.357} = 1.4294$.
5. *Result.* GM $= 1.4294$. Subtract 1 to get back to a percentage, $0.4294 = 42.94\%$.

Answer. Geometric mean, because the gains compound. 1.4294, so 42.94% per feature. Matches his main.tex.

Usual mistake. Taking the GM of the percentages (10, 50, ...) instead of the ratios, or forgetting to subtract 1 at the end.

By hand. Gains of 100, 300, 700% are ratios 2, 4, 8. *Plug in.* GM $= (2 \times 4 \times 8)^{1/3}$. *Work it out.* $2 \times 4 \times 8 = 64$ and $\sqrt[3]{64} = 4$. *Result.* $4 - 1 = 3$, so 300% per version. The AM of the ratios, $(2+4+8)/3 = 4.67$, would compound to $4.67^3 \approx 102$, not 64.

HW3 Q1b. Compiler times 150, 200, 120, 300, 270, 180 ms. Mean, and which?

Question. *b. The running time of a new compiler is measured with source programs of comparable size. In 6 experiments, the running times are measured as 150 ms, 200 ms, 120 ms, 300 ms, 270 ms, and 180 ms. What is the mean execution time? What type of mean did you use and why?*

In plain words. These are running times, and times add up into a total. The arithmetic mean, the usual sum divided by count, is the one whose total matches the real total time, so it's the right one.

1. These are times, so the mean should be proportional to total time. Arithmetic mean (Summ p.25, p.27).
2. *Formula.* $\bar{x} = \dfrac{1}{n} \sum x_i$, where x_i are the times and $n = 6$ runs.

3. *Plug in.* $\dfrac{150+200+120+300+270+180}{6}$.
4. *Work it out.* $150+200=350$, $+120=470$, $+300=770$, $+270=1040$, $+180=1220$. Then $1220/6$.
5. *Result.* 203.33 ms.

Answer. Arithmetic mean, because times add up to a total. 203.33 ms.

Usual mistake. Using the harmonic mean because it's "performance". Times take the AM.

Weighted version (Summ p.33). *Formula.* $\bar{x}_w = \sum w_i x_i$ with weights w_i adding to 1. *Plug in.* Times 2, 5, 10 s with weights .5, .3, .2 give $.5(2)+.3(5)+.2(10)$. *Work it out.* $1+1.5+2$. *Result.* 4.5 s.

HW3 Q1c. DB at 2500, 3500, 3700, 1800, 2000 tps. Mean, and which?

Question. *c. A new database system is implemented and its performance is tested. In five different experiments, the following performance (speed) values are obtained. 2500 transactions per second, 3500 transactions per second, 3700 transactions per second, 1800 transactions per second, 2000 transactions per second. What is the mean performance (in terms of transactions per second)? What type of mean did you use and why?*

In plain words. These are rates, things per second, and you can't just average rates. If every run did the same amount of work, the honest average rate is total work over total time, and that works out to the harmonic mean, n divided by the sum of the $1/x_i$.

1. Rates have time in the denominator, so the mean should be inversely proportional to total time. Harmonic (Summ p.25, p.28).
2. *Formula.* $x_H = \dfrac{n}{\sum 1/x_i}$ (Summ p.22), where x_i are the rates and $n=5$ runs.
3. *Plug in.* $x_H = \dfrac{5}{1/2500+1/3500+1/3700+1/1800+1/2000}$.
4. *Work it out.* The reciprocals are $0.000400+0.000286+0.000270+0.000556+0.000500=0.0020115$. Then $5/0.0020115$.
5. *Result.* $x_H = 2485.66$ tps.

Answer. Harmonic mean, because these are rates with equal work per run. 2485.66 tps.

Usual mistake. The AM of rates, $(2500+3500+3700+1800+2000)/5 = 13500/5 = 2700$, overstates. The HM is right only if every run did the same work F (Summ p.28).

By hand. Rates 2, 3, 6. *Plug in.* $x_H = \dfrac{3}{1/2+1/3+1/6}$. *Work it out.* $1/2+1/3+1/6 = 3/6+2/6+1/6 = 6/6 = 1$, so $3/1$. *Result.* $x_H = 3$. The AM would say $11/3 = 3.67$.

Exam variant, unequal work. Program 1 does 100 Mops at 50 Mops/s, program 2 does 300 Mops at 100 Mops/s. *Formula.* True rate $= \dfrac{\text{total work}}{\text{total time}}$, with time = work over rate. *Plug in.* Times $100/50 = 2$ s and $300/100 = 3$ s, so rate $= \dfrac{100+300}{2+3}$. *Work it out.* $400/5$. *Result.* 80 Mops/s. The plain HM gives $2/(1/50+1/100) = 2/0.03 = 66.7$, which is wrong because it assumes equal work per run (Summ p.26, p.28).

HW3 Q2a,b. For normal.txt ($n=1000$) and not-normal.txt ($n=870$), mean, variance, median, Q1, Q2, Q3

Question. *Q2. Consider the two data sets, normal.txt and not-normal.txt. For each data set, do the following exercises. a. (0.5 pts) Find the mean and variance of the sample data. b. (0.5 pts) Find the median, first, second, and third quartile of the sample data. Note, you can use a program such as Microsoft Excel/OpenOffice Calc/Matlab/Mathematica/etc. for doing these exercises, or you can do them without the help of any program.* [The files hold 1000 and 870 values.]

In plain words. Summarize each data set by its center (mean and median) and its spread (variance and quartiles). Quartiles are the values that cut the sorted data into four equal parts. The course has a fixed recipe for where each sorted value sits, $(i-0.5)/n$, and you interpolate between neighbours when a quartile falls between two values.

1. Mean. *Formula.* $\bar{x} = \frac{1}{n}\sum x_i$, where x_i are the data values and n the count. *Plug in.* normal has $\sum x = 2014.15$, so $2014.15/1000$. not-normal has $\sum x = 181.25$, so $181.25/870$. *Result.* 2.0142 and 0.2083.
2. Variance. *Formula.* Sample variance with $n-1$, one-pass form $s^2 = \dfrac{\sum x_i^2 - (\sum x_i)^2/n}{n-1}$ (Summ p.39). *Plug in.* normal has $\sum x^2 = 5101.51$, so $\dfrac{5101.51 - 2014.15^2/1000}{999}$. *Work it out.* $2014.15^2/1000 = 4056.82$, $5101.51 - 4056.82 = 1044.69$, then $1044.69/999$. *Result.* 1.0457. not-normal, $\dfrac{76.01 - 181.25^2/870}{869} = \dfrac{76.01-37.76}{869} = \dfrac{38.25}{869} = 0.0440$.
3. Quartiles. *Formula.* Sort the data. The i-th smallest value $x_{(i)}$ sits at fraction $f_i = (i-0.5)/n$. Solving for the position of quantile p gives $i = pn + 0.5$ (Summ p.59). If i isn't a whole number, go that far between the two neighbours.
4. normal ($n=1000$). Q_1 at $i = 0.25(1000)+0.5 = 250.5$, halfway between $x_{(250)} = 1.2646$ and $x_{(251)} = 1.2714$, so 1.2680. Median at $i = 500.5$, halfway between 2.0320 and 2.0324, so 2.0322. Q_3 at $i = 750.5$, halfway between 2.6905 and 2.6948, so 2.6927.
5. not-normal ($n=870$). Q_1 at $i = 0.25(870)+0.5 = 218$, exactly $x_{(218)} = 0.0587$. Median at $i = 435.5$, halfway between 0.1426 and 0.1432, so 0.1429. Q_3 at $i = 653$, exactly $x_{(653)} = 0.2865$.

Answer.

	normal	not-normal
Mean	2.0142	0.2083
Variance ($n-1$)	1.0457	0.0440
Min	−1.6554	0.0007
Q_1	1.2680	0.0587
Median $= Q_2$	2.0322	0.1429
Q_3	2.6927	0.2865
Max	5.2128	1.2957

not-normal is right-skewed. The mean 0.208 sits above the median 0.143, and $Q_3 - Q_2 = 0.2865 - 0.1429 = 0.144$ against $Q_2 - Q_1 = 0.1429 - 0.0587 = 0.084$ (Summ p.48). For skewed data report the SIQR. *Formula.* SIQR $= (Q_3 - Q_1)/2$, half the width of the middle 50%. *Plug in.* $(0.2865 - 0.0587)/2$. *Work it out.* 0.2278/2. *Result.* 0.114 (Summ p.45, p.49).

By hand. Data 7, 1, 11, 5, 9, 3. Sorted 1, 3, 5, 7, 9, 11.

1. Mean. $\bar{x} = (1+3+5+7+9+11)/6 = 36/6 = 6$.
2. Variance. $\sum x^2 = 1+9+25+49+81+121 = 286$. $(\sum x)^2/n = 36^2/6 = 1296/6 = 216$. $s^2 = (286-216)/5 = 70/5 = 14$, so $s = \sqrt{14} = 3.742$.
3. Quartiles. The f_i are .083, .25, .417, .583, .75, .917. Q_1 at $i = 0.25(6)+0.5 = 2$, so $x_{(2)} = 3$. Q_3 at $i = 0.75(6)+0.5 = 5$, so $x_{(5)} = 9$. Median at $i = 3.5$, halfway between 5 and 7, so 6. With $n=10$, Q_1 at $i=3$ and Q_3 at $i=8$ exactly.

Exam variant with interpolation. Data 2, 4, 4, 5, 7, 9, 10, 12 ($n = 8$). The f_i are .0625, .1875, .3125, .4375, .5625, .6875, .8125, .9375. Q_1 at $i = 0.25(8) + 0.5 = 2.5$, between $x_{(2)} = 4$ and $x_{(3)} = 4$, so 4. Median at $i = 4.5$, halfway between 5 and 7, so 6. Q_3 at $i = 6.5$, halfway between $x_{(6)} = 9$ and $x_{(7)} = 10$, so 9.5. SIQR $= (9.5 - 4)/2 = 5.5/2 = 2.75$.

Usual mistake. Dividing by n for a sample variance, or taking Q_1 as $x_{(n/4)}$ without the $(i - 0.5)/n$ positions.

HW3 Q2c. Histogram of each

Question. *c. (1 pts) Plot the sample data using a histogram.* [For each of normal.txt and not-normal.txt.]

In plain words. A histogram chops the range into equal-width bins and counts how many values land in each, so you can see the shape of the data. The only real decision is how many bins, and Sturges' rule gives a default from n.

1. Equal-width bins over the range, plot relative frequency (Summ p.54).
2. Number of bins. *Formula.* Sturges, $k = \lfloor 1 + \log_2 n \rfloor$ (Summ p.54), where k is the bin count and n the number of values. *Plug in.* $1 + \log_2 1000 = 1 + 9.97 = 10.97$ and $1 + \log_2 870 = 1 + 9.76 = 10.76$. *Work it out.* Round down. *Result.* $k = 10$ for both.
3. Bin width. *Formula.* width $= (\max - \min)/k$. *Plug in.* normal $(5.2128 - (-1.6554))/10$, not-normal $(1.2957 - 0.0007)/10$. *Work it out.* 6.868/10 and 1.295/10. *Result.* 0.687 and 0.130.
4. Counts with 10 bins. normal gives 1, 6, 54, 128, 214, 258, 217, 83, 33, 6. not-normal gives 402, 228, 94, 69, 33, 20, 12, 6, 2, 4.

Answer. normal.txt is a symmetric bell with its peak in the bin around 1.8 to 2.5, close to the mean 2.01. not-normal.txt piles up in the first bin near 0 and trails off in a long right tail out to about 1.3. So one looks normal and the other is right-skewed.

Check. His main.tex says 11 bins for both because analyze.py used round$(1 + \log_2 n)$, which rounds 10.97 and 10.76 up. The slide floors, giving 10. No conclusion changes, but use the floor on the exam.

By hand. Six points 1, 3, 5, 7, 9, 11. $k = \lfloor 1 + \log_2 6 \rfloor = \lfloor 1 + 2.58 \rfloor = 3$. Width $= (11 - 1)/3 = 3.33$. Bins $[1, 4.33)$, $[4.33, 7.67)$, $[7.67, 11]$ hold 2, 2, 2. Bin width changes the shape (Summ p.55 to 56), so judge shape with the Q-Q plot.

HW3 Q2d. Quantile plot

Question. *d. (2 pts) Draw the quantile plot for the sample data.* [For each of normal.txt and not-normal.txt.]

In plain words. A quantile plot is just the sorted data plotted against where each point sits, its fraction f_i. You can read any percentile straight off it, and the shape shows the tails. A long flat stretch then a sharp climb means a long right tail.

1. Plot $(f_i, x_{(i)})$ with $f_i = (i - 0.5)/n$, where $x_{(i)}$ is the i-th smallest value (Summ p.59).
2. normal.txt gives an S shape. not-normal.txt stays flat, then shoots up near $f = 1$, the long right tail.

Answer. The S shape for normal.txt and the flat-then-steep curve for not-normal.txt in step 2.

By hand. In the Summ p.60 table ($n = 10$), $Q(0.80)$ sits at $i = 0.80(10) + 0.5 = 8.5$, halfway between $x_{(8)} = 44.5$ and $x_{(9)} = 50.3$. *Work it out.* $(44.5 + 50.3)/2 = 94.8/2$. *Result.* 47.4.

HW3 Q2e. Compare with a normal using a Q-Q plot

Question. *e. (3 pts) Compare the distribution of the sample data against the theoretical normal distribution using a Q-Q plot.* [For each of normal.txt and not-normal.txt.]

In plain words. A Q-Q plot lines up your sorted data against what a perfect normal with the same mean and SD would give at the same positions. If the data are normal, the points fall on a straight line. Bends at the ends tell you which way the data aren't normal.

1. For each f_i find z_i with $\Phi(z_i) = f_i$, where z_i is the standard normal value with area f_i below it. Use the table or $z_i = 4.91[f_i^{0.14} - (1 - f_i)^{0.14}]$ (Summ p.69).
2. *Formula.* Rescale $x_i = m + sz_i$, where m is the sample mean and s the sample SD, so normal data falls on the 45° line. Unscaled, the intercept is the mean and the slope the SD (Summ p.70).
3. Plot theoretical against observed (Summ p.63) and read the shape (Summ p.71 to 73).

Answer. normal.txt follows the line with correlation 0.9993, so it's consistent with normal. not-normal.txt sits above the line at both ends and below in the middle, correlation 0.904. The normal predicts about -0.47 at the low end but the data never go below 0, and the top values are far too large. Right-skewed, not normal.

By hand. Data 1, 3, 5, 7, 9, 11 with $m = 6$, $s = 3.742$.

1. f_i for $i = 1, 2, 3$ are $0.5/6 = .083$, $1.5/6 = .25$, $2.5/6 = .417$. From the table $z = -1.38, -0.67, -0.21$, and the top three are the same with a plus sign.
2. *Plug in.* $x = 6 + 3.742z$. *Work it out.* $6 - 3.742(1.38) = 6 - 5.16 = 0.84$, $6 - 3.742(0.67) = 6 - 2.51 = 3.49$, $6 - 3.742(0.21) = 6 - 0.79 = 5.21$, then by symmetry 6.79, 8.51, 11.16.
3. *Result.* Pairs (0.84, 1), (3.49, 3) and so on are all within 0.5 of the line, so normal.

Skewed version 1, 1, 1, 2, 3, 10. $m = 18/6 = 3$. Deviations $-2, -2, -2, -1, 0, 7$, squares sum to $4 + 4 + 4 + 1 + 0 + 49 = 62$, $s^2 = 62/5 = 12.4$, $s = 3.52$. Theoretical $3 - 3.52(1.38) = -1.86$, $3 - 3.52(0.67) = 0.64$, $3 - 3.52(0.21) = 2.26$, then 3.74, 5.36, 7.86. Ends above the line and middle below is the right-skew pattern.

Usual mistake. Plotting f_i instead of z_i on the x-axis, or reading "longer tails" (Summ p.71) when the shape is asymmetric (Summ p.73).

HW4. Confidence intervals

HW4 1.1 and 2.1. The file is the population. Population mean and variance

Question. *State which tool you used (Excel, Python, R) for each part. Problem 1, Normally distributed data. Use the population data in the file normal.txt. This data is normally distributed. 1. Find the population mean and variance. Problem 2, Data that is not normally distributed. Repeat parts 1 to 3 of Problem 1 for the population data in the file not-normal.txt. This data is not normally distributed.*

In plain words. The homework pretends the whole file is the entire population, so you know the true answer. Later you check whether your sample-based intervals actually caught it. For a whole population you divide by N, the number of values, not $N - 1$.

1. Mean. *Formula.* $\mu = \frac{1}{N}\sum x_i$, where N is the population size. *Plug in.* 2014.15/1000 and 181.25/870. *Result.* 2.0142 and 0.2083.
2. Variance. *Formula.* $\sigma^2 = \dfrac{\sum x_i^2 - (\sum x_i)^2/N}{N}$. Divide by N, not $N - 1$. The $n - 1$ is for estimating from a sample (Summ p.39).

Plug in. normal, 1044.69/1000. not-normal, 38.25/870. *Result.* 1.0447 and 0.0440.

3. Shortcut from HW3's value. $\sigma^2 = s^2 \times (N-1)/N = 1.0457 \times 999/1000 = 1.0447$.

Answer. normal has $\mu = 2.0142$, $\sigma^2 = 1.0447$. not-normal has $\mu = 0.2083$, $\sigma^2 = 0.0440$ (the same values as CI p.32).

HW4 1.2. Sample of 50 from normal.txt. 90, 95, 99% CIs. Do they contain μ?

Question. *2. Generate a random sample of size 50, that is, randomly select 50 numbers from the population. Find 90%, 95% and 99% confidence intervals for the mean. Is the population mean within the intervals?*

In plain words. A confidence interval is a range built from one sample that should catch the true mean most of the time. It's the sample mean plus or minus a margin. The margin is a critical value (bigger for more confidence) times the standard error $s/\sqrt{n}$, which is how much the sample mean would wobble if you redrew the sample.

1. $n > 30$, so use z (CI p.27, p.31). *Formula.* $\bar{x} \pm z[\alpha/2]\, s/\sqrt{n}$, where $\bar{x}$ is the sample mean, s the sample SD, n the sample size, and $z[\alpha/2]$ the normal value with area $\alpha/2$ above it. α is 1 minus the confidence, so 0.10, 0.05, 0.01.
2. From the sample, $\bar{x} = 2.2093$ and $s = 0.9443$.
3. SE. *Formula.* $\text{SE} = s/\sqrt{n}$. *Plug in.* $0.9443/\sqrt{50}$. *Work it out.* 0.9443/7.071. *Result.* 0.1335.
4. Margins. *Formula.* margin $= z[\alpha/2] \times \text{SE}$. *Plug in.* 1.645(0.1335), 1.960(0.1335), 2.576(0.1335). *Result.* 0.2196, 0.2617, 0.3439.
5. CIs. *Plug in.* 2.2093 ± 0.2196, 2.2093 ± 0.2617, 2.2093 ± 0.3439. *Result.* below.

Answer. 90% (1.9897, 2.4290), 95% (1.9476, 2.4710), 99% (1.8653, 2.5533). All contain $\mu = 2.0142$.

By hand. $n = 100$, $\bar{x} = 10$, $s = 2$. $\text{SE} = 2/\sqrt{100} = 2/10 = 0.2$. Margin $1.96 \times 0.2 = 0.392$. CI $10 \pm 0.392 = (9.61, 10.39)$.

Sample size for $\pm 5\%$ at 95% (CI p.44). *Formula.* $n = \left(\dfrac{z\, s}{r\, \bar{x}}\right)^2$, where $r = 0.05$ is the allowed error as a fraction of the mean. *Plug in.* $\left(\dfrac{1.96 \times 2}{0.05 \times 10}\right)^2$. *Work it out.* $3.92/0.5 = 7.84$, $7.84^2 = 61.5$. *Result.* Round up to 62 runs.

HW4 1.3. Sample of 16. Same three CIs

Question. *3. Generate a random sample of size 16. Find 90%, 95% and 99% confidence intervals for the mean. Is the population mean within the intervals?*

In plain words. Same thing with only 16 points. With a small sample s itself is shaky, so you use the t distribution, which has fatter tails and gives a bigger critical value to pay for that. Its shape depends on the degrees of freedom (df), which is $n - 1$.

1. Small n and a normal population. *Formula.* $\bar{x} \pm t[\alpha/2, n-1]\, s/\sqrt{n}$, where $t[\alpha/2, n-1]$ is the t value with area $\alpha/2$ above it and $n-1$ df (CI p.37). Here df $= 16 - 1 = 15$.
2. From the sample, $\bar{x} = 2.2911$, $s = 1.1332$.
3. SE. *Plug in.* $1.1332/\sqrt{16}$. *Work it out.* 1.1332/4. *Result.* 0.2833.
4. Margins. *Plug in.* 1.753(0.2833), 2.131(0.2833), 2.947(0.2833). *Result.* 0.4966, 0.6037, 0.8349.
5. CIs. *Plug in.* 2.2911 ± 0.4966, 2.2911 ± 0.6037, 2.2911 ± 0.8349.

Answer. 90% (1.7944, 2.7877), 95% (1.6872, 2.8949), 99% (1.4562, 3.1259). All contain $\mu = 2.0142$.

By hand. $n = 9$, $\bar{x} = 20$, $s = 6$. $\text{SE} = 6/\sqrt{9} = 6/3 = 2$. df $= 8$, $t[0.025, 8] = 2.306$. Margin $2.306 \times 2 = 4.612$. CI $20 \pm 4.612 = (15.39, 24.61)$. If σ is given, use z even for small n (CI p.27).

Exam variant from raw values. Data 8, 10, 12, 14.

1. Mean. $(8 + 10 + 12 + 14)/4 = 44/4 = 11$.
2. SD. *Formula.* $s^2 = \dfrac{\sum (x_i - \bar{x})^2}{n-1}$. *Plug in.* Deviations $-3, -1, 1, 3$, squares $9, 1, 1, 9$. *Work it out.* $20/3 = 6.67$, $s = \sqrt{6.67} = 2.582$.
3. SE. $2.582/\sqrt{4} = 2.582/2 = 1.291$.
4. CI. df $= 3$, $t[0.025, 3] = 3.182$. Margin $3.182 \times 1.291 = 4.11$. *Result.* $11 \pm 4.11 = (6.89, 15.11)$.

HW4 1.4. Compare widths from 1.2 and 1.3, explain with sample size and distribution

Question. *4. Compare the widths of the intervals from parts 2 and 3. Explain the difference in terms of both the sample size and the distribution you used for the critical value.*

In plain words. Why is the $n = 16$ interval so much wider? Width is critical value times $s/\sqrt{n}$, so three things can change it. Fewer points shrink $\sqrt{n}$, the t value is bigger than z, and the sample SD came out different just by luck.

1. *Formula.* Width $= 2 \times \text{crit} \times s/\sqrt{n}$.
2. *Plug in.* $n = 50$, $2 \times 1.960 \times 0.1335$. $n = 16$, $2 \times 2.131 \times 0.2833$. *Result.* 0.5235 and 1.2077 (unrounded).
3. Ratio. $1.2077/0.5235 = 2.31$.
4. Split the ratio into factors. Sample size $\sqrt{50/16} = \sqrt{3.125} = 1.77$ (CI p.39). Critical value $2.131/1.960 = 1.09$, from t having heavier tails (CI p.35). Luck in s, $1.1332/0.9443 = 1.20$.
5. Check, $1.77 \times 1.09 \times 1.20 = 2.31$.

Answer. The 95% interval at $n = 16$ is 2.31 times as wide as at $n = 50$. About 1.77x comes from the smaller sample, 1.09x from using t with 15 df instead of z, and 1.20x from the second sample happening to have a bigger SD.

Usual mistake. Blaming only n. The question asks for the critical value too. Halving a width takes 4 times the data.

HW4 2.2 and 2.3. Repeat with not-normal.txt

Question. *Problem 2. Repeat parts 1 to 3 of Problem 1 for the population data in the file not-normal.txt. This data is not normally distributed.* [So 2.2 is a sample of 50 and 2.3 a sample of 16, each with 90%, 95% and 99% CIs for the mean, and is the population mean within the intervals?]

In plain words. Same intervals, but on skewed data. The point is to see whether the normal-based recipes still work when the data aren't normal. At $n = 50$ the central limit theorem (CLT), which says averages of many values look normal even when the values don't, saves you. At $n = 16$ it's shakier.

1. $n = 50$. The data aren't normal but the CLT makes $\bar{x}$ close to normal, so z is fine (CI p.29, p.67). $\bar{x} = 0.2168$, $s = 0.2278$. $\text{SE} = 0.2278/\sqrt{50} = 0.2278/7.071 = 0.0322$. Margins $1.645(0.0322) = 0.0530$, $1.960(0.0322) = 0.0631$, $2.576(0.0322) = 0.0829$. CIs are 0.2168 plus or minus each.
2. $n = 16$, t with 15 df. $\bar{x} = 0.2406$, $s = 0.2040$. $\text{SE} = 0.2040/\sqrt{16} = 0.2040/4 = 0.0510$. Margins $1.753(0.0510) = 0.0894$, $2.131(0.0510) = 0.1087$, $2.947(0.0510) = 0.1503$. CIs are 0.2406 plus or minus each.

Answer.

Level	$n = 50$ (z)	$n = 16$ (t_{15})
90%	(0.1638, 0.2698)	(0.1512, 0.3300)
95%	(0.1537, 0.2800)	(0.1319, 0.3493)
99%	(0.1339, 0.2998)	(0.0903, 0.3909)

All six contain $\mu = 0.2083$.

Usual mistake. Trusting the t interval at $n = 16$ on skewed data. The CI p.67 example, $n = 16$ from a skewed population, covers only about 91% at nominal 95% and runs too narrow.

HW4 2a. Batching. 16 values in draw order into 4 batches of 4, CI from the batch means with t on 3 df

Question. *Then, for the smaller sample only (size 16), compute the 90%, 95% and 99% intervals two further ways. These correspond to the first and third responses on the lecture slides for when the normality assumption fails. 2a. Batching. Group your 16 observations into 4 batches of 4 and compute the mean of each batch. You now have 4 numbers. Compute the confidence intervals from those 4 batch means, using the t distribution with 3 degrees of freedom. Batch the observations in the order you drew them. Do not sort the sample first, sorting would group the small values together and the large values together, and the batch means would no longer be four draws from the same distribution.*

In plain words. Batching is a trick to make data look more normal. Averaging groups of 4 values gives batch means that are closer to normal than the raw values, the CLT done by hand. You then treat the 4 batch means as your whole sample, which leaves only $4 - 1 = 3$ degrees of freedom.

1. Average groups of 4 or more to push toward normal (CI p.68). Keep draw order, since sorting would make each batch a different slice.
2. Batch means 0.4212, 0.2368, 0.0904, 0.2139. *Formula.* $\bar{x} = \frac{1}{b}\sum \bar{x}_j$, where $\bar{x}_j$ is batch j's mean and $b = 4$ batches. *Plug in.* 0.9623/4. *Result.* 0.2406, same as the raw mean.
3. Variance of the batch means. *Formula.* $s^2 = \dfrac{\sum \bar{x}_j^2 - (\sum \bar{x}_j)^2/b}{b-1}$. *Plug in.* $\sum \bar{x}_j^2 = 0.2874$ and $(\sum \bar{x}_j)^2/4 = 0.9623^2/4 = 0.2315$. *Work it out.* $(0.2874 - 0.2315)/3 = 0.0559/3 = 0.0186$, $s = \sqrt{0.0186} = 0.1365$. *Result.* $\text{SE} = s/\sqrt{b} = 0.1365/2 = 0.0683$.
4. Margins with t on 3 df. $2.353(0.0683) = 0.161$, $3.182(0.0683) = 0.217$, $5.841(0.0683) = 0.399$. CIs are 0.2406 plus or minus each.

Answer. 90% (0.0799, 0.4012), 95% (0.0233, 0.4578), 99% (−0.1581, 0.6393). All contain μ.

By hand. 12 values into 3 batches with means 2, 4, 6. Mean $12/3 = 4$. $s^2 = ((2-4)^2 + (4-4)^2 + (6-4)^2)/2 = 8/2 = 4$, $s = 2$. $\text{SE} = 2/\sqrt{3} = 2/1.732 = 1.155$. df = 2, $t[0.025, 2] = 4.303$, margin $4.303 \times 1.155 = 4.97$. 95% CI $= 4 \pm 4.97 = (-0.97, 8.97)$. Batching needs lots of data.

HW4 2b. Bootstrap with $B = 10000$, seed 12345

Question. *2b. Bootstrap. Resample your 16 observations with replacement, recompute the mean, and repeat many times. The percentiles of those resampled means give the interval.* [The given NumPy script draws the 16 from not-normal.txt without replacement using seed 12345, makes $B = 10000$ resamples of size 16 with replacement, and prints the $a = (1 - \text{level})/2$ and $1 - a$ quantiles of the resampled means for levels 0.90, 0.95, 0.99.] *If you are working in Excel or R instead, you may use any equivalent implementation, but report B, your seed, and the tool you used.*

In plain words. The bootstrap pretends your 16 values are the whole population. You draw new samples of 16 from them with replacement, many times, and watch how much the mean jumps around. The middle 95% of those means is your 95% interval, with no normal assumption needed.

1. Treat the sample as the population. Draw 16 with replacement, take the mean, repeat B times, where $B = 10000$ is the number of resamples (CI p.69).
2. Sort the means. 95% runs from the 2.5th to the 97.5th percentile, which is about the 250th and the 9750th of 10000 sorted means. 90% runs from the 5th to the 95th percentile, 99% from the 0.5th to the 99.5th.

Answer. 90% (0.1618, 0.3242), 95% (0.1493, 0.3407), 99% (0.1230, 0.3705). All contain 0.2083.

By hand. Sample $\{2, 4, 9\}$. Resamples $\{2, 2, 9\}$ mean $13/3 = 4.33$, $\{4, 9, 9\}$ mean $22/3 = 7.33$, $\{2, 4, 4\}$ mean $10/3 = 3.33$, $\{9, 2, 4\}$ mean $15/3 = 5$. Repeat thousands of times, sort, read the percentiles.

Usual mistake. Resampling without replacement gives the same mean every time. If the 16 values missed the tail, every resample misses it too (CI p.69 to 70).

HW4 2c. Compare raw, batched and bootstrap at $n = 16$

Question. *2c. Compare.*

1. *Put the three sets of intervals for the sample of 16, raw, batched, and bootstrapped, into a single table, at all three confidence levels. State for each whether it contains the population mean.*
2. *Which method gives the widest intervals, and which the narrowest? Was that what you expected before computing them?*
3. *The batched interval is meant to correct for the data not being normal. Explain why the correction makes the interval wider rather than narrower, and say what you gave up to obtain it.*
4. *Look at your 99% batched interval. Does it contain any values that are impossible for this quantity? What does that tell you about the method at this sample size?*
5. *Your sample either did or did not contain the population mean. Explain why that single outcome does not, on its own, tell you whether a method's coverage is correct.*

In plain words. This lines up the three methods side by side and asks you to explain the pattern. Batching is widest because 3 df make the t value big. Bootstrap is narrowest because it can only see the spread inside the 16 values you already have.

Width = upper end minus lower end. For example raw 95% is $0.3493 - 0.1319 = 0.217$.

	Raw t	Batched	Bootstrap
90% width	0.179	0.321	0.162
95% width	0.217	0.435	0.191
99% width	0.301	0.797	0.248

Steps 1 to 5 answer parts 1 to 5 of 2c.

1. All nine contain 0.2083.
2. Batching is widest and bootstrap narrowest at every level. That's what we expected, since the slides predict it (CI p.68 says about 50% wider, CI p.70 calls bootstrap the narrowest). Here batching was about twice as wide ($0.435/0.217 = 2.0$) because the batch means were spread out.
3. Batching widens because df fall from 15 to 3, so t goes from 2.131 to 3.182, and here the SE also rose from 0.0510 to 0.0683. The cost is precision and any statement about individual values.
4. The 99% batched interval reaches −0.158, impossible for positive data. Too few df at high confidence.

5. Coverage is a long-run rate over repeated samples (CI p.33). One hit is consistent with 95% and with 91%, so you'd have to repeat the draw and count.

Which fix when. Correlated observations, batch. Median or a multiplicative process, transform. Mean with independent data, bootstrap (CI p.70).

The third fix by hand, transform. HW4 skipped it, but it's the other response on CI p.68. Data 1, 10, 100.

1. Take $\log_{10}$ to get 0, 1, 2. Mean $3/3 = 1$. $s^2 = ((0-1)^2 + 0 + (2-1)^2)/2 = 2/2 = 1$, $s = 1$.
2. SE $= 1/\sqrt{3} = 0.577$. With $t[0.025, 2] = 4.303$ the margin is $4.303 \times 0.577 = 2.48$.
3. Log-scale CI $1 \pm 2.48 = (-1.48, 3.48)$.
4. Back-transform with 10^x. $10^{-1.48} = 0.033$ and $10^{3.48} \approx 3050$, centred on $10^1 = 10$.

That's an interval for the median (here the geometric mean), not the mean, which is $111/3 = 37$ (CI p.68 trap).

HW4 Problem 3. Confidence intervals in the wild, 8 questions

Question. *Problem 3, A paper from the wild. Work through the homework exercise in Part III of the companion handout Confidence Intervals in the Wild. Read only the experimental setup and results sections of the paper you choose, and answer all eight questions there. Two or three sentences per question is enough. Give the full citation at the top. Choosing the paper is part of the exercise. Pick the entry from Part I of the handout that sits closest to your own research area, or any peer-reviewed paper from that area, ideally one you would cite anyway, that reports experimental results with confidence intervals, error bars, or any other statement of uncertainty. Papers that report results with no statement of uncertainty are also fair game, and often more revealing.* [Defaults if no area fits are entry 4, Georges et al., Statistically Rigorous Java Performance Evaluation, OOPSLA 2007, or entry 13, Henning et al., When Should I Run My Application Benchmark?, FSE Companion 2025. The eight handout questions are printed in italics at the start of each answer under "His paper" below.]

In plain words. You pick a real paper and audit how it reports uncertainty, using 8 fixed questions. The point is to spot when an error bar promises more than it should, usually because the observations weren't independent or the interval ignores systematic error. Two or three sentences per question is enough.

Checklist.

1. Paper and why.
2. Locate the first interval or error bar. Finding none is a finding.
3. Estimator. Mean, proportion, ratio, quantile, GM? Which lecture formula fits?
4. n. Justified, or just asserted?
5. Assumption. What does the interval assume? If skewed with small n, which of batch, transform, bootstrap (CI p.68)?
6. Independence. What's the unit, and are the units really independent (CI p.66)?
7. What the interval excludes. Systematic error, setup, population (CI p.17).
8. Interpretation. Do they claim "95% probability the true value is inside", which needs a credible interval?

His paper. Yan et al., Puffer, NSDI 2020.

1. *Say what you picked and why. Give the full citation and one or two sentences on how the paper relates to your own work.* Video streaming, already cited in HW2, and no Part I entry fits.
2. *Locate the interval. Quote the sentence or identify the figure where an interval or error bar first appears. If you cannot find one, say so, that is a finding, and the rest of the questions still apply to whatever the paper reports in its place.* Sec. 3.4, "95% CI on stall ratio between ±13% and ±21% of the mean". Fig. 9 error bars are 95% CIs.
3. *Identify the estimator. Mean, proportion, variance, quantile, geometric mean, or something else? Which formula from lecture applies, if any?* Stall ratio is a ratio, so bootstrap. Average SSIM is a duration-weighted mean with a z interval.
4. *Find n. How many measurements? Is the choice justified, or asserted?* 637,189 streams from 54,612 IPs, about 2.5 to 2.8 stream-years per scheme. That's whatever 8 months gave, not planned. The "2 stream-years to separate a 15% difference" is unexplained.
5. *Check the assumption. What distributional assumption does their interval require, and do the authors address it? If the response is plainly not normal and n is small, which of the three responses on slide 68 did they use, and if none, which would you have used?* Stalls are very skewed (4% of streams stalled). Bootstrap fits, but B and the interval type aren't stated.
6. *Check the independence. What is the unit being treated as an independent draw, a run, a request, a benchmark, a user, a seed? Are consecutive units plausibly independent, or is this an entry-15 situation dressed up as an entry-1 situation?* Randomized by session but the bootstrap resamples streams, which share a connection. The intervals are likely too narrow. Resample sessions.
7. *Check what the interval excludes. Name one source of error in their setup that the interval does not cover (pitfall 3).* One Stanford server, self-selected viewers, no Safari. Says nothing about other users or servers.
8. *Check the interpretation. Do the authors state a conclusion that requires a credible interval while computing a confidence interval?* No credible-interval claim. But they compare schemes by error-bar overlap, which isn't a test.

Exam scenario. "Mean RPC latency was 4.2 ms ± 0.1 ms (95% CI) from 10,000 requests in one run on one machine, so there is a 95% chance the true mean is between 4.1 and 4.3." Walk the checklist.

1. The claim is a single mean latency with a 95% CI, plus a probability reading of that CI. There's no paper to pick, so say what's being claimed.
2. The interval is 4.2 ± 0.1 ms, labelled 95%. Working backwards, $0.1 = 1.96\, s/\sqrt{10000}$ gives $s = 0.1 \times 100/1.96 = 5.1$ ms, so the per-request SD is about 5 ms, bigger than the mean. That fits skewed latency.
3. Estimator is a mean, z formula since n is large (CI p.31).
4. $n = 10{,}000$ is asserted, and it counts requests, not runs.
5. Latency is skewed, but for a mean at this n the CLT is fine (CI p.29).
6. The real problem. Consecutive requests share caches, queues and load, so they aren't independent and the interval is too narrow (CI p.66). Batch the requests or do several runs with one mean per run (CI p.70, HT p.22).
7. One machine, one run. Systematic error and the setup aren't in the interval (CI p.17).
8. "95% chance the true mean is inside" is a credible-interval claim. A CI only says the procedure covers μ in 95% of repeats (CI p.33). Honest headline. "In one run on one machine, mean RPC latency was 4.2 ms. The ±0.1 ms treats 10,000 back-to-back requests as independent, so the real uncertainty is larger, and we'd need several independent runs to give an honest inter-

val. Nothing here covers other machines."

HW5. Hypothesis testing

HW5 1.1. $\sigma = 0.8$ h known, $n = 36$, $\bar{x} = 2.74$. Test $H_0 : \mu = 2.5$ at $\alpha = 0.05$

Question. *Problem 1, Testing a mean with σ known. For a certain class of a new neural network architecture, we would like to test the hypothesis that the mean training time μ is equal to $\mu_0 = 2.5$ hours. The standard deviation is known to be 0.8 hours. 1. (7 points) Suppose we run experiments with 36 such systems and observe that the mean training time in our experiments is 2.74 hours. Test the null hypothesis $H_0 : \mu = 2.5$ hours at the level of significance $\alpha = 0.05$. Do you reject H_0 or fail to reject it?*

In plain words. A hypothesis test asks whether the data would be surprising if the boring claim H_0 (the mean is 2.5 h) were true. You measure how far the sample mean is from 2.5 in units of standard error, and that's Z. If $|Z|$ goes past 1.96, that would happen less than 5% of the time under H_0, so you reject it.

1. $H_0 : \mu = 2.5$, $H_1 : \mu \neq 2.5$. No direction is named, so two-sided (HT p.5).
2. *Formula.* σ is known, so $Z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ (HT p.8), where $\bar{x} = 2.74$ is the sample mean, $\mu_0 = 2.5$ the value in H_0, $\sigma = 0.8$ the known SD and $n = 36$ the number of systems.
3. *Plug in.* $Z = \dfrac{2.74 - 2.5}{0.8/\sqrt{36}}$.
4. *Work it out.* SE $= 0.8/6 = 0.1333$. Top $= 0.24$. $Z = 0.24/0.1333$.
5. *Result.* $Z = 1.80$.
6. Decision rule. Two-sided at 0.05, reject if $|Z| > z[0.025] = 1.960$. Since $1.80 < 1.96$, fail to reject.
7. Interval check (HT p.9). $2.74 \pm 1.96(0.1333) = 2.74 \pm 0.261 = (2.479, 3.001)$ contains 2.5, same verdict.

Answer. $Z = 1.80$, fail to reject H_0. Matches main.tex.

Small version. $n = 16$, $\sigma = 0.8$, $\bar{x} = 2.9$. SE $= 0.8/\sqrt{16} = 0.8/4 = 0.2$. $Z = (2.9 - 2.5)/0.2 = 0.4/0.2 = 2.0 > 1.96$, reject. The HT p.10 example gives $Z = -2.83$ and rejects.

Usual mistake. Writing "accept H_0". Failing to reject isn't evidence for H_0 (HT p.4).

HW5 1.2. p-value and conclusion

Question. *2. (5 points) For the same experiments, find the p-value. Using the p-value you computed, restate the conclusion you reached in part 1.*

In plain words. The p-value is the chance of getting a result at least this far from 2.5 if H_0 were true. A small p means the data would be surprising under H_0. You reject when $p < \alpha$, and that always gives the same verdict as the critical-value rule.

1. *Formula.* $p = P[\text{statistic this extreme} \mid H_0]$ (HT p.13). Two-sided, $p = 2\,(1 - \Phi(|Z|))$.
2. *Plug in.* $p = 2\,(1 - \Phi(1.80))$.
3. *Work it out.* $1 - 0.9641 = 0.0359$, then 2×0.0359.
4. *Result.* $p = 0.0718$.

Answer. $p = 0.072 > 0.05$, so fail to reject H_0. Same conclusion as part 1, the data don't show the mean differs from 2.5 h.

Usual mistake. Forgetting to double for two-sided, or saying "7% chance H_0 is true" (HT p.14).

HW5 1.3. $\alpha = 0.01$, $n = 36$. Smallest $\bar{x} > 2.5$ that rejects

Question. *3. (8 points) Suppose we reduce the level of significance to $\alpha = 0.01$. If we run experiments on 36 systems to test the null hypothesis that $\mu = 2.5$ hours, what is the minimum mean time (larger than 2.5 hours) we would need to observe to reject the hypothesis?*

In plain words. Flip the test around. Instead of asking whether a given mean rejects, find the cutoff mean where Z just hits the critical value. A smaller α means you demand stronger evidence, so the cutoff moves further from 2.5.

1. Two-sided at 0.01, so the critical value is $z[0.005] = 2.576$.
2. *Formula.* Reject when $\dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} > z[\alpha/2]$, so $\bar{x} > \mu_0 + z[\alpha/2]\,\sigma/\sqrt{n}$.
3. *Plug in.* $\bar{x} > 2.5 + 2.576 \times 0.8/6$.
4. *Work it out.* $0.8/6 = 0.1333$, $2.576 \times 0.1333 = 0.343$, $2.5 + 0.343$.
5. *Result.* $\bar{x} > 2.843$.
6. A one-sided $H_1 : \mu > 2.5$ would use $z[0.01] = 2.326$, so $2.5 + 2.326 \times 0.1333 = 2.5 + 0.310 = 2.810$.

Answer. About 2.84 h two-sided (2.81 one-sided). Matches main.tex.

Power version (HT p.19). How many runs to detect a shift of $\delta = 0.2$ h with 80% power at 0.05? *Formula.* $n = \left(\dfrac{(z[\alpha/2] + z[\beta])\,\sigma}{\delta}\right)^2$, where δ is the smallest shift you care about, power is $1 - \beta = 0.80$, and $z[0.20] = 0.84$. *Plug in.* $\left(\dfrac{(1.96 + 0.84) \times 0.8}{0.2}\right)^2$. *Work it out.* $2.80 \times 0.8 = 2.24$, $2.24/0.2 = 11.2$, $11.2^2 = 125.4$. *Result.* Round up to 126.

HW5 2.1. Suspect the mean time to first fault is below 740 h. H_0 and H_1?

Question. *Problem 2, Choosing the null hypothesis. A research group develops an autonomous driving system. Because of the specific sensor model adopted, they suspect that the mean time μ to experience the first fault in the sensor subsystem is less than $\mu_0 = 740$ hours. 1. If we use hypothesis testing to confirm this assertion, what should be the null hypothesis H_0? What should be the alternative hypothesis H_1? Carefully justify your choices for H_0 and H_1.*

In plain words. A test can only ever reject H_0, never prove it. So whatever you want to show goes in H_1, here that the mean is below 740. The test is one-sided because only one direction matters.

1. The claim to establish goes in H_1, because a test only supports a claim by rejecting H_0 (HT p.4).
2. $H_0 : \mu = 740$ (or ≥ 740), $H_1 : \mu < 740$, one-sided (HT p.5).

Answer. $H_0 : \mu = 740$ h, $H_1 : \mu < 740$ h, one-sided, because rejecting H_0 is the only way the data can confirm the suspicion.

Usual mistake. Putting the claim in H_0, so a small noisy experiment "supports" it by failing to reject. Picking the side after seeing the data is a disguised two-sided test at double α (HT p.5).

HW5 2.2. $n = 50$, $\bar{x} = 706$, $s = 96.1$, $\alpha = 0.01$. Can we confirm $\mu < 740$?

Question. *2. Suppose we test 50 such systems and obtain a mean time to first fault of 706 hours, with a standard deviation of 96.1 hours. If the target level of significance is $\alpha = 0.01$, can we confirm, based on these experiments, the assertion that the mean time to first fault is less than 740 hours?*

In plain words. Now actually run the test from 2.1. σ isn't known, but with 50 systems the sample SD s is a good stand-in, so the Z formula works. One-sided at 0.01 means you reject only if Z is below -2.326.

1. σ is unknown but $n = 50$, so s stands in and $t \approx Z$ (HT p.11).
2. *Formula.* $Z = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$, where $\bar{x} = 706$, $\mu_0 = 740$, $s = 96.1$ the sample SD, $n = 50$.
3. *Plug in.* $Z = \dfrac{706 - 740}{96.1/\sqrt{50}}$.
4. *Work it out.* SE $= 96.1/7.071 = 13.59$. Top $= -34$. $Z = -34/13.59$.
5. *Result.* $Z = -2.50$.
6. Decision. One-sided, reject if $Z < -z[0.01] = -2.326$. Since $-2.50 < -2.326$, reject.
7. p-value. *Formula.* one-sided lower tail $p = \Phi(Z)$. *Plug in.* $\Phi(-2.50) = 1 - \Phi(2.50)$. *Work it out.* $1 - 0.9938$. *Result.* $p = 0.0062 < 0.01$.

Answer. Reject H_0, so the data confirm $\mu < 740$ at 0.01. With t_{49} (crit 2.405) the verdict is the same.

Small version. $n = 36$, $\bar{x} = 716$, $s = 60$. SE $= 60/6 = 10$. $Z = (716 - 740)/10 = -24/10 = -2.4 < -2.326$, reject. $p = 1 - \Phi(2.4) = 1 - 0.9918 = 0.0082$.

Exam variant, small n and the side matters. $n = 9$, $\bar{x} = 12$, $s = 3$, $\mu_0 = 10$, population roughly normal. SE $= 3/\sqrt{9} = 3/3 = 1$. $t = (12 - 10)/1 = 2.0$ with $9 - 1 = 8$ df. Two-sided at 0.05 needs $t[0.025, 8] = 2.306$, so fail to reject. One-sided $H_1 \colon \mu > 10$ (stated before the data) needs $t[0.05, 8] = 1.860$, so reject. That's why the side has to be fixed in advance (HT p.5, HT p.11).

HW5 Problem 3. Hypothesis testing in the wild, questions 0 to 6

Question. *Problem 3, A paper from the wild (60 points). Work through the homework exercise in Part III of the companion handout Hypothesis Testing in the Wild. Read only the experimental setup and results sections of the paper you choose, and answer all seven questions there. Two or three sentences per question is enough, except for the last question, which asks you to rewrite the paper's key result as an effect size with a confidence interval. Give the full citation at the top. Choosing the paper is part of the exercise. Pick the entry from Part I of the handout that sits closest to your own research area, or any peer-reviewed paper from that area, ideally one you would cite anyway, that makes a claim of the form "A is better than B". Papers that make that claim with no test and no statement of uncertainty are also fair game, and often more revealing. Several entries (1, 3, 4, 7, 9 and 10) are about statistical practice rather than about one system beating another. If you pick one of them, apply the questions to the paper's own central empirical claim.* [Defaults if no area fits are Mytkowicz et al., Producing Wrong Data Without Doing Anything Obviously Wrong!, ASPLOS 2009, or Musgrave et al., A Metric Learning Reality Check, ECCV 2020. The seven handout questions, 0 to 6, are printed in italics at the start of each answer under "His paper" below.]

In plain words. Pick a paper that claims A beats B and audit its testing with 7 fixed questions. Mostly you're checking the things papers skip, like what counts as one independent observation, how many tests they ran, and whether they peeked. The last question rewrites the main result as an effect size, the actual size of the difference, with a CI.

Checklist.

0. Paper and why.
1. H_0, H_1, one or two-sided. Reconstruct if missing and say so.
2. Test and assumptions (normality, independence, pairing, equal variances). The unit, and does n count units or measurements (HT p.20)?
3. α, n, power. With no power analysis, the detectable effect is about 2.8×SE at $\alpha = 0.05$ and 80% power (HT p.19).
4. Count the tests. FWER $= 1 - (1 - \alpha)^m$ (HT p.27). Any correction, and FWER or FDR (HT p.28)?
5. Stopping rule (peeking, HT p.23 to 24), interference (HT p.22), right estimand.
6. Rewrite as an effect size with a CI (HT p.16, p.18). Big enough to matter?

His paper. Spang et al., Unbiased Experiments in Congested Networks, IMC 2021 (entry 8).

0. *Say what you picked and why. Give the full citation and one or two sentences on how the paper relates to your own work.* Closest to video streaming, and his comparisons share a bottleneck too.
1. *Write down H_0 and H_1. For the paper's central claim, state the null and alternative hypotheses, and whether the test is one- or two-sided. If the paper never states them, reconstruct them, and say so.* Not stated. Reconstructed $H_0 \colon \text{TTE} = 0$ against $H_1 \colon \text{TTE} \neq 0$, two-sided since the CIs are two-sided.
2. *Identify the test and its assumptions. Which test was used, if any? What does it assume (normality, independence, pairing, equal variances)? Do the authors check or address those assumptions? What is the experimental unit, and does n count units or measurements (slide 20)?* No named test. Regression on hourly means with Newey-West SEs. The unit is the session but n counts hour cells, since sessions in one hour share congestion. An A/A test and a baseline week check that the links match.
3. *Find α, n, and β. What significance level? How many observations, and what is an observation (a run, a seed, a benchmark, a user)? Is there any power analysis? If not, estimate roughly what effect size the experiment could have detected.* $\alpha = 0.05$ implied. About 14M sessions but only a few dozen hour cells. No power analysis. SE about 2.6, so the detectable effect is about $2.8 \times 2.6 \approx 7\%$.
4. *Count the tests. How many comparisons does the paper make in total? What is the family-wise error rate if none were corrected (slide 27)? Is there a correction, and does the choice between family-wise error and false discovery rate control fit the paper's claims (slide 28)?* 28 intervals in Fig. 5 (7 metrics × 4), plus 24 baseline metrics. FWER $= 1 - 0.95^{28} = 1 - 0.24 = 0.76$. No correction, and FDR fits the claim better.
5. *Check the stopping rule and the estimand. Was n fixed in advance, or could the authors have kept running until the result was significant (slides 23 to 25)? Could units in the experiment have interfered with each other (slide 22)? Does the test measure the quantity the paper's claim is about (pitfall 7)?* 5 days, not said to be fixed in advance. Interference is the paper's point. The naive A/B test measures the effect at 5% allocation, not the effect of deploying to everyone.
6. *Rewrite the result. Rewrite the paper's key sentence as an effect size with a confidence interval, as slides 16 and 18 recommend. Is the effect large enough to matter? Does the claim become stronger, weaker, or different?* Capping raised throughput

12% (95% CI about 7% to 17%), while the naive test said −5% (about −8% to −2%). The claim gets stronger since the sign flips, and the exact size is less certain.

By hand, FWER. *Formula.* FWER $= 1-(1-\alpha)^m$, the chance of at least one false positive when you run m tests at level α and every H_0 is true (HT p.27). *Plug in.* $m=10$, $\alpha=0.05$, so $1-0.95^{10}$. *Work it out.* $0.95^{10}=0.599$, $1-0.599$. *Result.* 0.40. Bonferroni (HT p.28) tests each at $\alpha/m=0.05/10=0.005$.

Where 2.8 comes from. $z[0.025]+z[0.20]=1.96+0.84=2.80$, the same sum as in the HW5 1.3 power version.

Exam scenario. "A new congestion controller ran on 10% of flows sharing links with the old one. We checked p daily and stopped on day 6 at $p=0.04$. Of 8 metrics, throughput was significantly better."

0. The claim is that the new controller gives better throughput, from a live test at 10% allocation. There's no paper to pick, so say what's being claimed.
1. No H_0 stated. Reconstruct H_0 no throughput difference, H_1 better, and ask if the side was fixed in advance (HT p.5).
2. Unit is the flow, but flows on one link aren't independent (HT p.20).
3. No power analysis, n wasn't fixed (HT p.19).
4. 8 metrics give FWER $=1-0.95^8=1-0.66=0.34$ with no correction (HT p.27 to 28). Bonferroni would need $p<0.05/8=0.00625$, and 0.04 fails that.
5. Daily peeking inflates the Type I error well past 5% (HT p.23 to 24). At 10% allocation the new flows interfere with the old, so this isn't the effect of deploying to everyone (HT p.22).
6. Report the throughput difference with a CI, not "significantly better" (HT p.16). Honest headline. "At 10% allocation, flows on the new controller had higher throughput by [difference] (95% CI [low, high]). This was 1 of 8 metrics, checked daily with no correction for peeking or multiple tests, so it's a lead to retest with a fixed sample size, not a finding."

HW6. Comparing two alternatives

HW6 P1. Same workload on routing A and B, 9 rows of delay. Is A better at 95%?

Question. *Problem 1, Two routing algorithms (20 points). Two packet routing algorithms are being evaluated. Experiments were conducted by injecting the same workload in the network and measuring the average end-to-end packet delay. The results of the various experiments are shown in Table 1. Can you say with 95% confidence that routing algorithm A is better than algorithm B?* [Table 1 is the A and B columns below, 9 experiments, rows 1 to 5 on the left half and 6 to 9 on the right.]

In plain words. The same workload ran on both, so each row is a matched pair. Take the difference in each row and build an ordinary one-sample CI on those 9 differences. If the CI includes 0, you can't say either one is better.

A	B	d	d^2	A	B	d	d^2
115.0	125.3	−10.3	106.09	120.3	87.9	32.4	1049.76
89.3	97.6	−8.3	68.89	98.9	104.2	−5.3	28.09
102.1	85.3	16.8	282.24	87.6	96.3	−8.7	75.69
94.3	101.3	−7.0	49.00	104.5	113.2	−8.7	75.69
117.3	112.7	4.6	21.16				

1. The same workload ran under both, so paired. Work on $d_i=A_i-B_i$, the delay of A minus the delay of B in row i. Negative favours A, since lower delay is better (Comp p.5, p.9).
2. Mean difference. *Formula.* $\bar{d}=\frac{1}{n}\sum d_i$, where $n=9$ pairs. *Plug in.* Positives $16.8+4.6+32.4=53.8$, negatives $-10.3-8.3-7.0-5.3-8.7-8.7=-48.3$. *Work it out.* $\sum d=53.8-48.3=5.5$, $\bar{d}=5.5/9$. *Result.* 0.611.
3. SD of the differences. *Formula.* $s_d^2=\dfrac{\sum d_i^2-(\sum d_i)^2/n}{n-1}$. *Plug in.* $\sum d^2=1756.61$ from the table, so $\dfrac{1756.61-5.5^2/9}{8}$. *Work it out.* $5.5^2/9=30.25/9=3.36$, $1756.61-3.36=1753.25$, $1753.25/8=219.16$. *Result.* $s_d=\sqrt{219.16}=14.80$.
4. SE. *Formula.* SE $=s_d/\sqrt{n}$. *Plug in.* $14.80/\sqrt{9}$. *Work it out.* $14.80/3$. *Result.* 4.935.
5. CI. *Formula.* $\bar{d}\pm t[\alpha/2,n-1]\times$SE, with df $=9-1=8$ pairs minus one. *Plug in.* $0.611\pm2.306\times4.935$. *Work it out.* $2.306\times4.935=11.38$. $0.611-11.38=-10.77$ and $0.611+11.38=11.99$. *Result.* CI $=(-10.77, 11.99)$, which contains 0.

Answer. No, you can't say A is better at 95%. Matches main.tex. It isn't evidence they're equal either, since the CI allows A 10.8 better or 12.0 worse.

Check. "Is A better than B?" is a one-sided question by HT p.5, and he used a two-sided interval. The verdict doesn't change. A is better only if $\bar{d}<0$, and $\bar{d}=+0.611$ points the other way, so the one-sided test with $t[0.05,8]=1.860$ can't reject either.

By hand. Four differences 2, 4, 3, 3. $\bar{d}=12/4=3$. Deviations $-1,1,0,0$, so $s_d^2=2/3=0.667$, $s_d=0.816$. SE $=0.816/\sqrt{4}=0.408$. df $=3$, margin $3.182\times0.408=1.30$. CI $3\pm1.30=(1.70,4.30)$ excludes 0.

Sign test on the HW6 data (Comp p.23). *Formula.* Under H_0 each difference is equally likely to be positive or negative, so the count of positives S follows a binomial with $n=9$ and chance 1/2. $P[S\le s]=\sum_{k=0}^{s}\binom{9}{k}/2^9$. *Plug in.* 3 positive out of 9, so $P[S\le3]=\dfrac{\binom{9}{0}+\binom{9}{1}+\binom{9}{2}+\binom{9}{3}}{512}$. *Work it out.* $\dfrac{1+9+36+84}{512}=\dfrac{130}{512}=0.254$. Two-sided doubles it. *Result.* $p=0.51$.

Usual mistake. Analysing paired data as unpaired. Comp p.8 shows the same data giving $(0.65, 3.00)$ paired and $(-22.0, 25.7)$ unpaired. The df come from pairs, $n-1=8$ (Comp p.9).

HW6 P2. CPU times, 30 on A and 24 on B. Is A faster at 90%?

Question. *Problem 2, Two computer systems (20 points). Two computer systems are being evaluated for their performance. Data corresponding to the CPU time for running some transactions on System A and System B are reported in Table 2. Is System A faster than System B at the 90% confidence level?* [Table 2, CPU time in seconds. System A (30 values) 1.3779, 0.0457, 1.4415, 1.3589, 0.6898, 0.7076, 3.9083, 2.2081, 0.9244, 6.5770, 4.4775, 4.6199, 0.1775, 0.3354, 0.3687, 0.0956, 1.2116, 3.8814, 0.2922, 3.0970, 3.2150, 1.2880, 3.0046, 6.8230, 0.7478, 0.3845, 0.2354, 0.9951, 0.3045, 1.6950. System B (24 values) 0.9584, 0.5726, 1.3915, 0.8075, 0.1344, 0.1277, 1.8032, 1.2048, 9.5949, 4.9107, 0.2834, 0.3841, 0.0812, 2.1362, 4.6396, 1.0127, 2.9434, 1.0844, 0.7860, 2.1536, 1.4948, 11.7346, 0.9436, 3.1111.]

In plain words. Different transactions ran on each system, in different counts, so there's nothing to pair. You compare the two means with the Welch method, which doesn't assume both systems have the same variance. It builds a CI on the difference in means and has its own formula for the degrees of freedom, called ν.

1. Different transactions and unequal sizes, so unpaired. Welch by default (Comp p.12 to 13).
2. From the data, A has $n_A=30$, $\bar{x}_A=1.8830$, $s_A=1.9017$. B has $n_B=24$, $\bar{x}_B=2.2623$, $s_B=2.9108$. Difference $\bar{x}_A-\bar{x}_B=1.8830-2.2623=-0.3793$.

3. SE of the difference. *Formula.* $s = \sqrt{s_A^2/n_A + s_B^2/n_B}$, where s_A^2 is A's sample variance and n_A its count, same for B. *Plug in.* $\sqrt{1.9017^2/30 + 2.9108^2/24}$. *Work it out.* $1.9017^2 = 3.6166$, $/30 = 0.12055$. $2.9108^2 = 8.4727$, $/24 = 0.35303$. Sum 0.47358. *Result.* $s = \sqrt{0.47358} = 0.6882$.
4. Welch df. *Formula.* $\nu = \dfrac{(s_A^2/n_A + s_B^2/n_B)^2}{\dfrac{(s_A^2/n_A)^2}{n_A - 1} + \dfrac{(s_B^2/n_B)^2}{n_B - 1}}$. *Plug in.* $\nu = \dfrac{0.47358^2}{0.12055^2/29 + 0.35303^2/23}$. *Work it out.* Top $0.47358^2 = 0.22428$. Bottom $0.014532/29 = 0.000501$ plus $0.124630/23 = 0.005419$, total 0.005920. $0.22428/0.005920$. *Result.* $\nu = 37.89$.
5. CI. *Formula.* $(\bar{x}_A - \bar{x}_B) \pm t[\alpha/2, \nu] \times s$, with 90% so $\alpha/2 = 0.05$. *Plug in.* $-0.3793 \pm 1.686 \times 0.6882$. *Work it out.* $1.686 \times 0.6882 = 1.160$. $-0.3793 - 1.160 = -1.540$ and $-0.3793 + 1.160 = 0.781$. *Result.* CI = $(-1.540, 0.781)$.
6. Test statistic for reference. $t = (\bar{x}_A - \bar{x}_B)/s = -0.3793/0.6882 = -0.55$, $p = 0.58$.

Answer. 0 is inside, so A isn't shown to be faster at 90%. Matches main.tex. Rounding ν down to 37 by hand gives 1.687 and $(-1.540, 0.782)$.

Check. The same one-sided point applies. "Is A faster" is one-sided. At 90% one-sided, $t[0.10, 37.9] = 1.304$ and the upper bound is $-0.379 + 1.304(0.688) = -0.379 + 0.897 = 0.518 > 0$. Still not shown, so the verdict holds, but say the test is one-sided.

By hand. A = 2, 4, 6, 8 and B = 6, 8, 10, 12, 14.

1. Means $20/4 = 5$ and $50/5 = 10$. Variances $s_A^2 = (9 + 1 + 1 + 9)/3 = 6.67$ and $s_B^2 = (16 + 4 + 0 + 4 + 16)/4 = 10$.
2. Pieces $6.67/4 = 1.667$ and $10/5 = 2$. $s = \sqrt{1.667 + 2} = \sqrt{3.667} = 1.915$.
3. $\nu = \dfrac{3.667^2}{1.667^2/3 + 2^2/4} = \dfrac{13.44}{0.926 + 1} = \dfrac{13.44}{1.926} = 6.98$, round down to 6.
4. 90% CI. $t[0.05, 6] = 1.943$, margin $1.943 \times 1.915 = 3.72$. $(5 - 10) \pm 3.72 = (-8.72, -1.28)$, which excludes 0, so A is smaller.

Usual mistake. Using $n - 1$ of one sample as the df, or reading overlap of two separate CIs as "no difference" (Comp p.15 to 16).

Slide twin, paired (Comp p.11). Cache hit ratio on 6 workloads under policies A and B

In plain words. The same paired recipe as HW6 P1, on a tiny example from the slides. The lesson is to check which t value the slide used, since that decides whether it's a 90% or a 95% interval.

1. Differences $d_i = A_i - B_i$ are +0.07, +0.09, −0.05, −0.06, +0.10, −0.03. *Formula.* $\bar{d} = \frac{1}{n}\sum d_i$. *Plug in.* $\sum d = 0.26 - 0.14 = 0.12$, $n = 6$. *Result.* $\bar{d} = 0.12/6 = 0.020$.
2. $\sum d^2 = 0.0049 + 0.0081 + 0.0025 + 0.0036 + 0.0100 + 0.0009 = 0.0300$.
3. *Formula.* $s_d^2 = \dfrac{\sum d^2 - (\sum d)^2/n}{n - 1}$. *Plug in.* $\dfrac{0.0300 - 0.12^2/6}{5}$. *Work it out.* $0.0144/6 = 0.0024$, $(0.0300 - 0.0024)/5 = 0.0276/5 = 0.00552$. *Result.* $s_d = 0.0743$.
4. SE $= 0.0743/\sqrt{6} = 0.0743/2.449 = 0.0303$.
5. The slide multiplies by $t[0.05, 5] = 2.015$, so margin $2.015 \times 0.0303 = 0.061$ and its interval $0.020 \pm 0.061 = (-0.041, +0.081)$ is a 90% interval. For 95% use $t[0.025, 5] = 2.571$, giving $0.020 \pm 0.078 = (-0.058, 0.098)$.

Answer. Contains 0, so no difference shown, and the width is the finding (Comp p.11).

Slide twin, unpaired Welch (Comp p.14). A has $n = 7$, mean 0.2657, SD 0.0934. B has $n = 9$, mean 0.4011, SD 0.1447

In plain words. The same Welch recipe as HW6 P2, with small numbers from the slides. This time the CI misses 0, so this one does show a difference.

1. *Formula.* $s = \sqrt{s_A^2/n_A + s_B^2/n_B}$. *Plug in.* $\sqrt{0.0934^2/7 + 0.1447^2/9}$. *Work it out.* $0.001246 + 0.002326 = 0.003572$. *Result.* $s = 0.0598$.
2. *Formula.* Welch ν as in HW6 P2. *Plug in.* $\nu = \dfrac{0.003572^2}{0.001246^2/6 + 0.002326^2/8}$. *Work it out.* Top 1.276×10^{-5}. Bottom $2.59 \times 10^{-7} + 6.76 \times 10^{-7} = 9.35 \times 10^{-7}$. *Result.* $\nu = 13.6$.
3. Difference $0.2657 - 0.4011 = -0.135$. $t[0.05, 13.6] = 1.765$, margin $1.765 \times 0.0598 = 0.106$.
4. 90% CI $= -0.135 \pm 0.106 = (-0.241, -0.029)$.

Answer. The CI excludes 0, so B has the higher hit ratio, by 3 to 24 points.

HW6 Problem 3. Comparing two systems in the wild, 8 questions

Question. *Problem 3, A paper from the wild (60 points). Work through the homework exercise in Part III of the companion handout Comparing Two Systems in the Wild. Read only the experimental setup and results sections of the paper you choose, and answer all eight questions there. Two or three sentences per question is enough, except for the last question, which asks you to rewrite the paper's headline comparison as an effect with a confidence interval and to say what the experiment could not have detected. Give the full citation at the top. Choosing the paper is part of the exercise. Pick the entry from Part I of the handout that sits closest to your own research area, or any peer-reviewed paper from that area, ideally one you would cite anyway, that compares exactly two systems, models, algorithms or configurations. "Ours" versus a baseline is ideal. Papers that make that comparison with no test and no statement of uncertainty are also fair game, and often more revealing. Several entries (4, 5, 9, 10 and 11) are guides to, or surveys of, statistical practice rather than a comparison of two systems. If you pick one of them, apply the questions to the paper's own central empirical claim.* [Defaults if no area fits are Curtsinger and Berger, STABILIZER, ASPLOS 2013 (unpaired t), Dietterich, Approximate Statistical Tests for Comparing Supervised Classification Learning Algorithms, 1998 (paired t), or Klees et al., Evaluating Fuzz Testing, CCS 2018 (Mann-Whitney U). The eight handout questions are printed in italics at the start of each answer under "His paper" below.]

In plain words. Pick a paper comparing two systems and audit it with 8 fixed questions. The big ones are whether the data should've been paired, whether error-bar overlap was used as a test, and whether the difference is big enough to matter. The last question rewrites the headline as a difference with a CI, plus the smallest difference the experiment could have caught.

Checklist.

1. Paper and why.
2. Design. Paired or unpaired, the unit, units per system (Comp p.5).
3. If unpaired, could it be paired? How much variance is between units (Comp p.6)?
4. Which test, and are df and an exact p reported? If no test, what instead?
5. Is the response normal, or a bounded score, ratio, quantile? Would a rank test or bootstrap be better (Comp p.17 to 18)?

6. Are error bars defined in the caption? Any conclusion from overlap (Comp p.15 to 16)?
7. Effect size in real units with an interval. Big enough, and who decides (Comp p.24)?
8. Rewrite the headline with an interval plus what it couldn't detect (Comp p.25).

His paper. Miller, Adding Error Bars to Evals, arXiv 2411.00640, 2024 (entry 3).

1. *Say what you picked and why. Full citation and a sentence on its relation to your work.* He compares VLMs on the same benchmark questions.
2. *Identify the design. Paired or unpaired? What is the unit, a benchmark, a test-set example, a run, a user? How many units per system?* Paired, with the question as the unit (the cluster for MGSM). Example models on 5000 MATH, 164 HumanEval, 2500 MGSM.
3. *If it is unpaired, could it have been paired? If yes, estimate what that would have bought. How much of the observed variance is between units rather than between systems?* Already paired. In Miller's example a correlation of 0.5 cuts the variance from 1/6 to 1/9, so the SE shrinks by $\sqrt{(1/9)/(1/6)} = \sqrt{2/3} = 0.82$, about 18%.
4. *Find the comparison. Which test, if any? Is the statistic reported with its degrees of freedom and an exact p? If there is no test, what is offered instead?* Paired z on per-question differences, ±1.96 SE. No df, no p. Significant if the CI excludes 0.
5. *Check the assumptions. Is the response plausibly normal, or is it a bounded score, a ratio, or a quantile? Would a rank-based test or a bootstrap have been more defensible (slides 17 to 18)?* Scores are 0 or 1, not normal, so it leans on the CLT. Fine for 5000, but a paired bootstrap or sign test is safer for 164.
6. *Look at the figure. If there are error bars, does the caption say what they are? Is any conclusion drawn from their overlap (slides 15 to 16)?* No figures, only SEs in tables. Single-model CIs on MATH overlap though the paired CI excludes 0.
7. *Find the effect size. What is the difference, in the units of the response, and does the paper give an interval for it? Is the difference large enough to matter, and who decides?* MATH difference +2.5 points, 95% CI (1.2, 3.8). Whether that matters is left to the user.
8. *Rewrite the claim. Restate the paper's headline comparison as an effect with an interval, and add the sentence about what the experiment could not have detected.* Galleon beat Dreadnought by 2.5 points on MATH (1.2 to 3.8). HumanEval −3.1 (−7.2, 1.0) and MGSM −2.7 (−6.1, 0.7) are consistent with no difference. HumanEval's SE is 2.1, so it couldn't reliably detect under about $2.8 \times 2.1 \approx 6$ points.

Exam scenario. "Compiler X and Y were each run on the same 8 benchmarks. An unpaired t test gave $p = 0.4$, and the error bars overlap, so the compilers perform the same."

1. The claim is that two compilers perform the same, based on a non-significant unpaired test and overlapping error bars over 8 benchmarks. There's no paper to pick, so say what's being claimed.
2. Same benchmarks under both, so it's paired with the benchmark as the unit (Comp p.5).
3. Unpaired throws away the correlation. Between-benchmark spread swamps the difference, as in Comp p.8 where the same data give $p = 0.87$ unpaired and $p = 0.008$ paired.
4. They ran an unpaired t and gave only p. Use a paired t on the 8 differences with $8 - 1 = 7$ df and report t, df and the exact p (Comp p.26).
5. Run times are skewed, so analyse log times or use the signed-rank test (Comp p.18, Comp p.21).
6. The bars aren't defined, and overlap isn't a test (Comp p.15).
7. Give the mean difference with a CI. "Perform the same" doesn't follow from a non-significant result.
8. Honest headline. "On the same 8 benchmarks, X's run time minus Y's averaged $\bar{d}$, with paired 95% CI $\bar{d} \pm 2.365\, s_d/\sqrt{8}$, using $t[0.025, 7] = 2.365$. With only 8 benchmarks we couldn't reliably detect a difference smaller than about $3.3\, s_d/\sqrt{8}$." The 3.3 is $t[0.025, 7] + t[0.20, 7] = 2.365 + 0.896$, the small-sample version of the 2.8×SE rule (HT p.19).

One checklist for any experiment on the exam

1. Claim and hypotheses, with the direction chosen before the data (HT p.4 to 5).
2. Unit and n. Count runs, not requests from one run (HT p.20, CI p.66).
3. Paired or unpaired, and could it be paired (Comp p.5)?
4. Test fits the response, whether z, t, Welch, rank or bootstrap (Comp p.17 to 18, CI p.67).
5. Power or detectable effect (HT p.19).
6. Number of tests, and peeking (HT p.23 to 28).
7. Effect size with an interval in real units (HT p.16, Comp p.24).
8. What the interval misses, such as systematic error and interference (CI p.17, HT p.22).
9. Reading. No "equivalent" from non-significance, no verdicts from overlap, no "95% probability" for a CI (HT p.14, Comp p.15, CI p.33).

Practice. "Our scheduler cuts mean latency from 12.1 to 11.4 ms, $p < 0.05$, from one run of each with 10,000 requests, with the best of three workloads shown." The unit is the run, so $n = 1$ per system and the p means nothing. Three workloads are a hidden family of tests. There's no interval on the 0.7 ms. Fix it with several independent runs per system on all three workloads, paired by workload, reporting the difference and its CI.

Homework-style mock problems, no calculator

Mock 1 (HW3 style). Three runs of equal work report 40, 60 and 120 req/s. Mean rate?

In plain words. Three rates from equal amounts of work, so the honest mean rate is total work over total time, which is the harmonic mean. Using fractions with a common denominator keeps it calculator-free.

1. Rates with equal work, so harmonic (Summ p.28). *Formula.* $x_H = \dfrac{n}{\sum 1/x_i}$, with $n = 3$ runs.
2. *Plug in.* $x_H = \dfrac{3}{1/40 + 1/60 + 1/120}$.
3. *Work it out.* Over 120, $3/120 + 2/120 + 1/120 = 6/120 = 0.05$. Then $3/0.05$.
4. *Result.* $x_H = 60$ req/s. The AM would say $(40 + 60 + 120)/3 = 220/3 = 73.3$.

Answer. 60 req/s, harmonic mean.

Mock 2 (HW4 style). $n = 25$, $\bar{x} = 50$, $s = 10$, σ unknown. 90% CI?

In plain words. Small sample and unknown σ means a t interval with $n - 1 = 24$ degrees of freedom. It's the mean plus or minus t times the standard error $s/\sqrt{n}$.

1. $n < 30$ and σ unknown, so t with 24 df, assuming roughly normal data (CI p.37). *Formula.* $\bar{x} \pm t[0.05, 24]\, s/\sqrt{n}$.
2. *Plug in.* SE $= 10/\sqrt{25} = 10/5 = 2$. $t[0.05, 24] = 1.711$.

3. *Work it out.* Margin $1.711 \times 2 = 3.42$. $50 - 3.42 = 46.58$ and $50 + 3.42 = 53.42$.
4. *Result.* CI = (46.58, 53.42).

Answer. 90% CI $(46.58, 53.42)$.

Mock 3 (HW5 style). $\sigma = 4$ known, $n = 64$, $\bar{x} = 51$, $H_0\colon \mu = 50$, two-sided, $\alpha = 0.05$

In plain words. σ is known, so it's a Z test. Count how many standard errors 51 is away from 50 and compare that with 1.96.

1. *Formula.* $Z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$. *Plug in.* $\dfrac{51 - 50}{4/\sqrt{64}}$. *Work it out.* SE $= 4/8 = 0.5$, $Z = 1/0.5$. *Result.* $Z = 2.0$.
2. $|Z| = 2.0 > 1.96$, reject. *Formula.* $p = 2(1 - \Phi(|Z|))$. *Plug in.* $2(1 - \Phi(2.0)) = 2(1 - 0.9772)$. *Work it out.* 2×0.0228. *Result.* $p = 0.0456$.
3. At $\alpha = 0.01$ (2.576) you'd fail to reject. Report p and the CI $51 \pm 1.96 \times 0.5 = 51 \pm 0.98 = (50.02, 51.98)$, not just the verdict (HT p.16).

Answer. $Z = 2.0$, $p = 0.046$, reject H_0 at 0.05.

Mock 4 (HW6 style). Paired differences 1, 3, 2, 4, 0 on 5 benchmarks. 95% CI on the mean difference

In plain words. The differences are already paired, so this is a one-sample t CI on the differences with $n - 1 = 4$ df. Then check whether 0 is inside.

1. Mean. $\bar{d} = (1 + 3 + 2 + 4 + 0)/5 = 10/5 = 2$.
2. SD. *Formula.* $s_d^2 = \dfrac{\sum(d_i - \bar{d})^2}{n - 1}$. *Plug in.* Deviations $-1, 1, 0, 2, -2$, squares $1, 1, 0, 4, 4$. *Work it out.* $10/4 = 2.5$. *Result.* $s_d = \sqrt{2.5} = 1.581$.
3. SE $= 1.581/\sqrt{5} = 1.581/2.236 = 0.707$.
4. CI. $t[0.025, 4] = 2.776$, margin $2.776 \times 0.707 = 1.963$. $2 - 1.963 = 0.04$ and $2 + 1.963 = 3.96$.

Answer. CI = (0.04, 3.96). It just excludes 0, so a difference is shown. The size is anywhere from almost nothing to about 4, which you should say.

Practice midterm

It covers the Intro, Metrics, Advisor, Summ, CS701, CI, Pub, Ethics, HT, Comp and Meas slides, plus the study guides and in-the-wild handouts for those lectures. The problems copy the style of HW3 to HW6, with fresh numbers. This is a full-length mock exam. Set a timer for two hours and put your phone and laptop away, because the real exam is open book and open notes but allows no electronics. Work with a pencil, your notes and the table values below, and write each answer down before you open its solution. A fair pace is about 30 minutes for Part A, 75 minutes for Part B and 15 minutes for Part C. Every solution shows each arithmetic step and cites the slide it comes from, and every number was computed twice.

These are the only critical values the exam uses. They're the values a printed t and normal table gives, so read them off your own table and check they match. The brackets are the slides' upper-tail notation, so $t[0.025, 7]$ is the t value with area 0.025 above it on 7 degrees of freedom (CI p.37, HT p.5). $\Phi(z)$ is the area below z in a normal table.

Upper-tail area	0.10	0.05	0.025	0.01	0.005
z	1.282	1.645	1.960	2.326	2.576
t, 5 df	1.476	2.015	2.571	3.365	4.032
t, 7 df	1.415	1.895	2.365	2.998	3.499
t, 8 df	1.397	1.860	2.306	2.896	3.355
t, 15 df	1.341	1.753	2.131	2.602	2.947

One normal-table value is also needed, $\Phi(2.5) = 0.9938$.

Part A. Concepts (17 short questions, about 30 minutes)

Answer each in two or three sentences. For true or false items, say which and give the reason, since the reason is where the marks are.

A1. True or false, with a reason. (a) If MIPS(A) > MIPS(B), machine A runs your program faster than machine B. (b) Linearity is one of the properties a good metric must have. (c) Is MFLOPS a means-based or an ends-based metric?

A2. A web search returns 25 pages. 15 of them are relevant, and it misses 35 other relevant pages. What are its precision and recall?

A3. Feitelson gives three meanings of "experimental computer science". Name them and say which one CS 700 is about.

A4. A first-year student says, "The advisor form isn't due until I've taken 24 credits, so I'll start looking that semester, and I'll open my first meeting by asking the professor to be my advisor." Give two reasons this plan is bad.

A5. True or false, with a reason. (a) Your CS 701 supervisor automatically becomes your PhD advisor. (b) If you can't find a CS 701 project this spring, you can take it next year. (c) If no faculty member agrees, the department will assign you one. (d) You should wait for one professor's answer before contacting a second.

A6. Which of these is acceptable for CS 701, and why? (i) A survey of 30 papers on federated learning. (ii) Reimplementing three published cache policies and evaluating them with a new metric. (iii) A deliverable that reads "paper accepted at a conference by May". (iv) A deliverable that reads "a draft paper to be revised and submitted for publication later".

A7. An author's papers have 15, 12, 7, 5, 5, 3 and 1 citations. What is the h-index?

A8. (a) A conference uses double-blind review. What must your submission do, and what happens if it doesn't? (b) True or false. It's fine to send the same paper to two conferences at once as long as you withdraw one when the other accepts.

A9. Classify each as fabrication, falsification, plagiarism or none. (a) You drop two runs that went against your trend and don't mention it. (b) You run out of time and type in plausible numbers for one benchmark. (c) You paste a sentence from a survey into your related work with no quotation marks or citation. (d) After publishing, you find a timer bug that inflated your speedup by 3%.

A10. (a) You used ChatGPT to draft part of the related work in a paper for an ACM venue. Can you list it as a co-author, and what must you do? (b) You're extending your IEEE conference paper into a journal paper. What does the journal version need?

A11. Which index of dispersion would you report for (a) CPU utilization in percent, (b) an unbounded, unimodal, symmetric quantity, (c) right-skewed request latencies with a few huge outliers?

A12. From one sample you get a 95% confidence interval of (47.6, 52.4) ms for the mean run time. Which statements are correct? (i) There's a 95% probability that μ lies in (47.6, 52.4). (ii) If you repeated the whole experiment many times, about 95% of the intervals built this way would contain μ. (iii) About 95% of individual runs take between 47.6 and 52.4 ms.

A13. True or false. Running a benchmark 1000 times instead of 10 makes the result more accurate.

A14. A paper compares two schedulers with 5 runs each, gets p = 0.30, and concludes that "the two schedulers perform the same." (a) What's wrong? (b) If the schedulers really differ, what kind of error did the paper make, and what is its probability called? (c) At a fixed n, what happens to that probability if you lower α?

A15. You test 20 benchmarks, each at $\alpha = 0.05$, and no real effect exists on any of them. (a) How many "significant" results do you expect? (b) What's the formula for the chance of at least one, and its value from the slides? (c) What per-test level would Bonferroni use?

A16. True or false. The 95% intervals for systems A and B overlap a little, so there's no significant difference between them.

A17. (a) Pick counting, tracing or sampling for each, with a reason. (i) Total L2 misses in a 60-second run. (ii) The code path behind the one-in-a-million request that takes 100 times the median. (iii) A two-hour job where you have no idea where the time goes. (b) True or false. 20 iterations inside one process give you n = 20 independent measurements.

Part B. Problems (8 problems, about 75 minutes)

Show every step, since that's how the homework was graded. State which formula you use and why before you plug in.

B1. Choosing a mean. For each part, say which mean (arithmetic, harmonic or geometric) you'd use and why, then compute it. (a) Five runs of a build take 120, 80, 150, 90 and 160 ms. What's the mean build time? (b) The same 120 MB file is sent over three paths at 40, 60 and 120 MB/s. What's the mean transfer rate? (c) Three features are added one on top of the other, and each improves solution quality over the previous version by 8%, 25% and 28%. What's the mean improvement per feature? You may use $1.2^3 = 1.728$.

B2. Quartiles. Twelve request latencies in ms, in the order measured.

23	15	60	18	31	12	25	20	14	28	22	17

Using the course convention $f_i = (i - 0.5)/n$ with linear interpolation, find (a) the five-number summary, (b) the IQR and SIQR, (c) the 90th percentile. (d) Which center and which index of dispersion would you report, and why?

B3. Confidence intervals for a mean. (a) Eight independent runs of a compile job take these times in seconds.

52	46	50	54	50	46	52	50

Find the mean, the sample standard deviation, and the 90%, 95% and 99% confidence intervals for the mean. (b) Treating these 8 runs as a pilot, how many runs would you need to be 95% confident the mean is within ±2% of the true mean? (c) A different job is run 64 times, with $\bar{x} = 120$ ms and $s = 16$ ms. Give the 95% and 99% intervals, then test $H_0 : \mu = 125$ against a two-sided alternative at $\alpha = 0.05$ and $\alpha = 0.01$, with the p-value. (d) If that job had been run only 16 times with the same $\bar{x}$ and s, what would the 95% half-width be, and why is it so much wider?

B4. A proportion from a sampling profiler. A profiler interrupts every 10 ms during a 4-second run, which gives 400 samples. In 80 of them the program counter is inside `parse()`. (a) Find 90% and 95% confidence intervals for the fraction of time spent in `parse()`, and check that the normal approximation is reasonable. (b) How many samples do you need to be 95% confident that the estimate is within ±5% of the true fraction (relative error, as on the slides), and how long must the run be? (c) How would the answer to (b) change if the question asked for ±5 percentage points instead?

B5. Hypothesis test for a mean. A team believes its new memory allocator brings the mean allocation latency below the old allocator's long-run mean of 40 µs. They measure 16 independent runs, each in a fresh process, and get $\bar{x} = 37$ µs with $s = 6$ µs. Assume the run means are roughly normal. (a) State H_0 and H_1 and justify the choice. (b) Test at $\alpha = 0.05$. (c) Bracket the p-value using the table. (d) What's the verdict at $\alpha = 0.01$? (e) At $\alpha = 0.01$, what is the largest sample mean that would let you reject H_0? (f) A colleague reruns the analysis, gets p = 0.064 and calls the result not significant. What did they do differently, and who is right?

B6. Paired comparison. Two routing algorithms are run on the same six injected workloads, and the average end-to-end delay (ms) is recorded for each. Can you say with 95% confidence that algorithm A gives lower delay than B?

Workload	**1**	**2**	**3**	**4**	**5**	**6**
A	42	35	51	28	60	47
B	41	37	54	31	65	53

B7. Unpaired comparison with Welch. CPU time (ms) is measured for different random transactions on two systems, 6 on A and 8 on B. Is System B faster than System A at the 90% confidence level?

A	52	57	46	54	48	55		
B	45	41	47	44	50	42	46	45

B8. Measurement. (a) Your clock has a resolution of 1 µs and the operation you want to time takes about 2.6 µs. What can one reading return, and how big is the error? (b) Instead you time 10,000 operations in a loop and get 27,412 µs, and an empty loop of 10,000 iterations takes 1,412 µs. What's the time per operation, what's the resolution of that figure, and what assumption does it need? (c) A clock read costs 20 ns, and you bracket each operation with two reads. What fraction of the measured time is timer overhead if the operation takes 60 ns? If it takes 4 µs? (d) A benchmark runs in 80 s untraced and 86 s with two trace probes on. Alone, the probes cost 2.5 s and 2.0 s. Report the perturbation and explain why 2.5 + 2.0 isn't 6. (e) The variance between processes is $\sigma_p^2 = 48$ ms^2 and between iterations inside a process is $\sigma_i^2 = 16$ ms^2. With a budget of 16 measurements, find the standard error of the overall mean for 1 process × 16 iterations, 2 × 8, and 16 × 1.

Part C. Paper audit (1 question, about 15 minutes)

This is the in-class version of Problem 3 from HW4 to HW6. Read the made-up results paragraph and answer the questions under it, two or three sentences each, except C3, which needs arithmetic.

"We compare FastKV with RocksDB on a single 16-core server. For each system we ran YCSB workload A five times and report the average throughput. The first 30 seconds of each run were discarded as warm-up, and throughput was timed with gettimeofday(). All FastKV runs were done on Monday, and the RocksDB runs on Tuesday after a kernel update. FastKV reaches 118.2 Kops/s against 104.6 Kops/s for RocksDB, a 13% improvement. Figure 4 shows both means with error bars of ±1 standard deviation (2.0 for FastKV and 4.0 for RocksDB). Because the bars do not overlap, the improvement is statistically significant. We repeated the comparison on 12 more workloads, and FastKV was significantly faster ($p < 0.05$) on 4 of them. Averaged over all 13 workloads, FastKV's arithmetic mean throughput is 9% higher. Finally, the two systems have equivalent p99 latency (t test, $p = 0.41$)."

C1. What is the experimental unit, what is n, and is the headline design paired or unpaired? Could any part of it have been paired? C2. Is the error-bar argument valid? C3. Write H_0 and H_1 for the headline claim, then rewrite the claim as a 95% confidence interval for the difference using Welch's method. C4. Is the arithmetic mean the right summary across workloads? C5. What's wrong with "significantly faster on 4 of 12 more workloads"? C6. Name two problems with the measurement setup that no interval can catch. C7. What's wrong with the p99 sentence?

Practice midterm, solutions

Find a question by topic

Question	Topic	Lecture
A1	Good metrics, MIPS and MFLOPS	Metrics (L2)
A2	Precision and recall	Metrics (L2)
A3	Meanings of experimental CS	Intro (L1)
A4	Choosing a PhD advisor	Advisor (L3)
A5	CS 701 rules	CS701 (L5)
A6	CS 701 projects and deliverables	CS701 (L5)
A7	h-index	Pub (L7)
A8	Double-blind review, dual submission	Pub (L7), Ethics (L8)
A9	Fabrication, falsification, plagiarism	Ethics (L8)
A10	AI tools as authors, self-plagiarism	Ethics (L8)
A11	Choosing an index of dispersion	Summ (L4)
A12	Meaning of a confidence interval	CI (L6)
A13	Accuracy versus precision	CI (L6), Meas (L11)
A14	Type I and Type II errors, power	HT (L9)
A15	Multiple comparisons, Bonferroni	HT (L9)
A16	Overlapping error bars	Comp (L10)
A17	Counting, tracing, sampling, independence	Meas (L11)

Question	Topic	Lecture
B1	Arithmetic, harmonic, geometric mean	Summ (L4)
B2	Quartiles, IQR, SIQR, percentiles	Summ (L4)
B3	Confidence interval for a mean, sample size, Z test	CI (L6), HT (L9)
B4	Confidence interval for a proportion, sample size	CI (L6)
B5	One-sided t test for a mean	HT (L9)
B6	Paired comparison	Comp (L10)
B7	Unpaired comparison, Welch's t	Comp (L10)
B8	Resolution, timer overhead, perturbation, nested variance	Meas (L11)
C	Paper audit	HT, Comp, Meas, Summ

Part A

A1.

In plain words. This checks whether you know what makes a performance metric trustworthy. A good metric has to track real speed, which means the time your program actually takes. MIPS (millions of instructions per second) and MFLOPS (millions of floating-point operations per second) count work done, not useful work finished, so they can mislead.

(a) False. MIPS counts instructions per second, and machines differ in how much work one instruction does, so a machine with more MIPS can still take longer on your program. That's exactly the slides' example of an unreliable metric (Metrics p.8, Metrics p.17).

(b) False. Linearity and ease of use are nice but not necessary. The required four are reliable, repeatable, consistent and independent (Metrics p.13).

(c) Means-based. It counts floating-point operations whether or not they did useful work, and it reports 0 for an integer program that still sorts or searches. On the slides' spectrum MFLOPS sits near the means-based end with clock rate and MIPS, while execution time is the ends-based metric because it only counts what actually got done (Metrics p.19, Metrics p.27-28).

Answer. (a) False. (b) False, the required four are reliable, repeatable, consistent and independent. (c) Means-based.

A2.

In plain words. Precision asks how much of what the search returned is useful. Recall asks how much of all the useful pages it managed to find. The only trap is the recall denominator, which is the 15 it found plus the 35 it missed.

Formula. $\text{Precision} = \dfrac{\text{relevant pages returned}}{\text{all pages returned}}$ and $\text{recall} = \dfrac{\text{relevant pages returned}}{\text{all relevant pages that exist}}$.

Plug in. Precision $= 15/25$. For recall, the relevant pages that exist are the 15 found plus the 35 missed, so recall $= 15/(15+35)$.

Work it out. $15/25 = 0.6$. $15 + 35 = 50$ and $15/50 = 0.3$.

Result. Precision 0.6 and recall 0.3. This is the same setup as the slides' example with 30 returned, 20 relevant and 40 missed (Metrics p.32).

Answer. Precision is 0.6 (60%) and recall is 0.3 (30%).

A3.

In plain words. It's a recall question from the first lecture. Feitelson (2006) points out that people use the phrase for three different things, and this course is about the one where you measure systems carefully, the way a scientist would.

The three are building systems as the counterpart to theory, experimentation as the feedback step in an engineering loop, and evaluating computer systems with the rigorous methods of the natural sciences. CS 700 is about the third one, which is why the course spends its time on measurement, error and statistics (Intro p.10).

Answer. Building systems, experimentation as a feedback step in engineering, and evaluating systems with scientific methods. CS 700 is about the third.

A4.

In plain words. This is the advisor lecture's "misconceptions" slide. Most faculty want to try working with you first, which the slides call the preliminary research co-operation phase (PRCP), before they agree to advise you. Starting late and asking straight away both skip that step.

First, the 24-credit point is a hard deadline, and most faculty want an informal preliminary research co-operation phase before they agree to advise anyone. That phase can take a semester or more, so starting in the last semester leaves no room for it (Advisor p.3, Advisor p.7-8).

Second, the slides say not to open with "can you be my PhD advisor?". You should explain your background and interests and ask how you can get involved in their research, because the faculty member has to see you work first and may already be over-committed (Advisor p.11-12).

Answer. The plan starts too late to leave time for the trial research phase, and it opens with the one question the slides say not to ask.

A5.

In plain words. Each statement is one of the CS 701 slides' misconceptions, and all four are false. The theme is that nothing happens automatically. You find a supervisor yourself, early, and you talk to several at once.

(a) False. Declaring the PhD advisor is a separate form and a separate process within the first 24 credits, and many students end up with a different advisor and area than their 701 project (CS701 p.16, CS701 p.18).

(b) False. The catalog rule doesn't allow it, and 701 is only offered in Spring (CS701 p.25, CS701 p.2).

(c) False. A faculty member has to agree to supervise, the department doesn't assign one (CS701 p.26).

(d) False. You should contact several faculty at once, because answers take time and the form is due Oct 31 (CS701 p.21, CS701 p.7).

Answer. All four are false.

A6.

In plain words. CS 701 is the first-year spring research course. The project needs hands-on work like code, experiments or simulations, and every deliverable has to be under your control. "Get accepted" can't be one, since acceptance depends on reviewers.

(i) Not acceptable on its own. A literature review is often part of a 701 project, but the project can't consist only of one, because it must have a hands-on component such as an implementation or experiments (CS701 p.10-11).

(ii) Acceptable. Implementing techniques from several papers and evaluating them, possibly with new metrics, is one of the slides' examples (CS701 p.12).

(iii) Not acceptable, because a conference or journal submission completed by the end of Spring can't be a deliverable (CS701 p.15).

(iv) Acceptable. The slides give this exact wording as the allowed alternative, and the report can then take the form of the draft paper (CS701 p.15).

Answer. (ii) and (iv) are acceptable, and (i) and (iii) aren't.

A7.

In plain words. The h-index measures how many well-cited papers someone has. Sort the citation counts from high to low and find the last rank where the count is still at least the rank.

Formula. The h-index is the largest h such that h papers each have at least h citations (Pub p.6). Sort the counts from high to low, number them by rank, and h is the last rank where count $\geq$ rank.

Plug in. The sorted counts are 15, 12, 7, 5, 5, 3, 1 at ranks 1 to 7.

Work it out. Rank 1 has $15 \geq 1$, rank 2 has $12 \geq 2$, rank 3 has $7 \geq 3$, rank 4 has $5 \geq 4$, and rank 5 has $5 \geq 5$. Rank 6 has 3, which is less than 6, so it stops there.

Result. The h-index is 5.

Answer. The h-index is 5.

A8.

In plain words. Double-blind means the reviewers don't know who wrote the paper and the authors don't know who reviewed it. Dual submission means sending the same paper to two places at once, and it's banned even if you plan to withdraw one later.

(a) You must leave out your names and affiliations and avoid indirect hints about who you are. A submission that breaks the double-blind rules is rejected right away, without review. The point is that reviewers judge the paper and not the authors' reputation (Pub p.21).

(b) False. A paper under review somewhere can't be submitted elsewhere, and the same holds for papers with substantial overlap. Dual submissions are often caught and draw severe sanctions (Pub p.20, Ethics p.14).

Answer. (a) Remove names, affiliations and identifying hints, or the paper is rejected without review. (b) False.

A9.

In plain words. The three kinds of research misconduct have exact definitions in GMU Policy 4007. Fabrication is making data up, falsification is changing or leaving out real data, and plagiarism is using someone else's words or ideas without credit. An honest mistake isn't misconduct.

(a) Falsification, because omitting data so the record misrepresents the research is the definition (Ethics p.13, Ethics p.20).

(b) Fabrication, because it means making up results and reporting them (Ethics p.20).

(c) Plagiarism. Verbatim text needs quotation marks and a citation, and even a paraphrase needs the citation (Ethics p.8-10).

(d) None. Research misconduct excludes honest error, though you should still correct the record (Ethics p.18).

Answer. (a) Falsification. (b) Fabrication. (c) Plagiarism. (d) None.

A10.

In plain words. Two policy questions from the integrity lecture. An AI tool can't be an author because it can't take responsibility for the work. Reusing your own published text is fine only when you extend a conference paper into a journal paper, and then the new version must clearly add material.

(a) No. ACM doesn't allow generative AI tools as authors. Using them to create content is allowed but must be fully disclosed, for example in the acknowledgements. Plain spelling, grammar and translation tools are the exception and don't need disclosure (Ethics p.24). You also take full responsibility for anything the tool produced, including plagiarism or fabricated facts (Ethics p.22).

(b) It needs substantially more material, typically 30% more for IEEE, and it must cite and attach the conference version with an explanation of the differences (Ethics p.14). Outside that case, copying your own published text counts as self-plagiarism (Ethics p.12).

Answer. (a) No, ChatGPT can't be a co-author, but you must disclose its use, for example in the acknowledgements. (b) Substantial new material, about 30% more for IEEE, plus a citation to the conference version and an explanation of what changed.

A11.

In plain words. An index of dispersion is a single number for how spread out the data are, like the range, the standard deviation or the IQR. Summ p.49 gives a three-step rule. Bounded data get the range, symmetric single-peak data get the COV (standard deviation divided by the mean), and everything else gets percentiles or the SIQR.

(a) The range, because the quantity is bounded between 0 and 100 (Summ p.49).

(b) The coefficient of variation $s/\bar{x}$, because for a unimodal symmetric distribution the standard deviation describes the spread well and dividing by the mean makes it unit-free (Summ p.41, Summ p.49).

(c) Percentiles or the SIQR. Outliers inflate the standard deviation the same way they drag the mean, while quartiles depend only on order (Summ p.45, Summ p.49). For the same reason you'd pair it with the median instead of the mean (Summ p.18).

Answer. (a) Range. (b) Coefficient of variation. (c) Percentiles or the SIQR.

A12.

In plain words. This tests what "95% confidence" really means. The 95% describes the method, not this one interval. Picture running the experiment 100 times and building 100 intervals, and about 95 of them catch the true mean.

Only (ii) is correct. The 95% belongs to the procedure. Out of 100 samples, about 95 of the intervals contain μ (CI p.33).

(i) is wrong because μ is a fixed number and this interval is fixed once computed, so it either contains μ or it doesn't (confidence-intervals study guide).

(iii) is wrong because the interval is for the mean, not for single runs. Its width shrinks like $1/\sqrt{n}$, so with enough runs it gets far narrower than the spread of individual runs (CI p.39).

Answer. Only (ii) is correct.

A13.

In plain words. Accuracy means being close to the true value. Precision means repeated readings agree with each other. More runs average out random noise, so they improve precision, but they can't fix a systematic bias that pushes every run the same way.

False. More runs narrow the confidence interval, and that's precision, which comes from random error. Accuracy depends on systematic error, such as clock drift or forgetting to clear the cache, and repetition doesn't touch it (CI p.7, CI p.17). You just get a tighter interval around a possibly wrong number (Meas p.20).

Answer. False. More runs improve precision, not accuracy.

A14.

In plain words. A p-value of 0.30 only says the data didn't show a clear difference. It doesn't say there is none, and with 5 runs a real difference can easily hide in the noise. A Type I error is a false alarm and a Type II error is a missed real effect.

(a) Failing to reject H_0 isn't evidence for H_0. With n = 5 the test had little chance of seeing a difference, so absence of evidence isn't evidence of absence (HT p.4, HT p.14). The paper should report the interval for the difference, which would show how wide the plausible range is (HT p.16).

(b) A Type II error, failing to reject a false H_0. Its probability is β, and $1-\beta$ is the power (HT p.6).

(c) β goes up. At a fixed n, lowering α raises β, and the only way to lower both is to collect more data (HT p.6).

Answer. (a) Not rejecting H_0 doesn't show the schedulers are the same. (b) A Type II error, with probability β. (c) β goes up.

A15.

In plain words. Every test at $\alpha = 0.05$ has a 5% chance of a false alarm even when nothing is going on. Run 20 of them and the false alarms add up. Bonferroni fixes this by making each test stricter, dividing α by the number of tests.

(a) *Formula.* Expected false positives $= m\alpha$, where m is the number of tests and α is the chance each one gives a false positive. *Plug in.* 20×0.05. *Result.* 1 false positive expected.

(b) *Formula.* For independent tests the family-wise error rate, the chance of at least one false positive, is $\text{FWER} = 1-(1-\alpha)^m$. It's 1 minus the chance that all m tests come out clean. *Plug in.* $1-0.95^{20}$. *Work it out.* Square repeatedly by hand. $0.95^2 = 0.9025$, $0.95^4 = 0.9025^2 \approx 0.8145$, $0.95^5 = 0.8145 \times 0.95 \approx 0.7738$, $0.95^{10} = 0.7738^2 \approx 0.5987$, and $0.95^{20} = 0.5987^2 \approx 0.3585$. Then $1-0.3585 = 0.6415$. *Result.* About 64%, which matches the slides (HT p.27).

(c) *Formula.* Bonferroni tests each hypothesis at α/m. *Plug in.* 0.05/20. *Result.* 0.0025 per test, which keeps the family-wise rate at or below 0.05 under any dependence (HT p.28).

Answer. (a) 1 expected false positive. (b) $1-(1-\alpha)^m = 1-0.95^{20} \approx 64\%$. (c) 0.0025 per test.

A16.

In plain words. Eyeballing error bars only works in one direction. If two 95% intervals don't overlap there's a real difference, but a small overlap can still hide a significant one. The safe move is to build an interval on the difference itself.

False. Non-overlap implies a difference, but overlap implies nothing. For 95% intervals, an overlap of up to about half a margin of error can still be significant at 0.05, because the standard error of the difference is smaller than the sum of the two half-widths. The right move is an interval on the difference (Comp p.15-16).

Answer. False.

A17.

In plain words. Counting means bumping a counter at every event. Tracing means recording every event with its details. Sampling means peeking at the program at regular intervals. Pick by what you need, which is a total, the full story of a rare event, or a cheap overview. For (b), iterations in one process share memory layout and JIT state, so they aren't independent.

(a)(i) Counting. You need only an exact total, and a counter gives it with tiny overhead (Meas p.7-8).

(a)(ii) Tracing. You need context about a rare event, and sampling would almost surely miss it (Meas p.8).

(a)(iii) Sampling. Its overhead is set by your sampling rate and not by the workload, which matters for a long run you can't afford to slow down (Meas p.7).

(b) False. Repetition has nested levels, and everything that varies between processes, like memory layout or JIT decisions, is frozen across those 20 iterations. n counts independent experimental units, here processes, not measurements (Meas p.25, HT p.20).

Part B

Answer. (a)(i) Counting, (ii) tracing, (iii) sampling. (b) False, all 20 iterations come from one process, so n = 1.

B1. Choosing a mean.

In plain words. The kind of number decides the mean. Times add up, so use the arithmetic mean. Rates for the same amount of work combine through total time, so use the harmonic mean. Percent changes that stack on each other multiply, so use the geometric mean.

Every calculation below goes in the same four steps. The formula with its symbols named, the numbers plugged in, the arithmetic, then the result.

(a) Arithmetic mean, because these are times and the sum of the times is the total time, so the mean should be directly proportional to it (Summ p.25, Summ p.27).

Formula. $\bar{x}_A = \frac{1}{n}\sum_{i=1}^{n} x_i$, where x_i is the time of run i and n is the number of runs.

Plug in. $n = 5$, so $\bar{x}_A = \dfrac{120+80+150+90+160}{5}$.

Work it out. $120+80=200$, $200+150=350$, $350+90=440$, $440+160=600$, and $600/5=120$.

Result. $\bar{x}_A = 120$ ms.

(b) Harmonic mean, because these are rates for the same amount of work, and a mean rate has to be inversely proportional to the total time (Summ p.28, Summ p.32).

Formula. $\bar{x}_H = \dfrac{n}{\sum_{i=1}^{n} 1/x_i}$, where x_i is the rate on path i and n is the number of paths.

Plug in. $\bar{x}_H = \dfrac{3}{\frac{1}{40}+\frac{1}{60}+\frac{1}{120}}$.

Work it out. Put the fractions over the common denominator 120. That gives $\frac{3}{120}+\frac{2}{120}+\frac{1}{120}=\frac{6}{120}$. Then $\bar{x}_H = 3 \div \frac{6}{120} = \frac{3\times 120}{6} = \frac{360}{6} = 60$.

Result. $\bar{x}_H = 60$ MB/s.

Check it the long way. Time is size over rate, so the transfers take $120/40 = 3$ s, $120/60 = 2$ s and $120/120 = 1$ s. That's $3 \times 120 = 360$ MB in $3+2+1 = 6$ s, which is $360/6 = 60$ MB/s. The arithmetic mean $(40+60+120)/3 = 220/3 = 73.3$ MB/s would overstate the real rate because it ignores that the slow path takes the most time.

(c) Geometric mean, because each feature builds on the previous one, so the improvements multiply (Summ p.30).

Formula. $\bar{x}_G = \left(\prod_{i=1}^{n} x_i\right)^{1/n}$, where x_i is the ratio new over old for feature i (an 8% gain is $x = 1.08$) and n is the number of features.

Plug in. $\bar{x}_G = (1.08 \times 1.25 \times 1.28)^{1/3}$.

Work it out. $1.08 \times 1.25 = 1.35$, then $1.35 \times 1.28 = 1.728$. The hint says $1.2^3 = 1.728$, so the cube root of 1.728 is 1.2.

Result. $\bar{x}_G = 1.2$, which is a mean improvement of $1.2 - 1 = 0.2 = 20\%$ per feature.

Check it by compounding, $1.2^3 = 1.728$, which is the real total gain of 72.8%. The arithmetic mean of 8, 25 and 28 is $61/3 = 20.33\%$, and compounding that gives $1.2033^3 \approx 1.742$, which overshoots the true total. This is HW3 Problem 1 with smaller numbers.

Answer. (a) 120 ms. (b) 60 MB/s. (c) 20% per feature.

B2. Quartiles.

In plain words. Quartiles split the sorted data into four equal chunks. After sorting, value number i sits at $p = (i-0.5)/n$, and anything in between is found by linear interpolation, meaning a straight line between the two neighbors. The IQR is $Q_3 - Q_1$, the width of the middle half of the data, and the SIQR is half of that.

Sort first. The ordered values $x_{(1)}, \ldots, x_{(12)}$ are

i	1	2	3	4	5	6	7	8	9	10	11	12
$x_{(i)}$	12	14	15	17	18	20	22	23	25	28	31	60

Formula. The quantile function puts $x_{(i)}$ at $p = (i-0.5)/n$ and interpolates in between (Summ p.59). Here $x_{(i)}$ is the ith smallest value, n is the sample size and p is the fraction you want (0.25 for Q_1). Solving $(i-0.5)/n = p$ for i gives the position

$$r = np + 0.5.$$

Let j be the whole-number part of r and $r - j$ the leftover fraction. Then

$$Q(p) = x_{(j)} + (r-j)\big(x_{(j+1)} - x_{(j)}\big),$$

which means start at value j and walk the fraction $r - j$ of the way to value $j + 1$.

(a) Plug in $n = 12$ for each quartile and work it out.

$$Q_1\colon\ r = 12(0.25) + 0.5 = 3 + 0.5 = 3.5,\ j = 3, \quad Q_1 = x_{(3)} + 0.5\big(x_{(4)} - x_{(3)}\big) = 15 + 0.5(17-15) = 15 + 1 = 16.$$

$$Q_2\colon\ r = 12(0.5) + 0.5 = 6 + 0.5 = 6.5,\ j = 6, \quad Q_2 = x_{(6)} + 0.5\big(x_{(7)} - x_{(6)}\big) = 20 + 0.5(22-20) = 20 + 1 = 21.$$

$$Q_3\colon\ r = 12(0.75) + 0.5 = 9 + 0.5 = 9.5,\ j = 9, \quad Q_3 = x_{(9)} + 0.5\big(x_{(10)} - x_{(9)}\big) = 25 + 0.5(28-25) = 25 + 1.5 = 26.5.$$

Result. The five-number summary is min 12, $Q_1 = 16$, median 21, $Q_3 = 26.5$, max 60 ms (Summ p.50).

(b) *Formula.* IQR $= Q_3 - Q_1$ and SIQR $= (Q_3 - Q_1)/2$, where Q_1 and Q_3 are the first and third quartiles (Summ p.45).

Plug in and work it out. IQR $= 26.5 - 16 = 10.5$ and SIQR $= 10.5/2 = 5.25$.

Result. IQR = 10.5 ms and SIQR = 5.25 ms.

(c) *Formula.* The same $r = np + 0.5$ and $Q(p)$ as above, with $p = 0.9$.

Plug in. $r = 12(0.9) + 0.5$, and $Q(0.9) = x_{(11)} + (r-11)\big(x_{(12)} - x_{(11)}\big)$.

Work it out. $12 \times 0.9 = 10.8$, so $r = 10.8 + 0.5 = 11.3$, $j = 11$ and $r - j = 0.3$. Then $Q(0.9) = 31 + 0.3(60-31) = 31 + 0.3(29) = 31 + 8.7 = 39.7$.

Result. The 90th percentile is 39.7 ms.

(d) *Formula.* $\bar{x} = \frac{1}{n}\sum x_i$. *Plug in and work it out.* The sum is $12+14+15+17+18+20+22+23+25+28+31+60 = 285$, so $\bar{x} = 285/12 = 23.75$ ms. That sits above 8 of the 12 values because the single 60 pulls it up. Report the median of 21 ms and the SIQR of 5.25 ms, because the data are skewed with an outlier and both are based on order, so the outlier can't move them (Summ p.18, Summ p.45, Summ p.49).

Excel's PERCENTILE interpolates at position $1 + (n-1)p$ instead and would give $Q_1 = 16.5$ and $Q_3 = 25.75$. Use the course convention on the exam, since that's what HW3 Problem 2 used.

Answer. (a) 12, 16, 21, 26.5, 60 ms. (b) IQR 10.5 ms and SIQR 5.25 ms. (c) 39.7 ms. (d) The median of 21 ms with the SIQR of 5.25 ms.

B3. Confidence intervals for a mean.

In plain words. A confidence interval is the mean plus or minus a critical value times the standard error $s/\sqrt{n}$, which measures how much the sample mean wobbles from sample to sample. Small samples use t with $n-1$ degrees of freedom and large ones use z. For "how many runs", set the half-width equal to the error you want and solve for n.

(a) Step 1, the mean. *Formula.* $\bar{x} = \frac{1}{n}\sum x_i$, where x_i is run i and n is the number of runs. *Plug in.* $\bar{x} = (52+46+50+54+50+46+52+50)/8$. *Work it out.* The sum is 400, and $400/8 = 50$. *Result.* $\bar{x} = 50$ s.

Step 2, the sample variance and standard deviation. *Formula.*

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}, \qquad s = \sqrt{s^2},$$

where $x_i - \bar{x}$ is how far run i is from the mean and $n-1$ is the course convention for the sample variance. *Plug in.* The deviations from 50 are 2, −4, 0, 4, 0, −4, 2, 0. *Work it out.* Their squares are 4, 16, 0, 16, 0, 16, 4, 0, which sum to 56. So $s^2 = 56/7 = 8$ and $s = \sqrt{8} \approx 2.828$. *Result.* $s \approx 2.828$ s.

Step 3, the standard error. *Formula.* SE $= s/\sqrt{n}$. *Plug in.* SE $= \sqrt{8}/\sqrt{8}$. *Result.* SE = 1 s.

Step 4, the intervals. With $n = 8$ and σ (the true standard deviation) unknown, you need the t distribution with $n - 1 = 7$ degrees of freedom, assuming the run times are roughly normal (CI p.27, CI p.37). *Formula.*

$$\bar{x} \pm t[\alpha/2,\ n-1]\,\frac{s}{\sqrt{n}},$$

where α is 1 minus the confidence level and $t[\alpha/2,\ n-1]$ is the t value with area $\alpha/2$ above it. For 90% $\alpha/2 = 0.05$, for 95% it's 0.025 and for 99% it's 0.005. *Plug in and work it out* with the 7 df row.

$$90\%\colon\ 50 \pm 1.895(1) = (50 - 1.895,\ 50 + 1.895) = (48.105,\ 51.895) \text{ s}$$

$$95\%\colon\ 50 \pm 2.365(1) = (50 - 2.365,\ 50 + 2.365) = (47.635,\ 52.365) \text{ s}$$

$$99\%\colon\ 50 \pm 3.499(1) = (50 - 3.499,\ 50 + 3.499) = (46.501,\ 53.499) \text{ s}$$

Higher confidence means a wider interval, because you need to cover more of the t distribution (CI p.32).

(b) The slides set the half-width equal to a fraction e of the mean and solve for n, using z (CI p.40). *Formula.*

$$n = \left(\frac{z[\alpha/2]\,s}{\bar{x}\,e}\right)^2,$$

where $z[\alpha/2]$ is the normal critical value, s and $\bar{x}$ come from the pilot runs, and e is the relative error as a decimal. *Plug in.* $z[0.025] = 1.96$, $s = 2.828$, $\bar{x} = 50$, $e = 0.02$.

$$n = \left(\frac{1.96 \times 2.828}{50 \times 0.02} \right)^2 .$$

Work it out. The top is $1.96 \times 2.828 = 5.543$ and the bottom is $50 \times 0.02 = 1$, so $n = 5.543^2 \approx 30.73$. A cleaner way by hand squares each piece first, $n = 1.96^2 \times s^2/(\bar{x}e)^2 = 3.8416 \times 8/1^2 = 30.73$. *Result.* Round up to 31 runs.

The pilot gives s, which is the slides' way around needing s before you have data (CI p.41). Read the wording carefully, though. On CI p.42-43 "within 7%" was turned into ±3.5%, so check whether a question gives the total width or the half-width.

(c) With $n = 64 > 30$, s is a good stand-in for σ and the normal distribution is fine (CI p.27, CI p.31).

Step 1, the standard error. *Formula.* $\text{SE} = s/\sqrt{n}$. *Plug in.* $16/\sqrt{64}$. *Work it out.* $\sqrt{64} = 8$ and $16/8 = 2$. *Result.* SE = 2 ms.

Step 2, the intervals. *Formula.* $\bar{x} \pm z[\alpha/2]\,\text{SE}$, where $z[\alpha/2]$ is the normal critical value. *Plug in and work it out.*

$$95\%\text{:}\ \ 120 \pm 1.960(2) = 120 \pm 3.92 = (116.08,\ 123.92) \text{ ms}$$

$$99\%\text{:}\ \ 120 \pm 2.576(2) = 120 \pm 5.152 = (114.848,\ 125.152) \text{ ms}$$

Step 3, the test. The hypotheses are $H_0\colon \mu = 125$ and $H_1\colon \mu \neq 125$. *Formula.*

$$Z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}},$$

where $\mu_0 = 125$ is the value H_0 claims. *Plug in.* $Z = (120 - 125)/2$. *Work it out.* $120 - 125 = -5$ and $-5/2 = -2.5$. *Result.* $Z = -2.5$.

Step 4, the decision. Two-sided means you compare $|Z|$ with $z[\alpha/2]$. At $\alpha = 0.05$, $|Z| = 2.5 > 1.960$, so reject H_0. At $\alpha = 0.01$, $|Z| = 2.5 < 2.576$, so don't reject.

Step 5, the p-value. *Formula.* $p = 2\big(1 - \Phi(|Z|)\big)$, where Φ is the area below a value in the normal table, and the 2 counts both tails. *Plug in.* $p = 2\big(1 - \Phi(2.5)\big) = 2(1 - 0.9938)$. *Work it out.* $1 - 0.9938 = 0.0062$ and $2 \times 0.0062 = 0.0124$. *Result.* $p = 0.0124$, which sits between 0.01 and 0.05 and gives the same two verdicts. The intervals agree too, since 125 is outside the 95% interval and inside the 99% one. That's the slides' point that the non-rejection region is the confidence interval (HT p.8, HT p.13).

(d) Now $n = 16 < 30$, so use t with $16 - 1 = 15$ df. *Formula.* Half-width $= t[0.025,\, n-1]\, s/\sqrt{n}$. *Plug in.* $2.131 \times 16/\sqrt{16}$. *Work it out.* $\sqrt{16} = 4$, so SE $= 16/4 = 4$ ms and the half-width is $2.131 \times 4 = 8.524$. *Result.* The half-width is 8.524 ms, against 3.92 ms before, and $8.524/3.92 \approx 2.2$ times wider.

Two things did it. The standard error doubled because $\sqrt{64}/\sqrt{16} = 8/4 = 2$, and the critical value grew from 1.960 to 2.131 because t has heavier tails to pay for estimating σ from a small sample (CI p.35, CI p.39). This is HW4 Problem 1 part 4.

Answer. (a) Mean 50 s, $s \approx 2.83$ s, 90% CI (48.1, 51.9) s, 95% CI (47.6, 52.4) s, 99% CI (46.5, 53.5) s. (b) 31 runs. (c) 95% CI (116.08, 123.92) ms and 99% CI (114.85, 125.15) ms. $Z = -2.5$ and $p = 0.0124$, so reject at 0.05 but not at 0.01. (d) 8.52 ms, wider because the standard error doubles and t is bigger than z.

B4. A proportion from a sampling profiler.

In plain words. A sampling profiler is basically a yes-or-no survey, since each sample is either inside `parse()` or not. So the answer is a proportion p, and its standard error is $\sqrt{p(1-p)/n}$. For the sample size, "±5%" on the slides means 5% of p (relative error), not 5 percentage points.

(a) A sampling profiler reports a proportion, so the proportion interval applies (CI p.51-53, measurement study guide).

Step 1, the proportion. *Formula.* $p = m/n$, where m is the number of samples inside `parse()` and n is the total number of samples. *Plug in.* $p = 80/400$. *Result.* $p = 0.2$.

Step 2, the standard error. *Formula.* $\text{SE} = \sqrt{p(1-p)/n}$. *Plug in.* $\sqrt{0.2 \times 0.8/400}$. *Work it out.* $0.2 \times 0.8 = 0.16$, $0.16/400 = 0.0004$, and $\sqrt{0.0004} = 0.02$. *Result.* SE = 0.02.

Step 3, the intervals. *Formula.* $p \pm z[\alpha/2]\sqrt{p(1-p)/n}$ (CI p.48). *Plug in and work it out.*

$$90\%\text{:}\ \ 0.2 \pm 1.645(0.02) = 0.2 \pm 0.0329 = (0.1671,\ 0.2329)$$

$$95\%\text{:}\ \ 0.2 \pm 1.960(0.02) = 0.2 \pm 0.0392 = (0.1608,\ 0.2392)$$

Step 4, the check. *Formula.* The normal approximation needs both np and $n(1-p)$ to be large, at least 10 (CI p.47, confidence-intervals study guide). *Plug in.* $np = 400 \times 0.2 = 80$ and $n(1-p) = 400 \times 0.8 = 320$. *Result.* Both are well above 10, so the approximation is fine. If either were small, the Wald interval would be unreliable and you'd switch to Wilson or Agresti-Coull (CI p.49). So the profiler's "20%" really means somewhere between about 16% and 24%.

(b) The slides write the half-width as a fraction e of p (CI p.54). *Formula.*

$$n = \frac{z[\alpha/2]^2\, p(1-p)}{(ep)^2},$$

where $z[\alpha/2]$ is the normal critical value, p is the pilot proportion and e is the relative error. *Plug in.* The target half-width is $ep = 0.05 \times 0.2 = 0.01$, so

$$n = \frac{1.96^2 \times 0.2 \times 0.8}{(0.01)^2}.$$

Work it out. $1.96^2 = 3.8416$, $3.8416 \times 0.16 = 0.614656$, $0.01^2 = 0.0001$, and $0.614656/0.0001 = 6146.56$. *Result.* Round up to 6147 samples.

For the run length, *formula* time = samples × sampling interval, *plug in* 6147×0.01 s, *result* 61.47 s, about a minute. This is the same calculation as the slides' OS-time example, which needed 3.46 hours for ±0.5% relative error (CI p.55-57).

(c) "±5 percentage points" is an absolute half-width of 0.05. *Formula.* $n = z[\alpha/2]^2\, p(1-p)/h^2$, where h is the absolute half-width. *Plug in.* $n = 0.614656/0.05^2$. *Work it out.* $0.05^2 = 0.0025$ and $0.614656/0.0025 = 245.9$. *Result.* Round up to 246 samples, which is $246 \times 0.01 = 2.46$ s of run time.

The relative version needs 25 times more samples here, because $0.05/0.01 = 5$ and n grows with the square, $5^2 = 25$. The measurement study guide uses the absolute form, so always check which one the question means.

Answer. (a) 90% CI (0.167, 0.233) and 95% CI (0.161, 0.239), and the normal approximation is fine. (b) 6147 samples, a run of about 61.5 s. (c) 246 samples, about 2.5 s, which is 25 times fewer.

B5. Hypothesis test for a mean.

In plain words. A hypothesis test assumes the boring claim H_0 (nothing changed) and asks whether the data would be too unlikely if it were true. The claim you want to prove goes in H_1, and since that claim is "below 40" the test is one-sided. Because s is estimated from 16 runs, you use t with 15 degrees of freedom.

(a) The team wants to establish that the mean is below 40, so its negation goes in the null. $H_0\colon \mu = 40$ (meaning $\mu \geq 40$) and $H_1\colon \mu < 40$. If the test rejects H_0, that's evidence for the claim, while failing to reject would prove nothing (HT p.4). The direction was stated before the data were collected, so a one-sided test is legitimate (HT p.5). This mirrors HW5 Problem 2.

(b) σ is unknown and $n = 16$ is small, so use t with $n - 1 = 16 - 1 = 15$ degrees of freedom (HT p.11).

Step 1, the standard error. *Formula.* $\text{SE} = s/\sqrt{n}$, where s is the sample standard deviation and n the number of runs. *Plug in.* $6/\sqrt{16}$. *Work it out.* $\sqrt{16} = 4$ and $6/4 = 1.5$. *Result.* SE = 1.5 µs.

Step 2, the test statistic. *Formula.*

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}},$$

where $\bar{x}$ is the sample mean and $\mu_0 = 40$ is the value in H_0. *Plug in.* $t = (37 - 40)/1.5$. *Work it out.* $37 - 40 = -3$ and $-3/1.5 = -2.0$. *Result.* $t = -2.0$.

Step 3, the decision. For a one-sided test in the lower direction, reject if $t < -t[\alpha,\, n-1]$. With $\alpha = 0.05$ that cutoff is $-t[0.05,\, 15] = -1.753$. Since $-2.0 < -1.753$, reject H_0. At the 5% level the data support a mean below 40 µs.

(c) The p-value is the chance of a statistic at least this extreme if H_0 is true (HT p.13). On the 15 df row, 2.0 falls between $t[0.05,\, 15] = 1.753$ and $t[0.025,\, 15] = 2.131$. So the one-sided p-value is between 0.025 and 0.05 (software gives 0.032).

(d) Now the cutoff is $-t[0.01,\, 15] = -2.602$, and -2.0 isn't beyond it, so don't reject. That matches (c), since $p > 0.025 > 0.01$.

(e) *Formula.* Reject when $(\bar{x} - \mu_0)/\text{SE} \leq -t[0.01,\, 15]$, which rearranges to $\bar{x} \leq \mu_0 - t[0.01,\, 15] \times \text{SE}$. *Plug in.* $\bar{x} \leq 40 - 2.602 \times 1.5$. *Work it out.* $2.602 \times 1.5 = 3.903$ and $40 - 3.903 = 36.097$. *Result.*

$$\bar{x} \leq 36.097 \text{ µs}.$$

Any sample mean of about 36.1 µs or less would reject at the 1% level. This is HW5 Problem 1 part 3 with t instead of z.

(f) The colleague used a two-sided test. *Formula.* Two-sided p = 2× one-sided p. *Plug in.* One-sided p is between 0.025 and 0.05, so two-sided p is between $2 \times 0.025 = 0.05$ and $2 \times 0.05 = 0.10$ (software gives $2 \times 0.032 = 0.064$). The team is right here, because the direction was fixed by the claim before any data existed. The trap runs the other way. Picking one-sided after seeing which way the data went is a two-sided test at double the α in disguise (HT p.5).

Whichever test you use, also report the effect, a drop of $40 - 37 = 3$ µs, with its interval (HT p.16, HT p.18). *Formula.* One-sided 95% upper bound $= \bar{x} + t[0.05,\, 15] \times \text{SE}$. *Plug in.* $37 + 1.753 \times 1.5$. *Work it out.* $1.753 \times 1.5 = 2.6295$ and $37 + 2.6295 = 39.63$. *Result.* 39.63 µs, which is below 40, matching the rejection in (b).

Answer. (a) $H_0\colon \mu = 40$, $H_1\colon \mu < 40$. (b) $t = -2.0 < -1.753$, so reject. (c) $0.025 < p < 0.05$. (d) Don't reject at 0.01. (e) $\bar{x} \leq 36.097$ µs, about 36.1 µs. (f) They ran a two-sided test, and the team's one-sided test is the right one.

B6. Paired comparison.

In plain words. Paired means the same workload was run on both algorithms, so you can subtract and get one difference per workload. That cancels out how hard each workload is. After that it's just a one-sample t interval on the six differences.

Each workload is measured under both algorithms, so the data are paired and you analyze the six differences (Comp p.5). Take $d_i = B_i - A_i$, so a positive d favors A.

Workload	**1**	**2**	**3**	**4**	**5**	**6**
$d = B - A$	−1	2	3	3	5	6
$d - \bar{d}$	−4	−1	0	0	2	3
$(d - \bar{d})^2$	16	1	0	0	4	9

Step 1, the mean difference. *Formula.* $\bar{d} = \frac{1}{n}\sum d_i$, where d_i is the difference for workload i and $n = 6$ is the number of pairs. *Plug in.* $\bar{d} = (-1 + 2 + 3 + 3 + 5 + 6)/6$. *Work it out.* The sum is 18 and $18/6 = 3$. *Result.* $\bar{d} = 3$ ms.

Step 2, the standard deviation of the differences. *Formula.* $s_d^2 = \sum(d_i - \bar{d})^2/(n-1)$. *Plug in.* The squared deviations in the table are 16, 1, 0, 0, 4, 9. *Work it out.* They sum to 30, so $s_d^2 = 30/5 = 6$ and $s_d = \sqrt{6} \approx 2.449$. *Result.* $s_d \approx 2.449$ ms.

Step 3, the standard error. *Formula.* $\text{SE} = s_d/\sqrt{n}$. *Plug in.* $\sqrt{6}/\sqrt{6}$. *Result.* SE = 1 ms.

Step 4, the interval. The paired interval is a one-sample t interval on the differences with $n - 1 = 6 - 1 = 5$ degrees of freedom, since df counts pairs (Comp p.9). *Formula.*

$$\bar{d} \pm t[\alpha/2,\, n-1]\, \frac{s_d}{\sqrt{n}}.$$

Plug in and work it out.

$$3 \pm t[0.025,\ 5](1) = 3 \pm 2.571 = (3 - 2.571,\ 3 + 2.571) = (0.429,\ 5.571) \text{ ms.}$$

Result. The interval excludes 0 and lies entirely above it, so yes, with 95% confidence A has lower delay than B, by somewhere between 0.43 and 5.57 ms. The mean delay under A is $(42 + 35 + 51 + 28 + 60 + 47)/6 = 263/6 \approx 43.8$ ms, and $3/43.8 \approx 0.07$, so that's roughly a 7% improvement with a wide range around it.

As a test, $H_0\colon \mu_d = 0$ and $H_1\colon \mu_d > 0$. *Formula.* $t = \bar{d}/(s_d/\sqrt{n})$. *Plug in.* $t = 3/1$. *Result.* $t = 3.0$. On the 5 df row, 3.0 lies between 2.571 and 3.365, so the one-sided p-value is between 0.01 and 0.025 and the two-sided one is between 0.02 and 0.05 (software gives 0.030 two-sided). Both reject at 0.05.

Pairing matters here. Delays range from 28 to 65 ms across workloads, far more than the 3 ms effect. An unpaired Welch test on these same 12 numbers would leave that workload spread in the standard error and give $t \approx 0.43$ and $p \approx 0.68$ (computed by software, just to show the contrast), which misses the effect completely (Comp p.6-8). Note that workload 1 favors B, and that's fine, since the claim is about the mean difference. This is HW6 Problem 1 with six workloads.

Answer. Yes. The 95% CI for the mean of $B - A$ is (0.43, 5.57) ms, entirely above 0, so A has lower delay.

B7. Unpaired comparison with Welch.

In plain words. Unpaired means the two samples are separate groups with nothing matching up. Welch's t compares the two means without assuming equal variances. Its degrees-of-freedom formula gives a fraction, which you round down to use the table.

The transactions differ between systems and the counts differ, so no value of A goes with a value of B. The data are unpaired, and Welch's t is the default (Comp p.5, Comp p.12-13).

Step 1, each system's mean and variance. *Formulas.* $\bar{x} = \frac{1}{n}\sum x_i$ and $s^2 = \sum(x_i - \bar{x})^2/(n-1)$, done once per system.

System A, $n_A = 6$. *Plug in.* $\bar{x}_A = (52 + 57 + 46 + 54 + 48 + 55)/6$. *Work it out.* The sum is 312 and $312/6 = 52$. The deviations from 52 are 0, 5, −6, 2, −4, 3, with squares 0, 25, 36, 4, 16, 9, summing to 90. So

$$s_A^2 = \frac{90}{6-1} = \frac{90}{5} = 18, \qquad \frac{s_A^2}{n_A} = \frac{18}{6} = 3.$$

System B, $n_B = 8$. *Plug in.* $\bar{x}_B = (45+41+47+44+50+42+46+45)/8$. *Work it out.* The sum is 360 and $360/8 = 45$. The deviations from 45 are 0, −4, 2, −1, 5, −3, 1, 0, with squares 0, 16, 4, 1, 25, 9, 1, 0, summing to 56. So

$$s_B^2 = \frac{56}{8-1} = \frac{56}{7} = 8, \qquad \frac{s_B^2}{n_B} = \frac{8}{8} = 1.$$

Step 2, the standard error of the difference. *Formula.*

$$s = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}},$$

where each fraction is one system's variance divided by its sample size. *Plug in.* $s = \sqrt{3+1}$. *Work it out.* $\sqrt{4} = 2$. *Result.* $s = 2$ ms.

Step 3, the Welch degrees of freedom (Comp p.12). *Formula.*

$$\nu = \frac{\left(\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}\right)^2}{\frac{(s_A^2/n_A)^2}{n_A - 1} + \frac{(s_B^2/n_B)^2}{n_B - 1}}.$$

Plug in.

$$\nu = \frac{(3+1)^2}{\frac{3^2}{6-1} + \frac{1^2}{8-1}} = \frac{(3+1)^2}{\frac{9}{5} + \frac{1}{7}}.$$

Work it out. The top is $4^2 = 16$. The bottom is $1.8 + 0.1429 = 1.9429$. So $\nu = 16/1.9429 \approx 8.24$. *Result.* Round down to $\nu = 8$, since the table has whole rows and rounding down is the safe direction. As a sanity check, ν must land between $\min(n_A, n_B) - 1 = 5$ and $n_A + n_B - 2 = 12$, and 8.24 does (Comp p.12).

Step 4, the 90% interval for $\mu_A - \mu_B$. *Formula.*

$$(\bar{x}_A - \bar{x}_B) \pm t[\alpha/2,\ \nu]\, s,$$

with $\alpha = 0.10$, so $\alpha/2 = 0.05$. *Plug in.* $\bar{x}_A - \bar{x}_B = 52 - 45 = 7$ and $t[0.05,\ 8] = 1.860$. *Work it out.*

$$7 \pm 1.860(2) = 7 \pm 3.72 = (7 - 3.72,\ 7 + 3.72) = (3.28,\ 10.72) \text{ ms.}$$

Result. The interval excludes 0 and is positive, so at the 90% level A takes longer and B is faster, by somewhere between 3.3 and 10.7 ms of CPU time.

As a test, *formula* $t = (\bar{x}_A - \bar{x}_B)/s$, *plug in* $t = 7/2$, *result* $t = 3.5$. That's above $t[0.005,\ 8] = 3.355$, so the one-sided p-value is below 0.005.

The variance ratio is $18/8 = 2.25$, under 4, so the old rule of thumb would have pooled. The current advice is Welch by default, chosen before you see any p-value, because picking whichever version clears the cutoff is p-hacking (Comp p.13). This is HW6 Problem 2 with small samples.

Answer. Yes. The 90% CI for $\mu_A - \mu_B$ is (3.28, 10.72) ms, which excludes 0, so B is faster.

B8. Measurement.

In plain words. These are the practical limits of timing things. Resolution is the clock's smallest tick, timer overhead is the cost of reading the clock, and perturbation is how much measuring slows the program down. You beat the first two by timing many operations together, and you beat run-to-run noise by repeating at the level where the variance lives, which here is processes.

(a) One reading counts whole ticks, so it returns either 2 µs or 3 µs depending on where the operation starts relative to a tick, and never 2.6. The quantization error is up to one tick, 1 µs. *Formula.* Relative error = tick / true time. *Plug in.* 1/2.6. *Result.* About 0.385, so about 38% of the thing you're measuring. That's the slides' $X \pm \Delta$ quantization error, and it's random (CI p.9-10).

(b) Subtract the loop's own cost, then divide by the count (measurement study guide, Part B). *Formula.* Time per operation = $(T_{\text{loop}} - T_{\text{empty}})/N$, where T_{loop} is the loop with the operation, T_{empty} the empty loop and N the number of iterations. *Plug in.*

$$\frac{27{,}412 - 1{,}412}{10{,}000}.$$

Work it out. $27{,}412 - 1{,}412 = 26{,}000$ and $26{,}000/10{,}000 = 2.6$. *Result.* 2.6 µs per operation.

Formula. Effective resolution = clock tick / N. *Plug in.* 1/10,000. *Result.* 0.0001 µs, which is 0.1 ns. The assumption is that the operation behaves the same when repeated. It often doesn't, because the loop warms the cache and the branch predictor, and the compiler may even drop a loop body whose result is unused (measurement study guide, question 3).

(c) *Formula.* Overhead fraction = $2c/(T + 2c)$, where c is the cost of one clock read and T is the true operation time. Two reads add $2c = 2 \times 20 = 40$ ns. *Plug in and work it out.*

$$T = 60 \text{ ns}\colon\ \frac{40}{60+40} = \frac{40}{100} = 0.40, \qquad T = 4000 \text{ ns}\colon\ \frac{40}{4000+40} = \frac{40}{4040} \approx 0.0099.$$

Result. 40% for the 60 ns operation and about 1% for the 4 µs one.

The rule is that if an operation isn't one or two orders of magnitude slower than a clock read, you time a loop of them as in (b) instead (measurement study guide, question 4).

(d) *Formula.* Perturbation = $(T_{\text{traced}} - T_{\text{untraced}})/T_{\text{untraced}}$. *Plug in.* $(86 - 80)/80$. *Work it out.* $86 - 80 = 6$ and $6/80 = 0.075$. *Result.* 7.5%.

The probes measured alone sum to $2.5 + 2.0 = 4.5$ s, but together they cost 6 s, and that's not a contradiction, because perturbation isn't additive. Two probes' buffers can evict the program's working set and slow it down beyond their own cost, or the second probe can find data already cached and cost less. So you have to measure the configuration you actually ran, with and without instrumentation (Meas p.16-17).

(e) *Formula.* The variance of the overall mean with p processes of k iterations each is

$$\text{Var}(\bar{x}) = \frac{\sigma_p^2}{p} + \frac{\sigma_i^2}{pk}, \qquad \text{SE} = \sqrt{\text{Var}(\bar{x})},$$

where σ_p^2 is the between-process variance and σ_i^2 the within-process variance (measurement study guide, Part B). *Plug in and work it out* with $\sigma_p^2 = 48$, $\sigma_i^2 = 16$ and $pk = 16$ every time.

$$1 \times 16\colon\ \frac{48}{1} + \frac{16}{16} = 48 + 1 = 49, \quad \text{SE} = \sqrt{49} = 7 \text{ ms}$$

$$2 \times 8\colon\ \frac{48}{2} + \frac{16}{16} = 24 + 1 = 25, \quad \text{SE} = \sqrt{25} = 5 \text{ ms}$$

$$16 \times 1\colon\ \frac{48}{16} + \frac{16}{16} = 3 + 1 = 4, \quad \text{SE} = \sqrt{4} = 2 \text{ ms}$$

Result. Same 16 measurements, but the standard error drops from 7 ms to 2 ms. Only more processes shrink the first term, so extra iterations inside one process can't buy down the bigger source of variance. Spend the budget at the level where the variance is (Meas p.25).

Answer. (a) 2 or 3 µs, with error up to 1 µs (about 38%). (b) 2.6 µs per operation, resolution 0.0001 µs (0.1 ns), assuming every repetition behaves the same. (c) 40% and about 1%. (d) 7.5% perturbation, and probe costs don't add. (e) Standard errors of 7, 5 and 2 ms.

Part C

In plain words. You're the reviewer of a made-up results paragraph, spotting its statistics and measurement mistakes. Each question points at one mistake from the lectures, like treating standard deviation bars as confidence intervals, running many tests with no correction, or calling a big p-value proof that two things are the same.

C1. The unit is one run, so n = 5 per system, and a "run" here is a separate measurement of one system, not a matched pair (HT p.20, Comp p.4). The headline comparison is unpaired, because run i of FastKV has nothing in common with run i of RocksDB, and they weren't even on the same day. The 13-workload comparison could have been paired by workload, since both systems can run the same workload, and pairing would cancel the large differences between workloads (Comp p.5-6).

C2. No. The bars are ±1 standard deviation, which describes the spread of single runs and doesn't shrink with n, so it says nothing direct about how precisely the means are known. The overlap rule on the slides is for confidence intervals, and even then non-overlap is the only safe direction (Comp p.15). The claim needs an interval on the difference, which C3 builds. Each bar is mean ± one standard deviation. FastKV's runs from $118.2 - 2.0 = 116.2$ to $118.2 + 2.0 = 120.2$, and RocksDB's runs from $104.6 - 4.0 = 100.6$ to $104.6 + 4.0 = 108.6$. So they don't overlap, but that isn't the evidence.

C3. The claim is that FastKV has higher mean throughput, so $H_0 : \mu_F = \mu_R$ and $H_1 : \mu_F > \mu_R$, one-sided (HT p.4-5). For the interval, use Welch with $n = 5$ each (Comp p.12).

Step 1, the standard error. *Formula.* $s = \sqrt{s_F^2/n_F + s_R^2/n_R}$, where $s_F = 2.0$ and $s_R = 4.0$ are the two standard deviations and $n_F = n_R = 5$ the run counts. *Plug in and work it out.*

$$\frac{s_F^2}{n_F} = \frac{2.0^2}{5} = \frac{4}{5} = 0.8, \qquad \frac{s_R^2}{n_R} = \frac{4.0^2}{5} = \frac{16}{5} = 3.2, \qquad s = \sqrt{0.8 + 3.2} = \sqrt{4} = 2.$$

Step 2, the Welch degrees of freedom. *Formula.* The same ν formula as in B7. *Plug in and work it out.*

$$\nu = \frac{(0.8 + 3.2)^2}{\frac{0.8^2}{5-1} + \frac{3.2^2}{5-1}} = \frac{4^2}{\frac{0.64}{4} + \frac{10.24}{4}} = \frac{16}{0.16 + 2.56} = \frac{16}{2.72} \approx 5.88.$$

Result. Round down to $\nu = 5$.

Step 3, the interval. *Formula.* $(\bar{x}_F - \bar{x}_R) \pm t[0.025,\, \nu]\, s$. *Plug in.* $\bar{x}_F - \bar{x}_R = 118.2 - 104.6 = 13.6$ and $t[0.025,\, 5] = 2.571$. *Work it out.*

$$13.6 \pm 2.571(2) = 13.6 \pm 5.142 = (13.6 - 5.142,\ 13.6 + 5.142) = (8.46,\ 18.74) \text{ Kops/s}.$$

The rewrite reads "FastKV's mean throughput on workload A was 13.6 Kops/s higher than RocksDB's (95% CI 8.5 to 18.7 Kops/s, Welch t, 5 runs each), which is roughly 8% to 18% of RocksDB's 104.6." The percentages come from dividing by the RocksDB mean, $8.46/104.6 \approx 0.081$ and $18.74/104.6 \approx 0.179$, and the headline 13% is $13.6/104.6 \approx 0.130$. The interval excludes 0, so the conclusion happens to survive, but by correct reasoning instead of the bar picture (Comp p.24, Comp p.26). It's still only a statement about sampling error, and C6 is why that matters.

C4. No. Throughput is a rate, and the mean of rates for equal work should be the harmonic mean, because it equals total work over total time (Summ p.28, Summ p.32). An arithmetic mean of rates gives too much weight to the fast workloads. Better still, report the per-workload numbers so readers can pick their own summary.

C5. That's 13 tests at $\alpha = 0.05$ with no correction. Suppose FastKV made no difference anywhere. *Formula.* Expected false positives $= m\alpha$, where m is the number of tests and α the per-test level. *Plug in.* 13×0.05. *Result.* 0.65. *Formula.* The chance of at least one is FWER $= 1 - (1 - \alpha)^m$. *Plug in.* $1 - 0.95^{13}$. *Work it out.* $0.95^{13} \approx 0.513$, so $1 - 0.513 = 0.487$. *Result.* About 49%, roughly a coin flip (HT p.26-27). Four out of 13 is more than chance alone would give, but you can't say which four are real without a correction. Bonferroni tests each at $\alpha/m = 0.05/13 \approx 0.0038$, and Holm is the other option (HT p.28). The paper should also report all 13 results, not just the ones that cleared 0.05 (Comp p.26).

C6. Any two of these. System and day are confounded, since every FastKV run was on Monday and every RocksDB run was on Tuesday after a kernel update. That's a systematic error that sits entirely outside the interval, and more runs would only tighten the interval around the biased difference. The fix is to interleave the two systems in randomized order on the same kernel (Meas p.20, Meas p.22-23). gettimeofday() isn't monotonic, so NTP can move it mid-run, and CLOCK_MONOTONIC is the right clock (Meas p.12). The 30-second warm-up cut is asserted, not justified, so you don't know how it was chosen or whether a steady state exists (Meas p.14).

C7. Two things. A large p-value isn't evidence of equivalence, especially with 5 runs, since the test may have had almost no power (HT p.14). And a t test compares means, while p99 is a quantile, so if the claim is about the tail the paper should compare p99s directly, for instance with a bootstrap interval on their difference (Comp p.18). An honest version would give the p99 difference with its interval and say how large a difference the experiment could have detected.

Answer. C1, the unit is one run, n = 5 per system, unpaired, and the 13-workload part could have been paired by workload. C2, not valid. C3, $H_0 : \mu_F = \mu_R$ and $H_1 : \mu_F > \mu_R$, with 95% CI (8.46, 18.74) Kops/s for the difference. C4, no, use the harmonic mean or report each workload. C5, 13 tests with no multiple-comparison correction. C6, any two of the Monday/Tuesday kernel confound, gettimeofday() and the unjustified warm-up. C7, a large p-value isn't proof of equivalence, and a t test compares means, not p99.

Practice set 2

A second mock exam on fresh numbers and on the topics set 1 left out, like weighted means, histograms, the variance interval, chi-square goodness of fit, two proportions, sample size and the sign test. Same rules as set 1. No electronics, pencil only, write each answer before you look at its solution. Aim for about 25 minutes on Part A and 70 minutes on Part B.

These are the only table values set 2 needs. The z and t brackets are upper-tail areas as before. The chi-square row gives the area *below* the value, which is how the slides label χ^2_L and χ^2_U (CI p.59-60).

Upper-tail area	**0.20**	**0.10**	**0.05**	**0.025**	**0.005**
z	0.842	1.282	1.645	1.960	2.576

Area below	**0.025**	**0.05**	**0.95**	**0.975**	**0.99**
χ^2, 3 df	0.22	0.35	7.81	9.35	11.34
χ^2, 9 df	2.70	3.33	16.92	19.02	21.67

Normal-table values $\Phi(2.0) = 0.9772$ and $\Phi(2.5) = 0.9938$. Powers of two $2^4 = 16$, $2^5 = 32$, $2^{10} = 1024$, $2^{16} = 65{,}536$ and $2^{32} = 4{,}294{,}967{,}296$.

Part A. Concepts (10 short questions, about 25 minutes)

Two or three sentences each, with the arithmetic where there is some.

A1. Three programs run at 100, 200 and 400 MFLOPS. (a) Each program runs for exactly 10 s. What's the mean MFLOPS? (b) Each program instead does the same 1000 million floating-point operations. What's the mean MFLOPS now? Say which mean you used each time and why they differ.

A2. True or false, with a reason. (a) A program that reports 0 MFLOPS does no useful work. (b) MFLOPS on machine A and machine B can be compared fairly because a FLOP is a FLOP.

A3. A paper compares old and new versions of a compiler on 5 benchmarks. The new version wins on all 5, and the paper reports "sign test, $p < 0.05$". Is that possible? What's the smallest two-sided p-value this design can produce?

A4. Five benchmarks give p-values 0.004, 0.011, 0.020, 0.035 and 0.30. At a family level of $\alpha = 0.05$, how many are significant with no correction, with Bonferroni, with Holm and with Benjamini-Hochberg?

A5. A free-running 32-bit counter ticks at 1 MHz. (a) How often does it roll over? (b) You read 4,294,966,296 at the start of an operation and 3,000 at the end. How long did it take, and what does a naive end minus start give? (c) With a 16-bit counter at the same rate, an operation really takes 100 ms. What elapsed time do you compute, and why can't you detect the mistake from the readings alone?

A6. (a) Why is the 95% confidence interval for σ^2 not centered on s^2? (b) The t interval for the mean gets rescued by the Central Limit Theorem when the data aren't normal. Does the variance interval get the same rescue?

A7. A power calculation says 20 runs per system will detect a 4 ms difference with power 0.80 at $\alpha = 0.05$. Your advisor wants to detect 2 ms instead. How many runs per system now? And after the experiment, a colleague offers to compute the power from your observed p-value. Is that useful?

A8. A sampling profiler takes 400 samples and only 3 land in `gc()`. Your friend reports the 95% interval

$$0.0075 \pm 1.96\sqrt{0.0075 \times 0.9925/400}.$$

What's wrong with relying on it, and what should be used instead?

A9. A CDN change cuts mean page load by 0.4 ms. The page-load standard deviation is 20 ms and the comparison used 10,000 loads. Compute Z and the two-sided p-value. Is the change significant, and is it important?

A10. Pick a test for each design and say why. (a) The same 12 inputs run on two versions, and the differences look symmetric but one is a big outlier. (b) The same 15 inputs on two versions, and the differences are strongly right-skewed. (c) 20 runs on machine A and 25 different runs on machine B, with long right tails. (d) 10 runs on each of two machines, roughly normal, but one machine's spread is three times the other's.

Part B. Problems (6 problems, about 70 minutes)

Show every step and name the formula before you plug in.

B1. A weighted harmonic mean. A simulation spends its floating-point work in three phases. Half of all FLOPs run in a dense phase at 400 MFLOPS, a quarter in a sparse phase at 200 MFLOPS and a quarter in a communication-bound phase at 100 MFLOPS. (a) What is the program's overall MFLOPS? Use the weighted mean that fits and say why. (b) What do the unweighted arithmetic mean and the FLOP-weighted arithmetic mean give, and why are they wrong? (c) Check your answer to (a) by assuming 1000 million FLOPs in total and adding up the time of each phase. (d) What fraction of the run's time does each phase take, and what does the time-weighted arithmetic mean of the rates give?

B2. Sturges and a histogram. Twenty response times in ms, in the order measured.

23	17	31	12	19	25	16	28	20	14
35	18	22	10	27	19	24	17	30	21

(a) How many bins does Sturges' rule give? (b) Using that many equal-width bins over the range, with bins of the form lo $<$ x $\leq$ hi as on the slides and the first bin also taking the minimum, give the count and relative frequency of each bin. (c) Describe the shape, and check your description against the mean and median. (d) A colleague redraws it with 10 bins of width 2.5. What changes, and what does that tell you? (e) How many bins would Sturges give for 1000 points?

B3. Chi-square goodness of fit. A load balancer is supposed to spread requests evenly over four servers. In a sample of 120 requests the servers received the following.

Server	**1**	**2**	**3**	**4**
Requests	42	20	34	24

(a) State H_0 and H_1, compute the expected counts and the χ^2 statistic, and give the degrees of freedom. (b) Test at $\alpha = 0.05$ and at $\alpha = 0.01$, and bracket the p-value. (c) If the claimed distribution had a parameter that you estimated from these same 120 requests, how would the degrees of freedom change? (d) Why does the textbook merge cells before computing χ^2 when some expected counts are small? (e) If the test had not rejected, could you report that the load is balanced?

B4. Confidence interval for a variance. Ten runs of a query take these times in ms.

44	46	47	47	50	50	52	53	55	56

(a) Find $\bar{x}$ and s^2. (b) Give the 95% confidence interval for σ^2 and for σ. (c) Give the 90% interval for σ^2. (d) The vendor's spec sheet says the run-to-run standard deviation is 3 ms. Do the data contradict it at the 95% level? (e) What assumption does the interval rest on, and how badly does it fail if the assumption is wrong?

B5. Comparing two proportions. Under the old cache configuration 180 of 800 requests missed a 100 ms latency target. Under the new configuration, on a separate set of requests, 140 of 800 missed it. (a) State H_0 and H_1 for "the miss rate changed" and run the two-proportion Z test. (b) Give the verdicts at $\alpha = 0.05$ and $\alpha = 0.01$ and the p-value. (c) Give a 95% confidence interval for the drop in miss rate. (d) Why does the test use a pooled proportion but the interval doesn't? (e) Check that the normal approximation is reasonable.

B6. Sample size and the sign test. You're planning to compare an old and a new build. (a) A pilot shows run-to-run standard deviation $s = 10$ ms on each build. With unpaired runs, how many runs per build do you need to detect a 5 ms difference at $\alpha = 0.05$ (two-sided) with power 0.80? (b) If you instead run both builds on the same inputs and the pilot shows the paired differences have $s_d = 4$ ms, how many pairs do you need? (c) Back to the unpaired design, how many runs per build to detect 2.5 ms? (d) You end up with 11 paired inputs, and the differences old minus new (ms) are listed below. Run a sign test at $\alpha = 0.05$, two-sided. (e) Why is the sign test the right choice here rather than the paired t test?

Input	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
d	3.1	0.4	12.0	0	1.2	−0.8	2.5	0.9	25.0	1.7	0.3

Question	Topic	Lecture
A1	Arithmetic vs harmonic mean of rates	4, Summarizing data
A2	What's wrong with MFLOPS	2, Metrics
A3	Smallest p-value of a sign test	10, Comparing alternatives
A4	Multiple comparisons (Bonferroni, Holm, BH)	9, Hypothesis testing
A5	Counter rollover	11, Measurement
A6	Interval for a variance	6, Confidence intervals
A7	Sample size and post-hoc power	9 and 10
A8	Proportion interval with few hits	6, Confidence intervals
A9	Statistical vs practical significance	9, Hypothesis testing
A10	Choosing a test	10, Comparing alternatives
B1	Weighted harmonic mean	4, Summarizing data
B2	Sturges' rule and histograms	4, Summarizing data
B3	Chi-square goodness of fit	chi-square handout
B4	Interval for a variance	6, Confidence intervals
B5	Two-proportion Z test and interval	6 and 9
B6	Sample size and the sign test	9 and 10

Every calculation below runs in the same four steps. *Formula* gives the formula in symbols and says what each symbol means. *Plug in* puts this problem's numbers into it. *Work it out* does the arithmetic one step at a time. *Result* states the number.

Part A

A1. *In plain words.* The question is really about which kind of average matches what was held fixed. When every program runs for the same time, each rate counts equally, so the ordinary arithmetic mean works. When every program does the same work, the slow one runs longest and should count more, and the harmonic mean (n divided by the sum of $1/x_i$) does exactly that.

(a) Equal time means every rate gets the same weight, so the true overall rate is the arithmetic mean.

1. *Formula.* $\bar{x}_A = \frac{1}{n}\sum x_i$, where n is the number of programs and x_i is program i's rate in MFLOPS.
2. *Plug in.* $\bar{x}_A = \frac{100+200+400}{3}$.
3. *Work it out.* $100 + 200 + 400 = 700$, and $700/3 = 233.3$. Check with total work over total time. Each program does $10\times$ its rate in 10 s, so the work is $1000 + 2000 + 4000 = 7000$ million FLOPs in $3 \times 10 = 30$ s, and $7000/30 = 233.3$.
4. *Result.* 233.3 MFLOPS.

(b) Equal work is the setup the harmonic mean is built for, since it then equals total work over total time (Summ p.28, lecture 3 study guide).

1. *Formula.* $\bar{x}_H = \dfrac{n}{\sum 1/x_i}$, where n is the number of programs and x_i is program i's rate.
2. *Plug in.* $\bar{x}_H = \dfrac{3}{\frac{1}{100} + \frac{1}{200} + \frac{1}{400}}$.
3. *Work it out.* Put the fractions over 400. $\frac{1}{100} = \frac{4}{400}$, $\frac{1}{200} = \frac{2}{400}$ and $\frac{1}{400} = \frac{1}{400}$, which add to $\frac{7}{400}$. Then $3 \div \frac{7}{400} = \frac{3\times400}{7} = \frac{1200}{7} = 171.4$. Check the long way. Each program does 1000 million FLOPs, so the times are $1000/100 = 10$ s, $1000/200 = 5$ s and $1000/400 = 2.5$ s. That's 3000 million FLOPs in $10 + 5 + 2.5 = 17.5$ s, and $3000/17.5 = 171.4$.
4. *Result.* 171.4 MFLOPS.

The answers differ because with equal work the slow program runs longest and drags the rate down, while with equal time every rate counts for the same 10 s. A rate mean has to be inversely proportional to total time, and which mean does that depends on whether work or time was held fixed (Summ p.25).

Answer. (a) 233.3 MFLOPS by the arithmetic mean, (b) 171.4 MFLOPS by the harmonic mean, and they differ because equal work lets the slow program take more of the time.

A2. *In plain words.* MFLOPS means millions of floating-point operations per second, a count of decimal-math steps done each second. The slide lists two problems with it. Programs that do no decimal math score zero even when they do real work, and nobody agrees on how to count one FLOP.

(a) False. An integer program reports 0 MFLOPS but still does useful work like sorting and searching (Metrics p.19).

(b) False. There's too much freedom in how to count a FLOP, for example transcendental operations and square roots, so MFLOPS isn't consistent across machines (Metrics p.19).

Answer. Both are false.

A3. *In plain words.* The sign test only counts how many pairs went each way, so with 5 pairs there are only a handful of possible outcomes. Even the most extreme one, 5 wins out of 5, is like flipping a coin and getting 5 heads, which isn't rare enough. So the claim is impossible no matter how big the wins were.

Under H_0 each benchmark is a fair coin flip, so the number of wins S is Binomial(5, $\frac{1}{2}$) (Comp p.23, comparing-alternatives study guide).

1. *Formula.* Two-sided $p = 2 \times P(S \geq s)$, and $P(S = n) = (\frac{1}{2})^n$, where n is the number of pairs and s is the observed number of wins.
2. *Plug in.* The most extreme result is $s = n = 5$, so $p = 2 \times (\frac{1}{2})^5$.
3. *Work it out.* $(\frac{1}{2})^5 = \frac{1}{2^5} = \frac{1}{32}$, and $2 \times \frac{1}{32} = \frac{2}{32} = 0.0625$.
4. *Result.* The smallest possible p is 0.0625, which is above 0.05.

So no outcome can give $p < 0.05$. With 5 or fewer pairs the sign test can't reach significance at all.

Answer. Not possible, since the smallest two-sided p-value with 5 pairs is $2/32 = 0.0625$.

A4. *In plain words.* When you run many tests, some will look significant by luck, so corrections make each test harder to pass. Bonferroni is the strictest, Holm is a step-by-step version that's a bit looser, and Benjamini-Hochberg (BH) is the loosest because it only controls the share of false finds. You just compare each sorted p-value with its own cutoff.

Here $m = 5$ is the number of tests, $\alpha = 0.05$, and $p_{(k)}$ is the k-th smallest p-value (HT p.28).

1. *No correction.* Reject if $p < 0.05$. The values 0.004, 0.011, 0.020 and 0.035 pass and 0.30 fails, so 4.
2. *Bonferroni.* The formula is cutoff $= \alpha/m = 0.05/5 = 0.01$. Only $0.004 \le 0.01$, so 1.
3. *Holm.* The formula is cutoff$_k = \alpha/(m-k+1)$, stopping at the first failure. For $k = 1$, $0.05/5 = 0.01$ and $0.004 \le 0.01$, reject. For $k = 2$, $0.05/4 = 0.0125$ and $0.011 \le 0.0125$, reject. For $k = 3$, $0.05/3 = 0.0167$ and $0.020 > 0.0167$, so stop. That's 2.
4. *Benjamini-Hochberg.* The formula is cutoff$_k = k\alpha/m$. The cutoffs are $1(0.05)/5 = 0.01$, $2(0.05)/5 = 0.02$, 0.03, 0.04 and 0.05. Compare $0.004 \le 0.01$, $0.011 \le 0.02$, $0.020 \le 0.03$, $0.035 \le 0.04$ and $0.30 > 0.05$. The largest k that passes is 4, so reject the 4 smallest.

Same p-values, and the count runs from 1 to 4, which is the point of the slides' six-comparison table (HT p.29). BH controls the false discovery rate, not the chance of any false positive, so it isn't the same guarantee as the other two.

Answer. No correction 4, Bonferroni 1, Holm 2, BH 4.

A5. *In plain words.* A counter that counts up and wraps back to 0 is like a car odometer rolling over. If the end reading is smaller than the start, it wrapped, and you add back one full lap. The danger is wrapping more than once, because then the readings can't tell you how many laps you missed.

(a)

1. *Formula.* Period $= 2^b/f$, where b is the number of counter bits and f is the tick rate.
2. *Plug in.* Period $= 2^{32}/10^6$ per second, since 1 MHz is 10^6 ticks per second.
3. *Work it out.* $2^{32} = 4{,}294{,}967{,}296$ ticks, and one tick is 1 µs, so that's 4,294,967,296 µs = 4294.97 s. Divide by 60 to get $4294.97/60 =$ 71.6 minutes.
4. *Result.* About 71.6 minutes.

(b)

1. *Formula.* Elapsed ticks = (end − start) mod 2^{32}, which in practice means add 2^{32} when end is less than start.
2. *Plug in.* $3{,}000 - 4{,}294{,}966{,}296 + 4{,}294{,}967{,}296$.
3. *Work it out.* The naive part $3{,}000 - 4{,}294{,}966{,}296 = -4{,}294{,}963{,}296$, a huge negative number. Adding 2^{32} gives $-4{,}294{,}963{,}296 +$ $4{,}294{,}967{,}296 = 4{,}000$ ticks. Another way to see it is 1,000 ticks from the start reading up to the wrap, plus 3,000 after it.
4. *Result.* 4,000 µs, which is 4 ms.

Unsigned 32-bit subtraction does this mod for you, as long as the interval is shorter than one period.

(c)

1. *Formula.* Same as (a) and (b) with $b = 16$. Computed time = true time mod 2^{16} ticks.
2. *Plug in.* Period $= 2^{16} = 65{,}536$ µs. True time is 100 ms = 100,000 µs.
3. *Work it out.* 100,000 is more than one period but less than two, so it wrapped once. $100{,}000 - 65{,}536 = 34{,}464$ µs.
4. *Result.* You'd compute 34.464 ms instead of 100 ms.

Both readings look perfectly normal, so nothing in the data shows the lost period. The timer's period has to be longer than anything you time, or you have to count the overflows. It's one more reason to report which clock you used, the way the slides ask you to report its resolution (Meas p.12).

Answer. (a) Every 71.6 minutes. (b) 4 ms, while the naive subtraction gives a huge negative number. (c) 34.464 ms, and you can't tell because one silent wrap leaves both readings looking normal.

A6. *In plain words.* The variance interval uses the chi-square distribution, which is lopsided with a long right tail, so the interval is lopsided too. Also, the Central Limit Theorem (averages of many values look normal) helps intervals for means, but variance isn't a plain average of the data, so it gets no such help.

(a) The interval divides $(n-1)s^2$ by two chi-square critical values, and the chi-square distribution isn't symmetric (CI p.59-60). The lower critical value sits much closer to 0 than the upper one sits above the middle, so the upper end of the interval lands much farther from s^2 than the lower end does. B4 shows it, with $s^2 = 16$ and an interval of 7.6 to 53.3, which is $16 - 7.6 = 8.4$ below and $53.3 - 16 = 37.3$ above.

(b) No. The Central Limit Theorem is about sums and means of many values (CI p.64), so it rescues the t interval. The variance interval leans on the data themselves being normal, and the slides warn that for non-normal populations it isn't very accurate, especially for small samples (CI p.62).

Answer. (a) Because chi-square isn't symmetric. (b) No, it needs the data to be normal.

A7. *In plain words.* The runs you need grow with one over the difference squared. Half the difference means four times the runs. Power (the chance you catch a real effect) is something you plan before the experiment, and computing it afterwards from your p-value just restates the p-value.

1. *Formula.* $n = \big((z[\alpha/2] + z[\beta])\,\sigma/\delta\big)^2$ (HT p.19), where σ is the run-to-run spread and δ is the difference to detect. Everything but δ stays fixed, so $n_{\text{new}} = n_{\text{old}} \times (\delta_{\text{old}}/\delta_{\text{new}})^2$.
2. *Plug in.* $n_{\text{new}} = 20 \times (4/2)^2$.
3. *Work it out.* $4/2 = 2$, $2^2 = 4$, and $20 \times 4 = 80$.
4. *Result.* 80 runs per system.

The post-hoc power isn't useful. Power computed after the fact from your own p-value tells you nothing new, and the calculation belongs before the data are collected (HT p.19, Comp p.25).

Answer. 80 runs per system, and post-hoc power isn't useful.

A8. *In plain words.* The usual proportion interval (called Wald) leans on a normal curve, which only works when you have plenty of hits and misses. With just 3 hits, the curve is a bad fit, and here the interval even dips below zero, which is impossible for a fraction. The slides say to switch to the Wilson or Agresti-Coull interval.

The friend's interval is the Wald interval, and it needs np and $n(1-p)$ both comfortably large (CI p.47, confidence-intervals study guide). Here $np = 400 \times 0.0075 = 3$, far too small. It even breaks visibly.

1. *Formula.* $\hat{p} \pm z[\alpha/2]\sqrt{\hat{p}(1-\hat{p})/n}$, where $\hat{p}$ is the fraction of samples in `gc()` and n is the number of samples.
2. *Plug in.* $0.0075 \pm 1.96\sqrt{0.0075 \times 0.9925/400}$, with $\hat{p} = 3/400 = 0.0075$.
3. *Work it out.* $0.0075 \times 0.9925 = 0.00744$. Then $0.00744/400 = 0.0000186$, and $\sqrt{0.0000186} = 0.00431$. Then $1.96 \times 0.00431 = 0.0085$. Lower end $0.0075 - 0.0085 = -0.001$, upper end $0.0075 + 0.0085 = 0.016$.
4. *Result.* $(-0.001,\ 0.016)$, and the lower end is a negative fraction of time, which is impossible.

Use the Wilson or Agresti-Coull interval instead. The slides recommend them over Wald, Wilson for $n < 40$ and Agresti-Coull above (CI p.49). Or collect many more samples so the count in `gc()` isn't tiny.

Answer. With only 3 hits the Wald interval is unreliable and even goes negative, so use Wilson or Agresti-Coull, or collect more samples.

A9. *In plain words.* Significant means unlikely to be pure luck. Important means big enough to matter. With 10,000 loads even a tiny 0.4 ms change shows up as significant, but nobody would ever notice 0.4 ms out of a 20 ms spread.

1. *Formula.* $Z = \dfrac{\text{effect}}{\sigma/\sqrt{n}}$ and two-sided $p = 2\big(1 - \Phi(|Z|)\big)$ (HT p.17), where the effect is the change in mean, σ is the standard deviation, n is the number of loads and Φ is the normal table.
2. *Plug in.* $Z = \dfrac{0.4}{20/\sqrt{10{,}000}}$.
3. *Work it out.* $\sqrt{10{,}000} = 100$, so $20/100 = 0.2$, and $0.4/0.2 = 2.0$. Then $\Phi(2.0) = 0.9772$, so $1-0.9772 = 0.0228$, and $2\times0.0228 = 0.0456$.
4. *Result.* $Z = 2.0$ and $p = 0.0456$.

It's significant at 0.05 because $0.0456 < 0.05$. But the effect is 0.4 ms against a 20 ms spread, a standardized size of $0.4/20 = 0.02$. No user would notice that. With enough data every non-zero effect becomes significant, so significance is partly a statement about n (HT p.17). Report the 0.4 ms with its interval and let the reader judge whether it matters (HT p.18).

Answer. $Z = 2.0$ and $p \approx 0.046$, so it's significant at 0.05 but not practically important.

A10. *In plain words.* The slides give a short decision list. First ask if the same input was used for both versions (paired) or not (unpaired). Then ask if the data look normal, have outliers, or are skewed (one long tail), and pick the test that doesn't break under that problem.

(a) Wilcoxon signed-rank. The design is paired, and the rank test is robust to an outlier as long as the differences are roughly symmetric (Comp p.17, Comp p.21).

(b) The sign test or a bootstrap, or analyze log(A/B). Signed-rank assumes symmetric differences, so skew rules it out (Comp p.17-18, Comp p.21, Comp p.23).

(c) Mann-Whitney U or a bootstrap. The samples are unpaired and skewed (Comp p.17, Comp p.19).

(d) Welch's t. It's unpaired and roughly normal, and Welch doesn't care about unequal variances, which is why it's the default (Comp p.12-13, Comp p.17).

Answer. (a) Wilcoxon signed-rank, (b) sign test or bootstrap, (c) Mann-Whitney U or bootstrap, (d) Welch's t.

Part B

B1. A weighted harmonic mean. *In plain words.* You want the program's true overall speed, which is total FLOPs divided by total time. The weights here are shares of work, and when you weight rates by work, the harmonic mean gives total work over total time. The arithmetic mean would overrate the fast phase, because the slow phase actually eats most of the clock.

(a) These are rates, and the weights are shares of the work, not of the time. The mean rate has to be inversely proportional to total time, so use the weighted harmonic mean (Summ p.25, Summ p.28, Summ p.33).

1. *Formula.* $\bar{x}_H = \dfrac{1}{\sum w_i/x_i}$ with $\sum w_i = 1$, where w_i is phase i's share of the FLOPs and x_i is its rate.
2. *Plug in.* $\bar{x}_H = \dfrac{1}{\frac{0.5}{400} + \frac{0.25}{200} + \frac{0.25}{100}}$.
3. *Work it out.* $0.5/400 = 0.00125$, $0.25/200 = 0.00125$ and $0.25/100 = 0.0025$. They add to $0.00125 + 0.00125 + 0.0025 = 0.005$, and $1/0.005 = 200$.
4. *Result.* 200 MFLOPS.

(b) Unweighted arithmetic mean.

1. *Formula.* $\bar{x}_A = \frac{1}{n}\sum x_i$.
2. *Plug in.* $(400 + 200 + 100)/3$.
3. *Work it out.* $400 + 200 + 100 = 700$, and $700/3 = 233.3$.
4. *Result.* 233.3 MFLOPS.

FLOP-weighted arithmetic mean.

1. *Formula.* $\bar{x}_A = \sum w_i x_i$, with w_i the FLOP shares.
2. *Plug in.* $0.5(400) + 0.25(200) + 0.25(100)$.
3. *Work it out.* $0.5 \times 400 = 200$, $0.25 \times 200 = 50$, $0.25 \times 100 = 25$, and $200 + 50 + 25 = 275$.
4. *Result.* 275 MFLOPS.

Both overstate the real rate of 200 because they ignore that the slow phase eats the most time. An arithmetic mean of rates isn't

proportional to anything physical unless the weights are shares of time, which (d) shows.

(c)

1. *Formula.* Time of a phase = FLOPs$_i/x_i$, and overall rate = total FLOPs/total time.
2. *Plug in.* Take 1000 million FLOPs. The dense phase does $0.5 \times 1000 = 500$, the sparse phase $0.25 \times 1000 = 250$ and the communication phase 250. Times are 500/400, 250/200 and 250/100.
3. *Work it out.* $500/400 = 1.25$ s, $250/200 = 1.25$ s and $250/100 = 2.5$ s. Total time is $1.25 + 1.25 + 2.5 = 5$ s, and $1000/5 = 200$.
4. *Result.* 200 MFLOPS, matching (a).

That's the lecture 3 study guide's point that the work-weighted harmonic mean is just total work over total time written as a mean.

(d)

1. *Formula.* Time share t_i = time$_i$/total time, then $\bar{x}_A = \sum t_i x_i$.
2. *Plug in.* Shares 1.25/5, 1.25/5 and 2.5/5. Mean = $0.25(400) + 0.25(200) + 0.5(100)$.
3. *Work it out.* $1.25/5 = 0.25$, $1.25/5 = 0.25$, $2.5/5 = 0.5$. Then $0.25 \times 400 = 100$, $0.25 \times 200 = 50$, $0.5 \times 100 = 50$, and $100+50+50 = 200$.
4. *Result.* Time shares 0.25, 0.25 and 0.5, and a time-weighted mean of 200 MFLOPS.

So the arithmetic mean weighted by time and the harmonic mean weighted by work give the same answer, because both equal total work over total time. The mistake in (b) was weighting an arithmetic mean by work.

Answer. (a) 200 MFLOPS. (b) 233.3 and 275 MFLOPS, both too high. (c) 5 s total, so 200 MFLOPS. (d) Time shares 0.25, 0.25 and 0.5, giving 200 MFLOPS again.

B2. Sturges and a histogram. *In plain words.* A histogram groups the data into bins and counts how many land in each. Sturges' rule picks the number of bins from the sample size, $1 + \log_2 n$. Then you sort the data, count per bin, and read off the shape, and the point of (d) is that a different bin size can make the same data look different.

(a)

1. *Formula.* $k = 1 + \log_2 n$ (Summ p.54), where k is the number of bins and n is the number of data points.
2. *Plug in.* $k = 1 + \log_2 20$.
3. *Work it out.* $2^4 = 16$ and $2^5 = 32$, and 20 sits between them, so $\log_2 20$ is between 4 and 5, about 4.32. Then $k = 1 + 4.32 = 5.32$.
4. *Result.* The slide floors it and HW3 rounded it, and both give 5 bins.

(b)

1. *Formula.* Bin width = $(\max - \min)/k$, and relative frequency = count$/n$.
2. *Plug in.* Width = $(35 - 10)/5$.
3. *Work it out.* $35 - 10 = 25$ and $25/5 = 5$ ms, so the edges are 10, 15, 20, 25, 30, 35. Sorted, the data are 10, 12, 14, 16, 17, 17, 18, 19, 19, 20, 21, 22, 23, 24, 25, 27, 28, 30, 31, 35. Count each bin and divide by 20, for example $3/20 = 0.15 = 15\%$ and $7/20 = 0.35 = 35\%$.
4. *Result.* The table below.

Bin (ms)	$10 \le x \le 15$	$15 < x \le 20$	$20 < x \le 25$	$25 < x \le 30$	$30 < x \le 35$
Values	10, 12, 14	16, 17, 17, 18, 19, 19, 20	21, 22, 23, 24, 25	27, 28, 30	31, 35
Count	3	7	5	3	2
Relative frequency	$3/20 = 15\%$	$7/20 = 35\%$	$5/20 = 25\%$	$3/20 = 15\%$	$2/20 = 10\%$

The counts add to $3 + 7 + 5 + 3 + 2 = 20$ and the percentages to 100%, as the slides plot it (Summ p.54-55).

(c) It peaks in the 15 to 20 bin and tails off further to the right than to the left, so it's mildly right-skewed. Check with the mean and median.

1. *Formula.* $\bar{x} = \frac{1}{n}\sum x_i$. For even n the median is the average of the $(n/2)$-th and $(n/2+1)$-th sorted values (Summ p.14).
2. *Plug in.* $\bar{x}$ = sum/20, and the median is the average of the 10th and 11th sorted values, 20 and 21.
3. *Work it out.* By bin, the sums are $36 + 126 + 115 + 85 + 66 = 428$, so $\bar{x} = 428/20 = 21.4$. Median = $(20 + 21)/2 = 41/2 = 20.5$. The course convention $r = n(0.5) + 0.5 = 20(0.5) + 0.5 = 10.5$ points to the same spot.
4. *Result.* Mean 21.4 ms and median 20.5 ms.

The mean sits above the median, which is what a right tail does.

(d) With 2.5 ms bins the edges are 10, 12.5, 15, ..., 35, and counting each one gives 2, 1, 3, 4, 2, 3, 1, 2, 1, 1. The single peak breaks into bumps that are just noise from having about two points per bin. Same data, different cell size, different shape, which is why the slides say cell size is critical (Summ p.54, Summ p.56).

(e)

1. *Formula.* $k = 1 + \log_2 n$.
2. *Plug in.* $k = 1 + \log_2 1000$.
3. *Work it out.* $2^{10} = 1024$, which is just above 1000, so $\log_2 1000$ is just under 10, about 9.97. Then $k = 1 + 9.97 = 10.97$.
4. *Result.* The slide's floor gives 10 bins and rounding gives 11. Say which convention you used.

Answer. (a) 5 bins. (b) Counts 3, 7, 5, 3, 2, which is 15%, 35%, 25%, 15%, 10%. (c) Mildly right-skewed, mean 21.4 above median 20.5. (d) It turns bumpy and noisy, so bin size changes the picture. (e) 10 bins by the floor, 11 by rounding.

B3. Chi-square goodness of fit. *In plain words.* Goodness of fit asks if the counts you saw match the counts a claim predicts. Here the claim is an even split, so you'd expect 30 per server. The chi-square statistic adds up how far each count is from 30, scaled by 30, and a big total means the even-split claim doesn't fit.

(a) H_0 says each server gets a quarter of the requests, and H_1 says the shares aren't all equal.

1. *Formula.* Expected count $e_i = n \times p_i$, and $\chi^2 = \sum(o_i - e_i)^2/e_i$ (chi-square goodness-of-fit handout), where o_i is the observed count, e_i the expected count, n the total and p_i the claimed share. Degrees of freedom are $k - m$, where k is the number of cells and m the number of quantities taken from the data to get the e_i.

2. *Plug in.* $e_i = 120 \times \frac{1}{4}$ for every server, and

$$\chi^2 = \frac{(42-30)^2}{30} + \frac{(20-30)^2}{30} + \frac{(34-30)^2}{30} + \frac{(24-30)^2}{30}.$$

3. *Work it out.* $e_i = 120/4 = 30$, well above 5. The differences are 12, -10, 4 and -6, and their squares are 144, 100, 16 and 36. They add to $144 + 100 + 16 + 36 = 296$, and $296/30 = 9.87$. Only the total of 120 came from the data, so $m = 1$ and df $= 4 - 1 = 3$.
4. *Result.* Expected 30 each, $\chi^2 = 9.87$ on 3 df.

(b) The test rejects for large χ^2 only, so the cutoff is the value with area $1 - \alpha$ below it on 3 df.

1. *Rule.* Reject if χ^2 is bigger than the 3 df table value with area $1 - \alpha$ below it.
2. *Plug in.* At $\alpha = 0.05$ the cutoff is the 0.95 value, 7.81. At $\alpha = 0.01$ it's the 0.99 value, 11.34.
3. *Work it out.* $9.87 > 7.81$, so reject at 0.05. $9.87 < 11.34$, so don't reject at 0.01.
4. *Result.* The statistic falls between the 0.95 and 0.99 points, so $0.01 < p < 0.05$ (software gives 0.020).

(c) Each parameter estimated from the same data adds one to m. So $m = 1 + 1 = 2$ and df $= 4 - 2 = 2$ (handout).

(d) The χ^2 statistic only follows the chi-square distribution approximately, and the approximation is poor when expected counts are small. The handout's Poisson example merges the end cells so that no expected frequency is below 5.

(e) No. Failing to reject isn't evidence for H_0, only an absence of evidence against it (HT p.4, HT p.14). You'd report the observed shares and how big an imbalance 120 requests could have detected.

Answer. $\chi^2 \approx 9.87$ on 3 df, so reject an even split at 0.05 but not at 0.01, with $0.01 < p < 0.05$.

B4. Confidence interval for a variance. *In plain words.* We want a range that probably contains the true spread of run times, not just the spread in these 10 runs. The formula divides $(n-1)s^2$ by two chi-square table values. Since chi-square is lopsided, the range stretches much farther above s^2 than below it.

(a)

1. *Formula.* $\bar{x} = \frac{1}{n}\sum x_i$ and $s^2 = \dfrac{\sum(x_i - \bar{x})^2}{n-1}$, where n is the number of runs.
2. *Plug in.* $\bar{x} = (44 + 46 + 47 + 47 + 50 + 50 + 52 + 53 + 55 + 56)/10$.
3. *Work it out.* The sum is 500, so $\bar{x} = 500/10 = 50$. Subtract 50 from each value to get $-6, -4, -3, -3, 0, 0, 2, 3, 5, 6$. Square them to get 36, 16, 9, 9, 0, 0, 4, 9, 25, 36, which add to 144. Then $s^2 = 144/(10-1) = 144/9 = 16$, and $s = \sqrt{16} = 4$.
4. *Result.* $\bar{x} = 50$ ms, $s^2 = 16$ ms^2, $s = 4$ ms.

(b)

1. *Formula.* $\dfrac{(n-1)s^2}{\chi^2_U} \le \sigma^2 \le \dfrac{(n-1)s^2}{\chi^2_L}$ on $n-1$ df (CI p.59), where χ^2_L is the value with area $\alpha/2$ below it and χ^2_U the value with area $1 - \alpha/2$ below it.
2. *Plug in.* $n - 1 = 9$ and $(n-1)s^2 = 9 \times 16 = 144$. For 95%, $\alpha/2 = 0.025$, so from the 9 df row $\chi^2_L = 2.70$ and $\chi^2_U = 19.02$. That gives $\dfrac{144}{19.02} \le \sigma^2 \le \dfrac{144}{2.70}$.
3. *Work it out.* $144/19.02 = 7.57$ and $144/2.70 = 53.33$. For σ, take square roots, $\sqrt{7.57} = 2.75$ and $\sqrt{53.33} = 7.30$.
4. *Result.* $7.57 \le \sigma^2 \le 53.33$ ms^2, and $2.75 \le \sigma \le 7.30$ ms.

Note the upper end is divided by the small critical value, and it's lopsided, from $16 - 7.57 = 8.4$ below s^2 to $53.33 - 16 = 37.3$ above it (CI p.60).

(c)

1. *Formula.* Same as (b), with $\alpha = 0.10$, so the 0.05 and 0.95 points.
2. *Plug in.* $\chi^2_L = 3.33$ and $\chi^2_U = 16.92$, so $\dfrac{144}{16.92} \le \sigma^2 \le \dfrac{144}{3.33}$.
3. *Work it out.* $144/16.92 = 8.51$ and $144/3.33 = 43.24$.
4. *Result.* $8.51 \le \sigma^2 \le 43.24$ ms^2.

Software with unrounded critical values gives 43.31 for the upper end, so the gap is just table rounding.

(d) No. The spec's standard deviation of 3 ms means $\sigma^2 = 3^2 = 9$, and 9 lies inside (7.57, 53.33). Equivalently 3 is inside (2.75, 7.30). Ten runs pin σ down only to somewhere between about 2.8 and 7.3 ms, so they can't rule out the spec.

(e) It assumes the run times themselves are normal (CI p.59). If they aren't, the interval can be badly off, especially at small n (CI p.62), and unlike the t interval for the mean the Central Limit Theorem doesn't fix it. Heavy-tailed run times, which are common, make it too narrow.

Answer. $\bar{x} = 50$ ms and $s^2 = 16$ ms^2. The 95% interval is 7.57 to 53.33 ms^2 for σ^2 and 2.75 to 7.30 ms for σ. The 90% interval is 8.51 to 43.24 ms^2. The 3 ms spec isn't contradicted.

B5. Comparing two proportions. *In plain words.* Two separate groups each have a miss rate, and we ask if the gap between them is bigger than chance would give. The Z test divides the gap by its standard error (the typical size of random wobble). For the test we assume no change, so we merge both groups into one rate, but for the interval we keep them separate.

(a) H_0 says $p_{\text{old}} = p_{\text{new}}$ and H_1 says $p_{\text{old}} \neq p_{\text{new}}$, two-sided because the question is whether the rate changed.

1. *Formula.* $\hat{p}_1 = x_1/n_1$, $\hat{p}_2 = x_2/n_2$, pooled $\hat{p} = (x_1 + x_2)/(n_1 + n_2)$, and

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}}$$

(hypothesis-testing study guide), where x is the number of misses and n the number of requests in each group.

2. *Plug in.* $\hat{p}_1 = 180/800$, $\hat{p}_2 = 140/800$, $\hat{p} = (180+140)/(800+800)$, and $Z = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/800 + 1/800)}}$.
3. *Work it out.* $\hat{p}_1 = 0.225$, $\hat{p}_2 = 0.175$, and $\hat{p} = 320/1600 = 0.2$. Then $0.2 \times 0.8 = 0.16$ and $1/800 + 1/800 = 2/800 = 0.0025$. So SE $= \sqrt{0.16 \times 0.0025} = \sqrt{0.0004} = 0.02$. The gap is $0.225 - 0.175 = 0.05$, and $Z = 0.05/0.02 = 2.5$.
4. *Result.* $Z = 2.5$.

(b)

1. *Formula.* Reject if $|Z| > z[\alpha/2]$, and two-sided $p = 2\big(1 - \Phi(|Z|)\big)$.
2. *Plug in.* $z[0.025] = 1.960$, $z[0.005] = 2.576$, and $p = 2(1 - \Phi(2.5)) = 2(1 - 0.9938)$.
3. *Work it out.* $2.5 > 1.960$, so reject at 0.05. $2.5 < 2.576$, so don't reject at 0.01. $1 - 0.9938 = 0.0062$ and $2 \times 0.0062 = 0.0124$.
4. *Result.* Reject at 0.05, don't reject at 0.01, $p = 0.0124$.

(c) Each rate's variance is $p(1-p)/n$ as on CI p.47, and the variances of two independent estimates add, the same way they do in Welch's standard error (Comp p.12).

1. *Formula.* $(\hat{p}_1 - \hat{p}_2) \pm z[\alpha/2]\sqrt{\dfrac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \dfrac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$.
2. *Plug in.* $0.05 \pm 1.96\sqrt{\dfrac{0.225 \times 0.775}{800} + \dfrac{0.175 \times 0.825}{800}}$.
3. *Work it out.* $0.225 \times 0.775 = 0.174375$ and $0.175 \times 0.825 = 0.144375$. They add to 0.31875, and $0.31875/800 = 0.000398$. Then $\sqrt{0.000398} = 0.0200$, and $1.96 \times 0.0200 = 0.039$. Lower end $0.05 - 0.039 = 0.011$, upper end $0.05 + 0.039 = 0.089$.
4. *Result.* $(0.011,\ 0.089)$.

The new configuration cuts the miss rate by somewhere between 1.1 and 8.9 percentage points. The point estimate, 5 points out of 22.5, is a relative drop of $0.05/0.225 = 22\%$.

(d) The test computes the standard error assuming H_0 is true, and under H_0 there's one common rate, so the best estimate of it uses all 1600 requests. The interval estimates the difference without assuming it's zero, so each group keeps its own rate.

(e) Check that every count is big. Old has 180 misses and $800 - 180 = 620$ hits, new has 140 misses and $800 - 140 = 660$ hits. All four are far above 10, so the normal approximation is fine (CI p.47, confidence-intervals study guide). The two groups are separate sets of requests, so this is the independent two-proportion test. If both configurations had been judged on the same requests you'd need McNemar's test instead (hypothesis-testing study guide).

Answer. $Z = 2.5$ and $p \approx 0.012$, so reject at 0.05 but not at 0.01. The drop is 0.05 with 95% interval 0.011 to 0.089.

B6. Sample size and the sign test. *In plain words.* Parts (a) to (c) plug into the sample-size formula from the slides. Bigger noise or a smaller target difference means more runs, and pairing helps a lot because it removes input-to-input noise. Part (d) is the sign test, which just counts how many inputs got faster and asks if that's more than a coin flip would give.

(a) Power 0.80 means $\beta = 0.20$, and $z[0.20] = 0.842$, which the slides round to 0.84 (HT p.19).

1. *Formula.* n per group $= 2\big((z[\alpha/2] + z[\beta])\, s/\delta\big)^2$ (Comp p.25), where s is the run-to-run standard deviation and δ the difference to detect.
2. *Plug in.* $n = 2\left(\dfrac{(1.96 + 0.84) \times 10}{5}\right)^2$.
3. *Work it out.* $1.96 + 0.84 = 2.8$. Then $2.8 \times 10 = 28$, and $28/5 = 5.6$. Squaring gives $5.6^2 = 31.36$, and $2 \times 31.36 = 62.72$. Always round up.
4. *Result.* 63 runs per build, $2 \times 63 = 126$ in all.

(b)

1. *Formula.* $n = \big((z[\alpha/2] + z[\beta])\, s_d/\delta\big)^2$ (Comp p.25), where s_d is the standard deviation of the paired differences.
2. *Plug in.* $n = \left(\dfrac{2.8 \times 4}{5}\right)^2$.
3. *Work it out.* $2.8 \times 4 = 11.2$, $11.2/5 = 2.24$, and $2.24^2 = 5.02$. Round up.
4. *Result.* 6 pairs.

Pairing removes the input-to-input variation, so the spread that matters drops from 10 to 4 and the factor of 2 disappears. That's 6 pairs against 126 runs.

(c)

1. *Formula.* Same as (a). Halving δ multiplies n by four (HT p.19).
2. *Plug in.* $n = 2\left(\dfrac{2.8 \times 10}{2.5}\right)^2$.
3. *Work it out.* $2.8 \times 10 = 28$, $28/2.5 = 11.2$, $11.2^2 = 125.44$, and $2 \times 125.44 = 250.88$. Round up. As a check, $4 \times 62.72 = 250.88$.
4. *Result.* 251 runs per build.

(d) Under H_0 the median difference is 0, so each nonzero difference is a fair coin flip (Comp p.23).

1. *Formula.* Drop zeros, let n be the number left and S the number of positive differences. Under H_0, $S \sim \text{Binomial}(n, \frac{1}{2})$, so $P(S = j) = \binom{n}{j}/2^n$, and two-sided $p = 2\,P(S \geq s)$ when s is above $n/2$.
2. *Plug in.* Input 4 is 0, so drop it and $n = 11 - 1 = 10$. Only input 6 is negative, so $S = 10 - 1 = 9$. Then $p = 2\,\dfrac{\binom{10}{9} + \binom{10}{10}}{2^{10}}$.
3. *Work it out.* $\binom{10}{9} = 10$ and $\binom{10}{10} = 1$, so the top is $10 + 1 = 11$. $2^{10} = 1024$, so $P(S \geq 9) = 11/1024 = 0.0107$. Then $p = 2 \times 0.0107 = 22/1024 = 0.0215$.
4. *Result.* $p = 0.0215 < 0.05$, so reject.

The new build is faster on more inputs than chance would explain.

(e) The differences are strongly right-skewed. Two of them, 12.0 and 25.0, are far above the rest, which sit between -0.8 and 3.1. The mean difference is $46.3/11 = 4.2$ ms while the median (the 6th of 11 sorted values) is 1.2 ms. Paired t assumes roughly normal differences, and signed-rank assumes symmetric ones, so for skewed paired differences the slides send you to the sign test or a bootstrap (Comp p.17, Comp p.21). The two big values also inflate s_d, and a paired t on these 11 differences gives only $t \approx 1.81$, $p \approx 0.10$ (software, just to show the contrast). That pick has to come from the design and a plot before any p-value is computed, since choosing whichever test clears 0.05 is p-hacking (Comp p.18). The sign test's claim is also narrower. It says the new build wins on most inputs, not how much faster it is on average.

Answer. (a) 63 runs per build. (b) 6 pairs. (c) 251 runs per build. (d) $S = 9$ of 10, $p \approx 0.021$, so reject. (e) The differences are skewed, so the paired t test's normality assumption fails.